

Proof Theory of Public Announcement Logic

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Chapter 1

Introduction

1.1 Modal Logic and Epistemic Logic

As Blackburn et al. [3, p. viii] put it, ask three modal logicians what modal logic is, and you are likely to get at least three different answers. According to [3], modal logic aims to give an expression of various relational structures by a formal language. A *relational structure* is simply a set together with a collection of relations on that set [3, p. viii], ranging from simple numerical comparisons, like one number being greater than another, to more complex scenarios, such as yesterday being earlier than today.

In 1963, Saul Kripke made a groundbreaking contribution by introducing *Kripke semantics* [21]. This semantic framework, based on the concept of possible worlds, has since become one of the most widely adopted one in modal logic. It facilitates the interpretation of modal operators such as $\Box$ (necessity) and $\Diamond$ (possibility), by considering their truth values across different “possible worlds” and the accessibility relation R between these worlds. Kripke evaluated φ is *necessary* (notation: $\Box\varphi$) to be true in a world w if and only if φ is true in every world v such that wRv , i.e. v is *accessible to* w . Based on relational structures between possible worlds, semantics for modal logics can be defined. Moreover it opens up a wealth of possibilities for defining different types of model structures under various conditions on R [9]. For example, when R is reflexive, i.e. for every w , wRw holds, the axiom $\Box p \rightarrow p$ is obtained. The development and impact of modal logic, particularly in the context of Kripke semantics, is extensively documented in various key texts. For a more comprehensive historical perspective, see Chellas [5], Hughes and Cresswell [17], and Blackburn et al. [3].

The exploration of *Epistemic logic*, as detailed in sources such as [7] and [46], takes us into the realm of philosophical logic, focusing on how knowledge and belief can be formally represented and reasoned about. This field of study was significantly influenced by the work of G. H. von Wright, a Finish philosopher, who was one of the early pioneers in this field to offer a deductive logical system tailored with an intended epistemic interpretation [12].

In von Wright's epistemic system, he employed the modality K to symbolize knowledge, with Kp indicating that the proposition p is known. His system incorporated key modal axioms, specifically the following axioms (K) and (T) which were foundational to his epistemic logic framework:

$$\begin{aligned} \text{(K)} & K(p \rightarrow q) \rightarrow (Kp \rightarrow Kq); \\ \text{(T)} & Kp \rightarrow p. \end{aligned}$$

Here axiom K is read as “if p implies q is known, then p is known implies that q is known”, and axiom T as “if p is known, then p is true”. This system is now called **KT**. Additionally, von Wright explored the epistemic interpretations of other modal axioms, particularly the following axioms (4) and (5)¹:

$$\begin{aligned} \text{(4)} & Kp \rightarrow KKp; \\ \text{(5)} & \neg Kp \rightarrow K\neg Kp. \end{aligned}$$

Intuitively, axiom 4 expresses that “ p is known implies that the fact ‘ p is known’ is known”, while 5 says that “if p is not known, then ‘ p is not known’ is a known fact”. Axioms (4) and (5) are also called *Positive Introspection Axiom* and *Negative Introspection Axiom*, respectively. Reading epistemically, (4) indicates that knowing p implies knowing that of knowing p , while (5) means that not knowing p implies knowing the fact of not knowing p . The system known as **S4** expands **KT** with Axiom (4), and by adding axiom (5) to **S4**, we get system **S5**. Despite von Wright's contention that axiom (4) is doubtful and axiom (5) is plainly unacceptable [12, p.71 and p.77], these systems appear to be the most studied logics of knowledge (cf. [46]). While von Wright didn't mention the modalities of belief, the logician we are about to introduce in the next paragraph managed to fill the gap.

A decade after that, Hintikka built upon the logical foundation set by von Wright to provide a clear semantic analysis on concepts of knowledge and belief. While in von Wright's work, knowledge is treated as *absolute* without specifying

¹Instead of using the formations from von Wright's original texts, we use the formations from [3] to make them more accessible and easier to be understood.

the knowledge bearer, he suggested that a system with modalities such as “known to a ” can be developed [12, p. 35]. This is exactly one of the significant contributions made by Hintikka, which is the introduction of modal operators K_a (representing knowledge) under **S4** and B_a (representing belief) [15], where K_ap (or B_ap , respectively) means that the individual a knows (or believes, respectively) that p is the case. These operators managed to capture the essence of knowledge and belief with a simple and direct idea. A world v is *epistemically accessible* from a scenario w if and only if everything the agent knows in w is true in v [48, §8.2]. Then to say that the knower (or the believer) a knows (or believes) the proposition p in w is to say p is true in all worlds that are *epistemically accessible* to w ². The same outcome can be obtained by applying Kripke semantics to epistemic operators. As emphasized by van Ditmarsch et al. [7, p. 2],

Hintikka’s work not only ended an era of attempts in philosophy to have the means to formally discuss these notions, it also marked the start of a prosperous half-century of further developments in epistemics, often from surprisingly new angles.

Further developments here refer to various expansions of epistemic logic, including Public Announcement Logic, which is about to be introduced in the next section.

1.2 Public Announcement Logic as an expansion of Epistemic Logic

The multi-agent version of epistemic logic introduced by Hintikka [15] is capable of describing knowledge among different agents. However, as Plaza pointed out, it falls short in capturing the “dynamics of knowledge” [37]. To tackle this limitation, Plaza introduced Public Announcement Logic (**PAL** for short), a framework designed to analyze how knowledge evolves within multi-agent systems in response to public announcements [37].

Plaza’s inspiration for **PAL** was drawn from classic logical puzzles involving exchanges of information. One of the most famous one is the Muddy Children Puzzle, which is discussed by many logicians, see for instance, Parikh [34]. This puzzle, with its various iterations, serves as an excellent illustration of the dynam-

²Some may prefer to use “scenarios” rather than worlds (see [16]).

ics of knowledge. Here we proceed our discussion based on the following story from [46], which is particularly insightful:

Example 1. (Muddy Children) A father informs three children in a room that at least one of them has mud on their forehead. While each child can see the others' foreheads, they cannot see their own. The father then asks if they know whether they are dirty. Initially, no child can answer affirmatively, and all the three of them is aware of it. As the father repeats his question, by the third round, all children correctly identify their state at the same time. This situation reveals that all three children are, in fact, muddy.

No need to mention other preliminary assumptions here, such as every child is intelligent and truthful, etc. The intriguing aspect of Example 1 lies in the evolving knowledge of the children with the public announcement made by the father as well as the silence of other children in the first two rounds. More specifically, the father's announcement in Example 1 – “At least one of you has mud on your forehead” – initially appears to be trivial, since each child can already see the mud on the others' foreheads. However, what really happened after the announcement is that new information is created. The critical question that arises is: How can we express this process of new information generation using a formal language?

To model this dynamic knowledge progression, Plaza extended the multi-agent version of modal logic **S5** with a public announcement operator $[\varphi]\psi$, interpreted as “if φ is true, then after the public announcement of φ , ψ holds.”³ Alternatively, we may sometimes say “after *update* with φ , it holds that ψ ”. Also in 1997, Gerbrandy and Groeneveld [13] independently formulated results similar to Plaza's by introducing a modal operator $[\varphi]_a\psi$ specifying the maker of the announcement, interpreted as “after *a* consciously updates with φ , ψ is true.” Both their works involve the concept of model update, which will be further discussed in Chapter 2.

1.3 Proof Theory for PAL

1.3.1 Hilbert Systems for Modal Logic and PAL

Hilbert system, also recognized as Hilbert-style axiomatization, is a system of formal deduction named after German mathematician David Hilbert. In modern logic

³In Plaza's original concept, public announcement was represented as a binary operation $+$, which is equivalent to $\neg[\varphi]\neg\psi$.

studies, it usually refers to a proof system characterized by schemes of logical axioms and inference rules [20]. For instance, the axiom **K** mentioned in Section 1.1 is regarded as an axiom. And by combining several axioms we get an intended Hilbert system, which will be defined in Section 2.4. For those seeking a comprehensive understanding of the historical evolution and nuances of the Hilbert system, a detailed introduction can be found in [26].

Given a given Hilbert system, the *Completeness* and *soundness* of it are two key concepts that bridges the gap between syntactic and semantic perspectives [3]. Given a semantically specified logic Σ and its corresponding proof system, it is necessary to prove that the semantic validity coincides with the theoremhood [7]. Generally, the implication from theoremhood to semantic validity is called *soundness*, and the opposite direction is called *completeness*. In a stronger expression, completeness is also used to mean that the theoremhood and validity match perfectly, i.e. they are equivalent. In this work we use the separated notation.

Soundness is often easier to show(cf. [7]). When it comes to proving completeness in modal logic, one of the most widely used approach is by introducing *canonical models* [3]. The idea behind this approach is based on the concept of semantic validity. Given a formula φ , φ is valid on a model class if and only if there is no counter model with respect to the model class in which φ is invalid. And to build canonical models is to build models out of *maximal consistent sets of formulas*, such that for any formula φ , either φ or $\neg\varphi$ is determined to be found in some worlds. Readers may refer to [3, Chapter 4, Notes] for a more detailed introduction of the history and methodology of canonical models.

The approach to proving the semantic completeness of **PAL** is quite different, since the standard techniques do not directly apply to **PAL** [7, p. 209]. Yanjing Wang and Qinxiang Cao [47] have suggested an alternative axiomatization, whose completeness can be proved directly by using the canonical model. Alternatively, a proof by translation is originally given by Plaza [37], in which the formulas of **PAL** are reduced to that of **S5** by axioms of **PAL**, i.e. a formula of the form $[\varphi]\psi$ is equivalent to another **S5** formula γ . This is exactly the reason why the axioms for **PAL** are often called recursion axioms.

1.3.2 Sequent Calculi for Modal Logic

Sequent calculus, also called Gentzen system, is a logic system devised by Gerhard Gentzen in the 1930s, which opened a field of proof theory by structuring logical deductions [10, 11]. Given two multisets of formula Γ and Δ , the sequent $\Gamma \Rightarrow \Delta$ expresses the implication from premises Γ to conclusions Δ , i.e., if all

formulas in Γ hold then some formula in Δ holds. A crucial aspect of sequent calculus is the concept of cut-elimination (Gentzen’s *Hauptsatz*), which simplifies proofs by removing instances of the cut rule, thereby yielding more intuitive and transparent “cut-free” proofs, where the cut rule is of the following form:

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut).$$

This process uncovers the constructive content of proofs and contributes to our understanding of proof normalization and proof search [45].

When we turn our eyes on modal logics, sequent calculi were provided, for instance, by Ohnishi and Matsumoto [31]. For modal logics such as **KT** and **S4**, traditional sequent calculi are acknowledged to be sound, complete, and cut-free in [31]. On the other hand, for modal logic **S5**, the cut-elimination theorem fails for a sequent calculus **G(S5)** of modal logic **S5**, as explained in [32, p.116]. To recover the cut elimination theorem, it is necessary to either incorporate global side conditions, as in the work of Braüner [4], or extend the sequent format with additional structures such as hypersequents (for example, see Poggiolesi [38]), display calculus (for example, see Dosen [8]), or labelled sequents by Negri [27]. In particular, Negri [27] introduced a universal method for producing contraction- and cut-free *labelled* sequent calculi for a broad spectrum of normal modal logics. Instead of regaining the cut elimination, Takano [44] established that, while system **G(S5)** does not allow the elimination of cuts, it can be shown that every application of the cut rule in **G(S5)** can be replaced by an analytic application of the cut rule, where the cut rule is *analytic* if the cut formula φ is a subformula of the conclusion $\Gamma, \Pi \Rightarrow \Delta, \Sigma$ of the cut rule above.

1.3.3 Sequent Calculi for PAL

Many attempts have been made to construct a sequent calculi for **PAL**, and they can be divided into two categories, namely *with label* and *without label*. In 2005, Negri [27] introduced the labelled formalism of the following form:

$$x : \varphi,$$

where x : denotes a possible world. This approach has been employed by several authors in the construction of labelled *cut-free* sequent calculi for **PAL**. In 2011, Maffezoli and Negri introduced a Gentzen-style proof system for **PAL** with Gerbrandy and Groenveled announcements **G3PAL**, where the formulas were formed

as

$$x :^{\psi, A} \varphi,$$

in which ψ, A denote the announcement made by A [25]. Then Balbiani et al. proposed another sequent calculus with labels for **PAL**, since **G3PAL** proposed in [25] is not complete [36]. Also Nomura et al. fixed the flaw to provide **GPAL** and specify the validity notion for sequents [30]. At last, Wu et al. introduced their labelled sequent calculus for **PAL** in the form of

$$x : \varphi,$$

without stack for announcements [50]. Also their rules were from recursion axioms, and the sequent calculus is decidable.

Furthermore, sequent calculi have been defined for various logics in the field of dynamic epistemic logic, even if one insists on adhering to the conventional notion of sequents and avoiding labelled formalism. Ma and Pietarinen introduced a cut-free non-labelled sequent calculus for a dynamic logic of abductive inferences [24]. Hatano and Sano [14] introduced a non-labelled sequent calculus for a constructive variant of dynamic logic of relation changers by van Benthem and Liu [2]. They transformed each of the recursion axioms to left and right inference rules of the proposed sequent calculus and emphasized the importance of the complexity measure from [7]. Then Wirsing and Knapp [49] provided a non-labelled sequent calculus, denoted as **G4_{P,A}**, for action model logic with **S4** as its base logic instead of **S5**.

Recently, Liu and Sano [22] introduced sequent calculus **G(S5PAL)** for the public announcement expansions of multi-agent **S5**. Unlike labelled sequent calculi, their proposed calculus do not internalize accessibility relations or incorporate labels. Instead, like in [14, 49], their inference rules for public announcement operators are derived naturally from the recursion axioms specified in [7]. As shown in [22], although **G(S5PAL)** does not enjoy the subformula property (i.e., a sequent is derivable only in terms of sequents consisting of subformulas of the original sequent), it enjoys the extended subformula property by extending the notion of subformula to the notion of closure and using the complexity measure from [7]. However, as Hatano and Sano have pointed out, the complexity measure is not properly defined [14]. In this paper, following Hatano and Sano's approach [14], we give a proof of the extended subformula property by the restricted definition of complexity, in which case a stronger result is obtained.

Moreover, to link sequent calculus with semantics, it is necessary to show the theoremhood of a given Hilbert system and that of its corresponding sequent

calculus are equivalent, as documented in Gentzen’s work [10, 11]. That is to say, if a formula φ is provable in a given sequent calculus, it implies that it can also be proven in the Hilbert system corresponding to it (also the opposite direction). Hence due to the completeness property of that Hilbert system, this also implies that φ is valid in the associated Kripke model. Another key aspect of sequent calculus is the ‘decision problem,’ which will be discussed in detail in Sections 3.4 and 4.4. The resolution of this problem follows from cut-elimination and the extended subformula property.

1.4 What is New on This Paper?

In this work, we take a general review on some important semantic and syntactic properties mentioned above. Moreover we introduce sequent calculi named **G(K45PAL)** and **G(S5PAL)** for the public announcement expansions of multi-agent **K45** and **S5**, respectively. Unlike labelled sequent calculi proposed in [25, 35, 30, 50], our proposed calculi do not internalize accessibility relations or incorporate labels. Instead, like in [14, 49], our inference rules for public announcement operators are derived naturally from the recursion axioms specified in [7] (it is worth noting that our inference rules for public announcement operators have almost the same form as those in [50] when disregarding the labels). Also we show the provabilities of **G(K45PAL)** and **G(S5PAL)** and their corresponding Hilbert systems are equivalent.

To emphasize the proof-theoretic importance of the complexity measure introduced in [7], we first establish that **G(K45PAL)** enjoys the cut-elimination theorem. Although our sequent calculi do not enjoy the subformula property (i.e., a sequent is derivable only in terms of sequents consisting of subformulas of the original sequent), we naturally extend the notion of subformula to the notion of closure. With this notion of closure, we follow Takano’s approach [44] to syntactically establish that every application of the cut rule in **G(S5PAL)** can be replaced by such an application of the cut rule that the cut formula can be taken from the closure of the conclusion of the cut rule. In other words, our calculus **G(S5PAL)** remains analytic in terms of the notion of closure. We also emphasize that the complexity measure from [7] plays an important role in this syntactic argument. Decidability of **G(K45PAL)** and **G(S5PAL)** can be proved by reducing the formulas that contain public announcement operators to those that do not. In this thesis, we give an alternative proof of the decidability of the two pairs of sequent calculi adapted from [33, Section 4], which is corollaries of the cut-elimination theorem

of $G(\mathbf{K45PAL})$ and the extended subformula property of $G(\mathbf{S5PAL})$, respectively.

1.5 Outline

The structure of this paper is organized as follows: Chapter 2 reviews syntax and semantics of multi-agent version of $\mathbf{K45}$ and $\mathbf{S5}$, together with Hilbert Systems. Hilbert systems $H(\mathbf{K45})$, $H(\mathbf{S5})$, $H(\mathbf{K45PAL})$ and $H(\mathbf{S5PAL})$ are introduced in Section 2.4, whose axioms are from [7, Table 7.1 and Table 7.3]. We use $\Box_a$ instead of K_a and B_a , as it can represent both. Soundness and Completeness of Hilbert systems are discussed in Sections 2.5 and 2.6, respectively. Soundness of the four systems follows directly, while the completeness proof is a little complicated. For $H(\mathbf{K45})$ and $H(\mathbf{S5})$, we apply the canonical model approach in [3, p. 4.2]. As for completeness of $\mathbf{PAL}$, the proof is done by translation introduced in [7, p. 7.4]. Chapter 3 presents the sequent calculi $G(\mathbf{K45})$ and $G(\mathbf{S5})$, together with the cut-elimination theorem of $G(\mathbf{K45})$ and analytic cut property of $G(\mathbf{S5})$, in which the latter is adopted from Takano (1992).

While the chapters above gives detailed background of the research, new contributions of this paper are shown in Chapter 4, where we introduce $G(\mathbf{K45PAL})$ and $G(\mathbf{S5PAL})$ as public announcement extensions for $G(\mathbf{K45})$ and $G(\mathbf{S5})$, respectively. Section 4.2 presents the proof of cut-elimination for $G(\mathbf{K45PAL})$ in terms of the complexity measure introduced by van Ditmarsch et al. (2007). Section 3.3 adopts Takano's strategy (1992) to establish that the cut formula in $G(\mathbf{S5PAL})$ can be restricted to the set of suitably extended subformulas (i.e., closure) of the conclusion of the cut rule. Finally, we take a look into the decision problem of the two sequent calculi. shown in the end of Chapter 3 and Chapter 4, namely Sections 3.4 and 4.4, respectively. Finally, chapter 5 concludes the paper.

Chapter 2

Syntax, Semantics and Hilbert Systems

2.1 Syntax and Semantics

Let Prop be a countably infinite set of propositional variables. The syntax L for epistemic logic is defined inductively as follows:

$$L \ni \varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \rightarrow \psi \mid \Box_a \varphi$$

where $p \in \text{Prop}$. Moreover, the syntax L^+ for public announcement logic is defined inductively as follows:

$$L^+ \ni \varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \rightarrow \psi \mid \Box_a \varphi \mid [\varphi]\psi$$

We use the following abbreviations:

- $\varphi \vee \psi$ is the abbreviation for $\neg\varphi \rightarrow \psi$;
- $\varphi \leftrightarrow \psi$ is the abbreviation for $(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$;
- $\Diamond_a \varphi$ is the abbreviation for $\neg\Box_a \neg\varphi$;
- $\perp$ is the abbreviation of $\neg(p \rightarrow p)$.
- $\top$ is the abbreviation of $\perp \rightarrow \perp$.

Moreover for a finite set Δ , $\bigwedge \Delta$ and $\bigvee \Delta$ are the conjunction or disjunction of all formulas in Δ (when Δ is empty, $\bigwedge \Delta := \top$ and $\bigvee \Delta := \perp$). A $\Box_a$ -formula is a formula of the form $\Box_a \varphi$.

We move on to the semantics. A *frame* F is a tuple $(W, (R_a)_{a \in \text{Ag}})$ where W is a set of states and $R_a \subseteq W \times W$ is a binary relation on W for each $a \in \text{Ag}$. A *model* M is a tuple $(W, (R_a)_{a \in \text{Ag}}, V)$ where $(W, (R_a)_{a \in \text{Ag}})$ is a frame and V is a function from Prop to $\mathcal{P}(W)$. Let $M = (W, (R_a)_{a \in \text{Ag}}, V)$ be a model and $w \in W$. The notion of φ being true at w in M (notation: $M, w \models \varphi$) is defined inductively as follows:

$$\begin{aligned} M, w \models p & \quad \text{iff} \quad w \in V(p), \\ M, w \models \neg \varphi & \quad \text{iff} \quad M, w \not\models \varphi, \\ M, w \models \varphi \wedge \psi & \quad \text{iff} \quad M, w \models \varphi \text{ and } M, w \models \psi, \\ M, w \models \varphi \rightarrow \psi & \quad \text{iff} \quad M, w \not\models \varphi \text{ or } M, w \models \psi, \\ M, w \models \Box_a \varphi & \quad \text{iff} \quad \text{for all } v \in W, w R_a v \text{ implies } M, v \models \varphi, \\ M, w \models [\varphi] \psi & \quad \text{iff} \quad M, w \models \varphi \text{ implies } M^\varphi, w \models \psi, \end{aligned}$$

where $M^\varphi = (W^\varphi, (R_a^\varphi)_{a \in \text{Ag}}, V^\varphi)$, $W^\varphi = \{w \in W \mid M, w \models \varphi\}$, $R_a^\varphi = R_a \cap (W^\varphi \times W^\varphi)$ and $V^\varphi(p) = V(p) \cap W^\varphi$.

A formula φ is *valid* in a model (notation: $M \models \varphi$) iff $M, w \models \varphi$ for all $w \in W$. A formula φ is *valid* in a frame F (notation: $F \models \varphi$) if $(F, V) \models \varphi$ for all valuations V . Given a class $\mathbb{F}$ of frames, φ is *valid* in $\mathbb{F}$ if φ is valid in F for every $F \in \mathbb{F}$.

Definition 2. We define the following conditions on a binary relation R :

$$\begin{aligned} R \text{ is reflexive} & \quad \text{iff} \quad \forall w (w R w); \\ R \text{ is transitive} & \quad \text{iff} \quad \forall w v u (w R v \text{ and } v R u \text{ implies } w R u); \\ R \text{ is symmetric} & \quad \text{iff} \quad \forall w v (w R v \text{ implies } v R w); \\ R \text{ is Euclidean} & \quad \text{iff} \quad \forall w v u (w R v \text{ and } w R u \text{ implies } v R u); \\ R \text{ is serial} & \quad \text{iff} \quad \forall w \exists v (w R v). \end{aligned}$$

Moreover if R is reflexive, transitive and Euclidean (which implies that R is symmetric), we say that R is an *equivlent* relation.

Definition 3. We define a number of well-known classes of frames together with their corresponding classes of frames as listed in Table 2.1.

The classe of frames, for instance, $\mathbb{F}_{\mathbf{K45}}$ is defined to be the class of all transitive and Euclidean frames and $\mathbb{F}_{\mathbf{S5}}$ is to be the class of all reflexive, transitive and

Table 2.1: Classes of frames

$\mathbb{F}_K$	the class of all frames
$\mathbb{F}_{KT}$	the class of reflexive frames
$\mathbb{F}_{K4}$	the class of transitive frames
$\mathbb{F}_{KB}$	the class of symmetric frames
$\mathbb{F}_{K5}$	the class of Euclidean frames
$\mathbb{F}_{KD}$	the class of serial frames
$\mathbb{F}_{S4}$	the class of reflexive and transitive frames
$\mathbb{F}_{K45}$	the class of transitive and Euclidean frames
$\mathbb{F}_{S5}$	the class of reflexive, transitive and Euclidean frames

Euclidean frames, etc. A *logic* is a set of formulas that are valid in the corresponding class of frame in Table 2.1. For instance, **K45** and **S5** are defined to be the sets of all L -formulas that are valid in the class of $\mathbb{F}_{K45}$ and $\mathbb{F}_{S5}$, respectively. The public announcement expansion **K45PAL** of **K45** is the set of all L^+ -formulas that are valid in the class $\mathbb{F}_{K45}$. The public announcement expansion **S5PAL** of **S5** is the set of all L^+ -formulas that are valid in the class $\mathbb{F}_{S5}$.

Note that while **PAL** is also suitable for **K**, **T**, **4**, **5** and **B**, problems may emerge when we combine **D** and **PAL** together, since in the model M^φ updated by $[\varphi]$, the relation R_a^φ may no longer be serial.

2.2 Model Update

In this section we demonstrate the notion of model update with Example 1. To do so we need to build a model under **S5**, and see how it is updated by new information. Here to simplify the discussion, we use figures instead of detailed definitions and arrow for reflexive relations are omitted.

Let p (or q or r , respectively) represent the proposition “Child 1 (or 2 or 3, respectively) have a muddy forehead”. Then the initial knowledge of the three children can be expressed as in the model of Figure 2.1, where “ p, q, r ” is the actual world and R_1 , R_2 and R_3 represents the knowledge of child 1, 2 and 3, respectively. Since the model is under **S5**, R_1 , R_2 and R_3 are equivalent relations, i.e. reflexive, transitive and Euclidean.

Let $\varphi \equiv \neg(\neg p \wedge \neg q \wedge \neg r)$ represent the father’s announcement “At least one of you has mud on your forehead”. Then intuitively, after the announcement $[\varphi]$ is

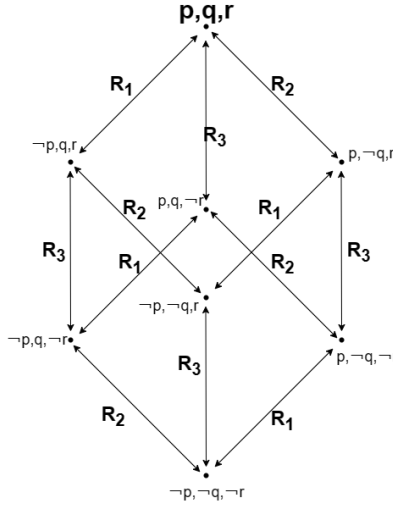


Figure 2.1: Before the public announcement

made, all the children immediately knows that the world “ $\neg p, \neg q, \neg r$ ” is no longer possible, as shown in Figure 2.2.

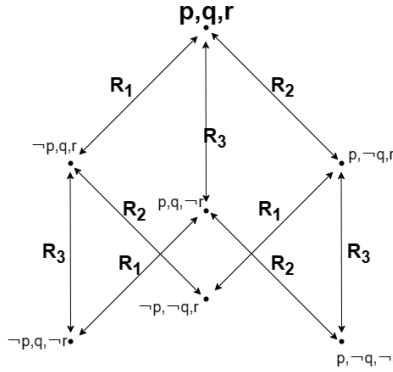


Figure 2.2: First round

Semantically, what happens between Figure 2.1 and Figure 2.2 is the original model M is updated to M^φ by the announcement φ , in which all the states where φ is not valid are erased.

Then after the father’s question and nobody have the answer, what is realized by the children is that none of the worlds “ $p, \neg q, \neg r$ ”, “ $\neg p, q, \neg r$ ” and “ $\neg p, \neg q, r$ ”

is possible. Since, for instance, if “ $p, \neg q, \neg r$ ” is the actual world, then child 1 would have known the answer already (which is, there *is* mud on his forehead), and the thinking process is similar for child 2 and 3. So what happens in the second round, as shown in Figure 2.3, is that all the three worlds mentioned above are erased.

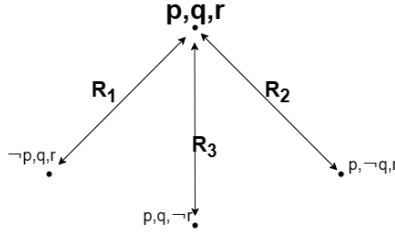


Figure 2.3: Second round

The semantic interpretation here is that the proposition

$$\neg(\Box_1 p \vee \Box_1 \neg p) \wedge \neg(\Box_2 q \vee \Box_2 \neg q) \wedge \neg(\Box_3 r \vee \Box_3 \neg r)$$

(i.e. nobody knows the answer) is valid. Then, for instance, since at the world “ $p, \neg q, \neg r$ ”, $\Box_1 p$ is valid (since it is reflexive), it follows that if this cannot be the actual world.

At last, since still no one have the answer, the similar thinking process is repeated, erasing the rest other three possible worlds and leaving only the actual world possible, as shown in Figure 2.4, which gives answer to all the three children.

p,q,r

Figure 2.4: Last round

2.3 Subformula, Complexity and Closure

The subformula $\text{Sub}(\varphi)$ of an L^+ formula φ is defined inductively as follows:

Definition 4.

$$\begin{aligned}
\text{Sub}(p) &:= \{p\} \\
\text{Sub}(\neg\varphi) &:= \text{Sub}(\varphi) \cup \{\neg\varphi\} \\
\text{Sub}(\varphi \bullet \psi) &:= \text{Sub}(\varphi) \cup \text{Sub}(\psi) \cup \{\varphi \bullet \psi\} \quad (\bullet \in \{\wedge, \rightarrow\}) \\
\text{Sub}(\Box_a\varphi) &:= \text{Sub}(\varphi) \cup \{\Box_a\varphi\}
\end{aligned}$$

The following definition and lemma are adapted from [7, Definition 7.21] and [7, Lemma 7.22], respectively and they are key ingredients for establishing semantic completeness of the public announcement logic in [7].

Definition 5. The complexity c of a formula φ in L^+ is inductively defined as

$$\begin{aligned}
c(p) &:= 1, \\
c(\neg\varphi) &:= 1 + c(\varphi), \\
c(\varphi_1 \bullet \varphi_2) &:= 1 + \max(c(\varphi_1), c(\varphi_2)) \quad (\bullet \in \{\wedge, \rightarrow\}), \\
c(\Box_a\varphi) &:= 1 + c(\varphi), \\
c([\varphi]\psi) &:= (2 + c(\varphi)) \cdot c(\psi).
\end{aligned}$$

The last definition may seem a bit weird but it plays an important syntactic role in the proof of several theorems, as shown in Sections 2.6, 4.2 and 3.3. We define the extended notion $\text{CL}(\varphi)$ of $\text{Sub}(\varphi)$ as follows (our notion is a simplification of [7, Definition 7.27] in the setting without the common knowledge operators). Inference rules for public announcements in our sequent calculi do not satisfy the subformula property but the *extended* version of the subformula property in terms of $\text{CL}(\varphi)$.

Definition 6. The *closure* $\text{CL}(\varphi)$ of an L^+ -formula φ is defined inductively as follows:

$$\begin{aligned}
\text{CL}(p) &:= \{p\} \\
\text{CL}(\neg\varphi) &:= \text{CL}(\varphi) \cup \{\neg\varphi\} \\
\text{CL}(\varphi \bullet \psi) &:= \text{CL}(\varphi) \cup \text{CL}(\psi) \cup \{\varphi \bullet \psi\} \quad (\bullet \in \{\wedge, \rightarrow\}) \\
\text{CL}(\Box_a\varphi) &:= \text{CL}(\varphi) \cup \{\Box_a\varphi\} \\
\text{CL}([\varphi]p) &:= \text{CL}(\varphi) \cup \{p, [\varphi]p\} \\
\text{CL}([\varphi]\neg\psi) &:= \text{CL}(\varphi) \cup \text{CL}([\varphi]\psi) \cup \{[\varphi]\neg\psi\} \\
\text{CL}([\varphi](\psi \bullet \gamma)) &:= \text{CL}([\varphi]\psi) \cup \text{CL}([\varphi]\gamma) \cup \{[\varphi](\psi \bullet \gamma)\} \quad (\bullet \in \{\wedge, \rightarrow\}) \\
\text{CL}([\varphi]\Box_a\psi) &:= \text{CL}(\Box_a[\varphi]\psi) \cup \text{CL}(\varphi) \cup \{[\varphi]\Box_a\psi\} \\
\text{CL}([\varphi][\psi]\gamma) &:= \text{CL}([\varphi \wedge [\varphi]\psi]\gamma) \cup \{[\varphi][\psi]\gamma\}
\end{aligned}$$

For a set Ξ of formulas, we define $\text{CL}(\Xi) := \bigcup_{\varphi \in \Xi} \text{CL}(\varphi)$.

Note that $\text{Sub}(\varphi) \subseteq \text{CL}(\varphi)$ and $\text{CL}(\varphi)$ is finite. Moreover, if a formula φ is from L , it follows that $\text{CL}(\varphi) = \text{Sub}(\varphi)$.

2.4 Hilbert Systems

In this section we introduce the Hilbert system $H(\mathbf{K})$ and its extensions(cf. [3, 7]). First we define the Hilbert system $H(\mathbf{K})$, as shown in Table 2.2.

Table 2.2: Hilbert Systems $H(\mathbf{K})$ and its extensions

All the Axioms and Rules of $H(\mathbf{K})$	
(Taut)	all instances of propositional tautologies
(K)	$\Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a\varphi \rightarrow \Box_a\psi)$
(MP)	From $\varphi \rightarrow \psi$ and φ infer ψ
(Nec)	From φ infer $\Box_a\varphi$
Additional Axiom Schemes	
(T)	$\Box_a\varphi \rightarrow \varphi$
(4)	$\Box_a\varphi \rightarrow \Box_a\Box_a\varphi$
(B)	$\varphi \rightarrow \Box_a\Diamond_a\varphi$
(5)	$\neg\Box_a\varphi \rightarrow \Box_a\neg\Box_a\varphi$
(D)	$\Box_a\varphi \rightarrow \Diamond_a\varphi$

Now we can define the extensions of $H(\mathbf{K})$, as in Table 2.3:

Definition 7. • $H(\mathbf{KT})$ consists of (Taut), (K), (MP), (Nec) and (T).

- $H(\mathbf{KB})$ consists of (Taut), (K), (MP), (Nec) and (B).
- $H(\mathbf{KD})$ consists of (Taut), (K), (MP), (Nec) and (D).
- $H(\mathbf{S4})$ consists of (Taut), (K), (MP), (Nec), (T) and (4).
- $H(\mathbf{K45})$ consists of (Taut), (K), (MP), (Nec), (4) and (5).
- $H(\mathbf{S5})$ consists of (Taut), (K), (MP), (Nec), (T), (4) and (5).
- Hilbert systems $H(\mathbf{K45PAL})$ and $H(\mathbf{S5PAL})$ expands Hilbert systems $H(\mathbf{K45})$ and $H(\mathbf{S5})$ with the rules for **PAL**, respectively.

Table 2.3: Additional Recursion Axioms for **PAL**

$([\Box]at)$	$[\Box]\varphi \leftrightarrow (\varphi \rightarrow \Box\varphi)$
$([\Box]\neg)$	$[\Box]\neg\psi \leftrightarrow (\Box\varphi \rightarrow \neg[\Box]\psi)$
$([\Box]\wedge)$	$[\Box](\psi \wedge \gamma) \leftrightarrow ([\Box]\psi \wedge [\Box]\gamma)$
$([\Box]\rightarrow)$	$[\Box](\psi \rightarrow \gamma) \leftrightarrow ([\Box]\psi \rightarrow [\Box]\gamma)$
$([\Box]\Box)$	$[\Box]\Box_a\psi \leftrightarrow (\Box\varphi \rightarrow \Box_a[\Box]\psi)$
$([\Box]\Box)$	$[\Box][\Box]\gamma \leftrightarrow [\Box] \wedge [\Box]\psi \gamma$

2.5 Soundness

In this section we prove the soundness of $H(\mathbf{K})$ and the extensions of it, which is a well-known fact. The proofs are based on [3, p. 4].

In what follows, we use Λ as an *axiomatic extension* of $H(\mathbf{K})$, i.e. an extension of $H(\mathbf{K})$ by new axiom schemes. A formula φ is a theorem of Λ if $\varphi \in \Lambda$ (notation: $\vdash_{\Lambda} \varphi$).

First we define the derivability of Hilbert systems as follows:

Definition 8. Let Σ be a possibly empty subset of $\{\mathbf{T}, \mathbf{B}, \mathbf{4}, \mathbf{5}, \mathbf{D}\}$ and $H(\mathbf{K}\Sigma)$ be the Hilbert system extending $H(\mathbf{K})$ with the axiom schemes corresponding to Σ . A L formula φ is *derivable* in $H(\mathbf{K}\Sigma)$ (notation: $H(\mathbf{K}\Sigma) \vdash \varphi$) there exists a list of L formulas $\psi_1, \psi_2, \psi_3, \dots, \psi_n$ such that $\psi \equiv \varphi$ and for each i such that $1 \leq i \leq n$, one of the followings hold:

- (1) ψ_i is an axiom;
- (2) ψ_i is obtained by (MP) from ψ_j and ψ_k where $1 \leq j < i$ and $1 \leq k < i$;
- (3) ψ_i is obtained by (NEC) from ψ_j where $1 \leq j < i$.

Definition 9. Let $\Lambda \in \{\mathbf{K45PAL}, \mathbf{S5PAL}\}$. A L^+ formula φ is *derivable* in $H(\Lambda)$ (notation: $H(\Lambda) \vdash \varphi$) there exists a list of L^+ formulas $\psi_1, \psi_2, \psi_3, \dots, \psi_n$ such that $\psi \equiv \varphi$ and for each i such that $1 \leq i \leq n$, one of the followings hold:

- (1) ψ_i is an axiom;
- (2) ψ_i is obtained by (MP) from ψ_j and ψ_k where $1 \leq j < i$ and $1 \leq k < i$;
- (3) ψ_i is obtained by (NEC) from ψ_j where $1 \leq j < i$.

Theorem 10 (Soundness). Let Σ be a possibly empty subset of $\{\mathbf{T}, \mathbf{B}, \mathbf{4}, \mathbf{5}, \mathbf{D}\}$ and $H(\mathbf{K}\Sigma)$ be the Hilbert system extending $H(\mathbf{K})$ with the axiom schemes corresponding to Σ . Then, for any L -formula φ , if φ is a theorem in $H(\mathbf{K}\Sigma)$, then φ is valid in *all* Kripke frames which satisfies the corresponding properties to Σ .

Proof. Fix any formula φ such that $H(\mathbf{K}\Sigma) \vdash \varphi$. Let $M^{\mathbf{K}\Sigma}$ be any Kripke model that satisfies the corresponding properties to Σ . Our goal is to show $M^{\mathbf{K}\Sigma} \models \varphi$. By Definition 8, in all cases it suffices to show that the axioms are valid, and that the rules (MP) and (Nec) preserve validity on the class of frames in question. So we divide the argument into the following cases:

- (Nec) Fix any Kripke model M . Suppose $M \models \psi$. We show $M \models \Box_a \psi$. It suffices to show for all $w \in W$, $M, w \models \Box_a \psi$. Fix any $w \in W$. We show $M, w \models \Box_a \psi$ i.e.

$$\forall u(wR_a u \text{ implies } M, u \models \psi).$$

So fix any $u \in W$. Suppose $wR_a u$. By $M \models \psi$, $M, u \models \psi$ holds clearly.

- (K) Fix any Kripke model M and $w \in W$. We show $M, w \models \Box_a(\psi \rightarrow \gamma) \rightarrow (\Box_a \psi \rightarrow \Box_a \gamma)$. Suppose $M, w \models \Box_a(\psi \rightarrow \gamma)$. We show $M, w \models \Box_a \psi \rightarrow \Box_a \gamma$. Suppose $M, w \models \Box_a \psi$. We show $M, w \models \Box_a \gamma$. It suffices to show

$$\forall u(wR_a u \text{ implies } M, u \models \gamma).$$

So fix any $u \in W$. Suppose $wR_a u$. By $M, w \models \Box_a(\psi \rightarrow \gamma)$, $M, u \models \psi \rightarrow \gamma$. And by $M, w \models \Box_a \psi$, $M, u \models \psi$. Therefore $M, u \models \gamma$ holds.

- (T) Fix any reflexive Kripke model M and $w \in W$. We show $M, w \models \Box_a \psi \rightarrow \psi$. Suppose $M, w \models \Box_a \psi$. We show $M, w \models \psi$. Since M is a reflexive model, $wR_a w$. Then by $M, w \models \Box_a \psi$, $M, w \models \psi$ holds.
- (4) Fix any transitive Kripke model M and $w \in W$. We show $M, w \models \Box_a \psi \rightarrow \Box_a \Box_a \psi$. Suppose $M, w \models \Box_a \psi$. We show $M \models \Box_a \Box_a \psi$. It suffices to show

$$\forall u(wR_a u \text{ implies } M, u \models \Box_a \psi).$$

So fix any $u \in W$. Suppose $wR_a u$. We show $M, u \models \Box_a \psi$. It suffices to show

$$\forall v(uR_a v \text{ implies } M, v \models \psi).$$

So fix any $v \in W$. Suppose $uR_a v$. Since M is a transitive model, $wR_a v$. Then by $M, w \models \Box_a \psi$, $M, v \models \psi$ holds.

- (B) Fix any symmetric Kripke model M and $w \in W$. We show $M, w \models \psi \rightarrow \Box_a \Diamond_a \psi$. Suppose $M, w \models \psi$. We show $M, w \models \Box_a \Diamond_a \psi$. It suffices to show

$$\forall u(wR_a u \text{ implies } M, u \models \Diamond_a \psi).$$

So fix any $u \in W$. Suppose $wR_a u$. Since M is a symmetric model, $wR_a u$ implies $uR_a w$. Then by $M, w \models \psi$, $M, u \models \Diamond_a \psi$ holds.

- (5) Fix any Euclidean Kripke model M and $w \in W$. We show $M, w \models \neg \Box_a \psi \rightarrow \Box_a \neg \Box_a \psi$. Suppose $M, w \models \neg \Box_a \psi$. We show $M, w \models \Box_a \neg \Box_a \psi$. It suffices to show

$$\forall u(wR_a u \text{ implies } M, u \models \neg \Box_a \psi).$$

So fix any $u \in W$. Suppose $wR_a u$. By $M, w \models \neg \Box_a \psi$,

$$\exists v(wR_a v \text{ and } M, v \models \neg \psi).$$

So fix such v . Since M is an Euclidean model, $wR_a u$ and $wR_a v$ implies $uR_a v$. Then by $M, v \models \neg \psi$, $M, u \models \neg \Box_a \psi$ holds.

- (D) Fix any serial Kripke model M and $w \in W$. We show $M, w \models \Box_a \psi \rightarrow \Diamond_a \psi$. Suppose $M, w \models \Box_a \psi$. We show $M, w \models \Diamond_a \psi$. It suffices to show

$$\exists u(wR_a u \text{ and } M, u \models \psi).$$

By $M, w \models \Box_a \psi$, we have

$$\forall v(wR_a v \text{ implies } M, v \models \psi).$$

Since M is a serial model, there exists v such that $wR_a v$. So let $u \equiv v$ and $M, u \models \psi$ holds clearly.

Therefore, we have finished establishing Theorem 10. $\square$

Theorem 11 (Soundness of **PAL**). Let $\Lambda \in \{\mathbf{K45PAL}, \mathbf{S5PAL}\}$. Then, for any L^+ -formula φ , if φ is a theorem in $H(\Lambda)$, then φ is valid in *all* Kripke frames which satisfies the corresponding properties to Λ .

Proof. Fix any formula φ such that $H(\Lambda) \vdash \varphi$. Let M^Λ be any Kripke model which satisfies the corresponding properties to Λ . By Definition 9, it suffices to show that the axioms are valid, and that the rules (MP) and (Nec) preserve validity on the class of frames in question. Since the other cases are discussed in the proof of Theorem 10, we only comment on the axioms for **PAL**. Again we divide the argument into the following cases:

- ($\Box$ at)

- ($\Rightarrow$) Suppose $M, w \models [\psi]p$. We show $M, w \models \psi \rightarrow p$. Suppose $M, w \models \psi$. We show $M, w \models p$. It suffices to show $w \in V(p)$. By $M, w \models [\psi]p$, $M, w \models \psi$ implies $M^\psi, w \models p$. Then we have $M^\psi, w \models p$ i.e. $w \in V(p) \cap W^\psi$. Then $w \in V(p)$ i.e. $M, w \models p$ holds.
- ($\Leftarrow$) Suppose $M, w \models \psi \rightarrow p$. We show $M, w \models [\psi]p$. It suffices to show $M, w \models \psi$ implies $M^\psi, w \models p$. Suppose $M, w \models \psi$. We show $M^\psi, w \models p$. It suffices to show $w \in V(p) \cap W^\psi$ i.e. $w \in V(p)$ and $w \in W^\psi$. By $M, w \models \psi \rightarrow p$, we have $w \in V(p)$. And by $M, w \models \psi$, $w \in W^\psi$ holds.

- ($\Box\neg$)

- ($\Rightarrow$) Suppose $M, w \models [\psi]\neg\gamma$. We show $M, w \models \psi \rightarrow \neg[\psi]\gamma$. Suppose $M, w \models \psi$. We show $M, w \models \neg[\psi]\gamma$. It suffices to show $M, w \not\models [\psi]\gamma$ i.e.

$$M, w \models \psi \text{ and } M^\psi, w \not\models \gamma.$$

Since $M, w \models \psi$ holds, we show $M^\psi, w \models \gamma$. By $M, w \models [\psi]\neg\gamma$, $M, w \models \psi$ implies $M^\psi, w \models \neg\gamma$. Therefore $M^\psi, w \not\models \gamma$ holds.

- ($\Leftarrow$) Suppose $M, w \models \psi \rightarrow \neg[\psi]\gamma$. We show $M, w \models [\psi]\neg\gamma$. It suffices to show

$$M, w \models \psi \text{ implies } M^\psi, w \models \neg\gamma.$$

Suppose $M, w \models \psi$. We show $M^\psi, w \models \neg\gamma$. By $M, w \models \psi \rightarrow \neg[\psi]\gamma$, we have $M, w \models \neg[\psi]\gamma$ i.e.

$$M, w \models \psi \text{ and } M^\psi, w \not\models \gamma.$$

Therefore $M^\psi, w \models \neg\gamma$ holds.

- ($\Box\wedge$)

- ($\Rightarrow$) Suppose $M, w \models [\psi](\gamma \wedge \chi)$. We show $M, w \models [\psi]\gamma \wedge [\psi]\chi$. It suffices to show $M, w \models [\psi]\gamma$ and $M, w \models [\psi]\chi$, i.e.

$$M, w \models \psi \text{ implies } M^\psi, w \models \gamma \text{ and } M^\psi, w \models \chi.$$

Suppose $M, w \models \psi$. By $M, w \models [\psi](\gamma \wedge \chi)$, we have $M^\psi, w \models \gamma \wedge \chi$, which implies $M^\psi, w \models \gamma$ and $M^\psi, w \models \chi$.

- ($\Leftarrow$) Suppose $M, w \models [\psi]\gamma \wedge [\psi]\chi$. We show $M, w \models [\psi](\gamma \wedge \chi)$. It suffices to show

$$M, w \models \psi \text{ implies } M, w \models \gamma \wedge \chi.$$

Suppose $M, w \models \psi$. We show $M, w \models \gamma \wedge \chi$ i.e. $M, w \models \gamma$ and $M, w \models \chi$. By $M, w \models [\psi]\gamma \wedge [\psi]\chi$, $M, w \models \psi$ implies $M, w \models \gamma$ and $M, w \models \chi$. Therefore $M, w \models \gamma \wedge \chi$ holds.

- ($\Box \rightarrow$)

- ($\Rightarrow$) Suppose $M, w \models [\psi](\gamma \rightarrow \chi)$. We show $M, w \models [\psi]\gamma \rightarrow [\psi]\chi$. It suffices to show $M, w \models [\psi]\gamma$ implies $M, w \models [\psi]\chi$. Suppose $M, w \models [\psi]\gamma$. We show $M, w \models [\psi]\chi$. It suffices to show

$$M, w \models \psi \text{ implies } M^\psi, w \models \chi.$$

Suppose $M, w \models \psi$. We show $M^\psi, w \models \chi$. By $M, w \models [\psi](\gamma \rightarrow \chi)$, $M, w \models \psi$ implies $M^\psi, w \models \gamma \rightarrow \chi$. Also by $M, w \models [\psi]\gamma$, $M, w \models \psi$ implies $M^\psi, w \models \gamma$. Hence $M^\psi, w \models \chi$ holds.

- ($\Leftarrow$) Suppose $M, w \models [\psi]\gamma \rightarrow [\psi]\chi$. We show $M, w \models [\psi](\gamma \rightarrow \chi)$. It suffices to show

$$M, w \models \psi \text{ implies } M^\psi, w \models \gamma \rightarrow \chi.$$

Suppose $M, w \models \psi$. We show $M^\psi, w \models \gamma \rightarrow \chi$ i.e.

$$M^\psi, w \models \gamma \text{ implies } M^\psi, w \models \chi.$$

Suppose $M^\psi, w \models \gamma$. We show $M^\psi, w \models \chi$. By $M, w \models [\psi]\gamma \rightarrow [\psi]\chi$, we have $M, w \models [\psi]\chi$. Then by $M, w \models \psi$, $M^\psi, w \models \chi$ holds.

- ($\Box\Box$)

- ($\Rightarrow$) Suppose $M, w \models [\psi]\Box_a\gamma$. We show $M, w \models \psi \rightarrow \Box_a[\psi]\gamma$. Suppose $M, w \models \psi$. We show $M, w \models \Box_a[\psi]\gamma$. It suffices to show

$$\forall u(wR_a u \text{ implies } M, u \models [\psi]\gamma).$$

Fix any $u \in W$. Suppose $wR_a u$. We show $M, u \models [\psi]\gamma$. Suppose $M, u \models \psi$. We show $M^\psi, u \models \gamma$. By $M, w \models [\psi]\Box_a\gamma$, we have $M^\psi, w \models \Box_a\gamma$. Then it suffices to show $wR_a^\psi u$. Since $R_a^\psi = R_a \cap (W^\psi \times W^\psi)$, it suffices to show $w, u \in W^\psi$ i.e. $M, w \models \psi$ and $M, u \models \psi$. By our suppositions, both of them hold.

- ($\Leftarrow$) Suppose $M, w \models \psi \rightarrow \Box_a[\psi]\gamma$. We show $M, w \models [\psi]\Box_a\gamma$. Suppose $M, w \models \psi$. We show $M^\psi, w \models \Box_a\gamma$. Fix any $u \in W$. Suppose $wR_a^\psi u$. We show $M^\psi, u \models \gamma$. By $M, w \models \psi \rightarrow \Box_a[\psi]\gamma$ and $M, w \models \psi$, we have $M, w \models \Box_a[\psi]\gamma$. Since $R_a^\psi = R_a \cap (W^\psi \times W^\psi)$, it follows that $R_a^\psi \subseteq R_a$. Therefore by $wR_a^\psi u$ we have $wR_a u$. Then by $M, w \models \Box_a[\psi]\gamma$, we have $M, u \models [\psi]\gamma$ i.e.

$$M, u \models \psi \text{ implies } M^\psi, u \models \gamma.$$

So it suffices to show $M, u \models \psi$. By $wR_a^\psi u$, $u \in W^\psi$ i.e. $M, u \models \psi$ holds.

• ($\Box\Box$)

- ($\Rightarrow$) Suppose $M, w \models [\psi][\gamma]\chi$. We show $M, w \models [\psi \wedge [\psi]\gamma]\chi$. Suppose $M, w \models \psi \wedge [\psi]\gamma$. We show $M^{\psi \wedge [\psi]\gamma}, w \models \chi$. By $M, w \models \psi \wedge [\psi]\gamma$, we have $M, w \models \psi$ and $M, w \models [\psi]\gamma$, i.e.

$$M, w \models \psi \text{ implies } M^\psi, w \models \gamma.$$

Therefore $M^\psi, w \models \gamma$. Also by $M, w \models [\psi][\gamma]\chi$, we have $M^\psi, w \models [\gamma]\chi$, i.e.

$$M^\psi, w \models \gamma \text{ implies } M^{\psi, \gamma}, w \models \chi.$$

Hence we have $M^{\psi, \gamma}, w \models \chi$. Our goal is to show $M^{\psi \wedge [\psi]\gamma}, w \models \chi$. Then it suffices to show $M^{\psi \wedge [\psi]\gamma} = M^{\psi, \gamma}$, i.e. $W^{\psi \wedge [\psi]\gamma} = W^{\psi, \gamma}$, $R^{\psi \wedge [\psi]\gamma} = R^{\psi, \gamma}$ and $V^{\psi \wedge [\psi]\gamma} = V^{\psi, \gamma}$. Since $R_a^\psi = R_a \cap (W^\psi \times W^\psi)$ and $V^\psi = V(p) \cap W^\psi$, it suffices to show $W^{\psi \wedge [\psi]\gamma} = W^{\psi, \gamma}$, where

$$\begin{aligned} W^{\psi \wedge [\psi]\gamma} &= \{ w \in W \mid M, w \models \psi \wedge [\psi]\gamma \} \\ &= \{ w \in W \mid M, w \models \psi \text{ and } M, w \models \psi \text{ implies } M^\psi, w \models \gamma \} \\ &= \{ w \in W \mid M, w \models \psi \text{ and } M^\psi, w \models \gamma \} \\ &= W^{\psi, \gamma}. \end{aligned}$$

Since $W^{\psi, \gamma}$ denotes the set of states where γ is valid *after* W^ψ , the last equivalence holds clearly.

- ($\Leftarrow$) Suppose $M, w \models [\psi \wedge [\psi]\gamma]\chi$. We show $M, w \models [\psi][\gamma]\chi$. Suppose $M, w \models \psi$. We show $M^\psi, w \models [\gamma]\chi$. Suppose $M^\psi, w \models \gamma$. We show $M^{\psi, \gamma}, w \models \chi$. By $M, w \models [\psi \wedge [\psi]\gamma]\chi$,

$$M, w \models \psi \wedge [\psi]\gamma \text{ implies } M^{\psi \wedge [\psi]\gamma}, w \models \chi.$$

Since $M^{\psi \wedge [\psi]\gamma} = M^{\psi, \gamma}$, it suffices to show $M, w \models \psi \wedge [\psi]\gamma$, i.e.

$$M, w \models \psi \text{ and } M^\psi, w \models \gamma,$$

which are both valid by our suppositions.

□

2.6 Completeness

2.6.1 Canonical Model

In this section we give a proof on completeness of $H(\mathbf{K})$ and the extensions of it, namely $H(\mathbf{K}\Sigma)$ where $\Sigma \subseteq \{\mathbf{T}, \mathbf{B}, \mathbf{4}, \mathbf{5}, \mathbf{D}\}$. The proof is adapted from [3, Chapter 4]. First we introduce some terminologies.

Definition 12. Let Σ be a possibly infinite set of formulas.

- A formula ψ is *derivable* from Σ in Λ (notation: $\Sigma \vdash_\Lambda \psi$) if there exists finitely many formulas $\varphi_1, \dots, \varphi_n$ from Σ such that $\vdash_\Lambda \bigwedge_{1 \leq i \leq n} \psi_i \rightarrow \varphi$.
- Σ is Λ -*inconsistent* if $\Sigma \vdash_\Lambda \perp$.
- Σ is Λ -*consistent* if Σ is not Λ -inconsistent, i.e. $\Sigma \not\vdash_\Lambda \perp$.
- Σ is *maximally Λ -consistent* (Λ -MCS for short) if Σ is Λ -consistent and for all set of formulas Δ such that $\Sigma \subsetneq \Delta$, Δ is Λ -inconsistent.

Proposition 13 (Properties of MCSs). Let Σ be a Λ -MCS. We have the followings:

- (1) $\Sigma \vdash_\Lambda \varphi$ iff $\varphi \in \Sigma$;
- (2) $\neg\varphi \in \Sigma$ iff $\varphi \notin \Sigma$;
- (3) $\Lambda \subseteq \Sigma$;
- (4) $\varphi \wedge \psi \in \Sigma$ iff $\varphi \in \Sigma$ and $\psi \in \Sigma$.
- (5) $\varphi \rightarrow \psi \in \Sigma$ iff $\varphi \in \Sigma$ implies $\psi \in \Sigma$.

Proof. (1) • ($\Rightarrow$) Suppose $\Sigma \vdash_{\Lambda} \varphi$. We show $\varphi \in \Sigma$. Suppose $\varphi \notin \Sigma$ for contradiction. Let $\Sigma' = \Sigma \cup \{\varphi\}$. Since Σ is a Λ -MCS, Σ' is Λ -inconsistent, i.e. $\Sigma \vdash_{\Lambda} \perp$. Then $\Sigma \cup \{\varphi\} \vdash_{\Lambda} \perp$, i.e. $\Sigma \vdash_{\Lambda} \neg\varphi$, which is contradicted to $\Sigma \vdash_{\Lambda} \varphi$. Therefore $\varphi \in \Sigma$.

• ($\Leftarrow$) Suppose $\varphi \in \Sigma$. We show $\Sigma \vdash_{\Lambda} \varphi$. It suffices to show that

$$\exists \varphi_1, \dots, \varphi_n \in \Sigma (\vdash_{\Lambda} \bigwedge_{1 \leq i \leq n} \psi_i \rightarrow \varphi).$$

By $\vdash_{\Lambda} \varphi \rightarrow \varphi$ and $\varphi \in \Sigma$, $\Sigma \vdash_{\Lambda} \varphi$ holds.

- (2) • ($\Rightarrow$) Suppose $\neg\varphi \in \Sigma$. We show $\varphi \notin \Sigma$. Suppose for contradiction that $\varphi \in \Sigma$. Then $\Sigma \vdash_{\Lambda} \perp$, which is contradicted to the fact that Σ is a Λ -MCS.
- ($\Leftarrow$) Suppose $\varphi \notin \Sigma$. We show $\neg\varphi \in \Sigma$. Let $\Sigma' = \Sigma \cup \{\varphi\}$. Then $\Sigma' \vdash_{\Lambda} \perp$, i.e. $\Sigma \cup \{\varphi\} \vdash_{\Lambda} \perp$ since Σ is a Λ -MCS. Therefore $\Sigma \vdash_{\Lambda} \neg\varphi$ holds.

(3) We show $\Lambda \subseteq \Sigma$. It suffices to show the following:

$$\forall \varphi (\varphi \in \Lambda \text{ implies } \varphi \in \Sigma).$$

Hence we fix any $\varphi \in \Lambda$, i.e., $\vdash_{\Lambda} \varphi$. Then $\Sigma \vdash_{\Lambda} \varphi$. Then by (1) $\varphi \in \Sigma$ holds.

- (4) • ($\Rightarrow$) Suppose $\varphi \wedge \psi \in \Sigma$. Then by $\vdash_{\Lambda} \varphi \wedge \psi \rightarrow \varphi$ and $\vdash_{\Lambda} \varphi \wedge \psi \rightarrow \psi$, $\Sigma \vdash_{\Lambda} \varphi$ and $\Sigma \vdash_{\Lambda} \psi$ holds. Hence by (1) $\varphi \in \Sigma$ and $\psi \in \Sigma$.
- ($\Leftarrow$) Suppose $\varphi \in \Sigma$ and $\psi \in \Sigma$. We show $\varphi \wedge \psi \in \Sigma$. By (1), it suffices to show $\Sigma \vdash_{\Lambda} \varphi \wedge \psi$. By $\varphi \in \Sigma$ and $\psi \in \Sigma$, $\Sigma \vdash_{\Lambda} \varphi \wedge \psi$ holds.
- (5) • ($\Rightarrow$) Suppose $\varphi \rightarrow \psi \in \Sigma$. We show $\varphi \in \Sigma$ implies $\psi \in \Sigma$. Suppose $\varphi \in \Sigma$. We show $\psi \in \Sigma$. By $\varphi \rightarrow \psi \in \Sigma$ and $\varphi \in \Sigma$ we have $\Sigma \vdash_{\Lambda} \psi$. Then by (1), $\psi \in \Sigma$ holds.
- ($\Leftarrow$) Suppose $\varphi \in \Sigma$ implies $\psi \in \Sigma$. We show $\varphi \rightarrow \psi \in \Sigma$. By (1), it suffices to show $\Sigma \vdash_{\Lambda} \varphi \rightarrow \psi$, which is followed by our supposition that $\varphi \in \Sigma$ implies $\psi \in \Sigma$.

□

Definition 14. Let Δ and Σ be two sets. Δ is the *finite subset* of Σ (notation: $\Delta \subseteq_{\omega} \Sigma$) if for every $\varphi \in \Delta$, $\varphi \in \Sigma$ and there exists a bijection from Δ to $\{1, 2, 3, \dots, n\}$ where $n \in \mathbb{N}$.

The following Lemma 15 shows that any consistent set of formulas can be extended to a maximal consistent one.

Lemma 15 (Lindenbaum's Lemma). Let Σ be an Λ -consistent set of formulas. Then there exists a set Σ^+ such that $\Sigma \subseteq \Sigma^+$ and Σ^+ is a Λ -MCS.

Proof. Let $\varphi_0, \varphi_1, \varphi_2, \dots$ be an enumeration of all the formulas of L . We define the set Σ^+ as the union of a chain of Λ -consistent sets as follows:

$$\begin{cases} \Sigma_0 = \Sigma \\ \Sigma_{n+1} = \begin{cases} \Sigma_n \cup \{\varphi_n\} & \text{if } \Sigma_n \cup \{\varphi_n\} \text{ is } \Lambda\text{-consistent} \\ \Sigma_n \cup \{\neg\varphi_n\} & \text{otherwise} \end{cases} \\ \Sigma^+ = \bigcup_{n \geq 0} \Sigma_n \end{cases}$$

Then it suffices to show Σ^+ has the following properties:

- (1) Σ_n is Λ -consistent for all n ;
 - (2) For every formula φ , exactly one of φ and $\neg\varphi$ is in Σ^+ ;
 - (3) If $\Sigma^+ \vdash_{\Lambda} \varphi$, then $\varphi \in \Sigma^+$;
 - (4) Σ^+ is a Λ -MCS.
- (1) By induction on n .

For the base step, let $n = 0$. Then $\Sigma_0 = \Sigma$ is Λ -consistent by definition.

For inductive step, let Σ_{n+1} be a set of formulas. If $\Sigma_n \cup \{\varphi_n\}$ is Λ -consistent, then our goal is done by letting $\Sigma_{n+1} = \Sigma_n \cup \{\varphi_n\}$. Hence we suppose $\Sigma_n \cup \{\varphi_n\}$ is Λ -inconsistent. Then it suffices to show $\Sigma_n \cup \{\neg\varphi_n\}$ is Λ -consistent. By induction hypothesis, Σ_n is Λ -consistent. Suppose for contradiction that $\Sigma_n \cup \{\neg\varphi_n\}$ is Λ -inconsistent. It follows that $\Sigma_n \vdash_{\Lambda} \neg\varphi_n$. Also since $\Sigma_n \cup \{\varphi_n\}$ is Λ -inconsistent, we have $\Sigma_n \vdash_{\Lambda} \varphi_n$. Then $\Sigma_n \vdash_{\Lambda} \perp$, which is contradicted to the fact that Σ_n is Λ -consistent.

- (2) By the definition of Σ^+ . For every formula φ_n ($n \geq 0$), either $\Sigma_n \cup \{\varphi_n\} \subseteq \Sigma^+$ or $\Sigma_n \cup \{\neg\varphi_n\} \subseteq \Sigma^+$.
- (3) Suppose $\Sigma^+ \vdash_{\Lambda} \varphi$. Then there exists a finite subset Δ of Σ^+ such that $\vdash_{\Lambda} \bigwedge \Delta \rightarrow \varphi$. Since $\Sigma^+ = \bigcup_{n \geq 0} \Sigma_n$, $\Delta \subseteq_{\omega} \Sigma_m$ for some m . Then $\Sigma_m \vdash_{\Lambda} \varphi$, which implies $\varphi \in \Sigma_m \subseteq \Sigma^+$.

(4) We show:

(4-1) Σ^+ is Λ -consistent.

(4-2) For any Σ^* such that $\Sigma^+ \subsetneq \Sigma^*$, Σ^* is Λ -inconsistent.

(4-1) It suffices to show for any $\Sigma' \subsetneq_\omega \Sigma^+$, Σ' is Λ -consistent. Fix any $\Sigma' \subsetneq_\omega \Sigma^+$. Let $\Sigma' = \{\psi_0, \psi_1, \psi_2, \dots, \psi_m\} \subsetneq \Sigma^+ = \bigcup_{n \in \mathbb{N}} \Sigma_n$. Then for every $\psi_i \in \Sigma'$, $\psi_i \in \bigcup_{n \in \mathbb{N}} \Sigma_n$, i.e. $\exists n_i \in \mathbb{N} (\psi_i \in \Sigma_{n_i})$. It follows by the definition of Σ_n that for any i, j , $i \leq j$ implies $\Sigma_i \subseteq \Sigma_j$. Then for $k = \max \{n_i \mid 1 \leq i \leq m\}$, we obtain $\Sigma' \subseteq \Sigma_k$, which is Λ -consistent. Therefore Σ' is Λ -consistent.

(4-2) Fix any Σ^* such that $\Sigma^+ \subsetneq \Sigma^*$. Then $\psi \in \Sigma^*$ but $\psi \notin \Sigma^+$ for some ψ . Fix such ψ . It follows by $\psi \notin \Sigma^+$ that $\neg\psi \in \Sigma^+ \subsetneq \Sigma^*$. Then $\Sigma^* \vdash_\Lambda \perp$, i.e., Σ^* is Λ -inconsistent.

□

We are now ready to build models out of MCSs.

Definition 16. The *canonical model* M^Λ for Λ is a tuple $(W^\Lambda, (R_a^\Lambda)_{a \in Ag}, V^\Lambda)$ where:

- $W^\Lambda = \{\Gamma \mid \Gamma \text{ is a } \Lambda\text{-MCS}\}$, i.e. W^Λ is the set of all Λ -MCSs;
- $\Gamma R_a^\Lambda \Sigma$ iff for all formulas φ , $\Box_a \varphi \in \Gamma$ implies $\varphi \in \Sigma$;
- $\Gamma \in V^\Lambda(p)$ iff $p \in \Gamma$ ($p \in \text{Prop}$).

It follows that $\Gamma R_a^\Lambda \Sigma$ iff $\{\varphi \mid \Box_a \varphi \in \Gamma\} \subseteq \Sigma$.

Next we establish the following lemmas.

Lemma 17. Let Γ be a Λ -MCS. If $\Box_a \psi \notin \Gamma$, then $\{\neg\psi\} \cup \{\gamma \mid \Box_a \gamma \in \Gamma\}$ is Λ -consistent.

Proof. Suppose $\Box_a \psi \notin \Gamma$. Let $\Sigma = \{\neg\psi\} \cup \{\gamma \mid \Box_a \gamma \in \Gamma\}$. We show Σ is Λ -consistent. Suppose for contradiction that Σ is Λ -inconsistent. Then $\Sigma \vdash_\Lambda \perp$, i.e.

$$\exists \Delta \subseteq_\omega \Sigma \text{ such that } \vdash_\Lambda \bigwedge \Delta \rightarrow \perp.$$

Then we divide the argument into the following two cases: $\neg\psi \in \Delta$ and $\neg\psi \notin \Delta$:

- Suppose $\neg\psi \in \Delta$. Let $\Delta = \Delta' \cup \{\neg\psi\}$, then we have

$$\vdash_{\Lambda} \bigwedge \Delta' \wedge \neg\psi \rightarrow \perp.$$

- Suppose $\neg\psi \notin \Delta$. Then $\Delta \subseteq \{\varphi \mid \Box_a \varphi \in \Gamma\}$. Hence by

$$\vdash_{\Lambda} \bigwedge \Delta \wedge \neg\psi \rightarrow \bigwedge \Delta,$$

we obtain

$$\vdash_{\Lambda} \bigwedge \Delta \wedge \neg\psi \rightarrow \perp.$$

Therefore we have

$$\exists \Pi (\Pi \subseteq_{\omega} \{\gamma \mid \Box_a \gamma \in \Gamma\} \text{ and } \vdash_{\Lambda} \bigwedge \Pi \wedge \neg\psi \rightarrow \perp).$$

And $\vdash_{\Lambda} \bigwedge \Pi \wedge \neg\psi \rightarrow \perp$ implies

$$\vdash_{\Lambda} \bigwedge \Pi \rightarrow (\neg\psi \rightarrow \perp),$$

i.e.

$$\vdash_{\Lambda} \bigwedge \Pi \rightarrow \psi.$$

By (Nec), we obtain

$$\vdash_{\Lambda} \Box_a (\bigwedge \Pi \rightarrow \psi).$$

Then by (K), we have

$$\vdash_{\Lambda} \bigwedge \Box_a \Pi \rightarrow \Box_a \psi.$$

Since $\Pi \subseteq \{\gamma \mid \Box_a \gamma \in \Gamma\}$, $\Box_a \Pi \subseteq \Gamma$. Then $\Box_a \psi \in \Gamma$ since Γ is a Λ -MCS, which is contradicted to the supposition. $\square$

Lemma 18 (Truth Lemma). Let Σ be a Λ -MCS. For any formula φ ,

$$\varphi \in \Sigma \text{ iff } M^{\Lambda}, \Sigma \models \varphi.$$

Proof. By induction on φ .

For base step, let $\varphi \equiv p$. Then $p \in \Sigma$ iff $M^{\Lambda}, \Sigma \models p$ follows from the property of Σ , since Σ is a Λ -MCS.

For inductive step, we divide the argument into the following cases:

- $(\varphi \equiv \neg\psi)$

- ($\Rightarrow$) Suppose $\neg\psi \in \Sigma$. We show $M^\Lambda, \Sigma \models \neg\psi$, i.e. $M^\Lambda, \Sigma \not\models \psi$. By Proposition 13 (2), $\neg\psi \in \Sigma$ implies $\psi \notin \Sigma$. Then by induction hypothesis, $M^\Lambda, \Sigma \not\models \psi$ holds.
- ($\Leftarrow$) Suppose $M^\Lambda, \Sigma \models \neg\psi$. We show $\neg\psi \in \Sigma$. By Proposition 13 (2), it suffices to show $\psi \notin \Sigma$. By $M^\Lambda, \Sigma \models \neg\psi$ we have $M^\Lambda, \Sigma \not\models \psi$. Then by induction hypothesis, $\psi \notin \Sigma$ holds.
- ($\varphi \equiv \psi \wedge \gamma$)
 - ($\Rightarrow$) Suppose $\psi \wedge \gamma \in \Sigma$. We show $M^\Lambda, \Sigma \models \psi \wedge \gamma$. It suffices to show $M^\Lambda, \Sigma \models \psi$ and $M^\Lambda, \Sigma \models \gamma$. By Proposition 13 (4), $\psi \wedge \gamma \in \Sigma$ implies $\psi \in \Sigma$ and $\gamma \in \Sigma$. Then by induction hypothesis, $M^\Lambda, \Sigma \models \psi$ and $M^\Lambda, \Sigma \models \gamma$ holds.
 - ($\Leftarrow$) Suppose $M^\Lambda, \Sigma \models \psi \wedge \gamma$. We show $\psi \wedge \gamma \in \Sigma$. By Proposition 13 (4), it suffices to show $\psi \in \Sigma$ and $\gamma \in \Sigma$. By $M^\Lambda, \Sigma \models \psi \wedge \gamma$, we have $M^\Lambda, \Sigma \models \psi$ and $M^\Lambda, \Sigma \models \gamma$. Then by induction hypothesis, $\psi \in \Sigma$ and $\gamma \in \Sigma$ holds.
- ($\varphi \equiv \psi \rightarrow \gamma$)
 - ($\Rightarrow$) Suppose $\psi \rightarrow \gamma \in \Sigma$. We show $M^\Lambda, \Sigma \models \psi \rightarrow \gamma$. Suppose $M^\Lambda, \Sigma \models \psi$. We show $M^\Lambda, \Sigma \models \gamma$. By Proposition 13 (5), we have $\psi \in \Sigma$ implies $\gamma \in \Sigma$. By induction hypothesis, $M^\Lambda, \Sigma \models \psi$ implies $\psi \in \Sigma$. Hence we have $\gamma \in \Sigma$. Therefore by induction hypothesis, $M^\Lambda, \Sigma \models \gamma$ holds.
 - ($\Leftarrow$) Suppose $M^\Lambda, \Sigma \models \psi \rightarrow \gamma$. We show $\psi \rightarrow \gamma \in \Sigma$. By Proposition 13 (5), it suffices to show $\psi \in \Sigma$ implies $\gamma \in \Sigma$. By $M^\Lambda, \Sigma \models \psi \rightarrow \gamma$, we have $M^\Lambda, \Sigma \models \psi$ implies $M^\Lambda, \Sigma \models \gamma$. Hence by induction hypothesis, $\psi \in \Sigma$ implies $\gamma \in \Sigma$.
- ($\varphi \equiv \Box_a \psi$)
 - ($\Rightarrow$) Suppose $\Box_a \psi \in \Sigma$. We show $M^\Lambda, \Sigma \models \Box_a \psi$. It suffices to show

$$\forall \Delta (\Sigma R_a^\Lambda \Delta \text{ implies } M^\Lambda, \Delta \models \psi).$$

So fix any Λ -MCS Δ . Suppose $\Sigma R_a^\Lambda \Delta$. By Definition 16, $\Gamma R_a^\Lambda \Delta$ iff $\{\varphi \mid \Box_a \varphi \in \Sigma\} \subseteq \Delta$. Hence we have $\psi \in \Delta$. Then by induction hypothesis, $M^\Lambda, \Delta \models \psi$ holds.

- ($\Leftarrow$) Suppose $M^\Lambda, \Sigma \models \Box_a \psi$. We show $\Box_a \psi \in \Sigma$. Suppose $\Box_a \psi \notin \Sigma$ for contradiction. By Lemma 17, $\{\neg\psi\} \cup \{\gamma \mid \Box_a \gamma \in \Sigma\}$ is Λ -consistent. Let $\Delta = \{\neg\psi\} \cup \{\gamma \mid \Box_a \gamma \in \Sigma\}$. By Lemma 15, there exists a Λ -MCS Δ^+ such that $\Delta \subseteq \Delta^+$. Fix such Δ^+ . It follows that $\Sigma R_a^\Lambda \Delta$. By induction hypothesis, $\neg\psi \in \Delta^+$ implies $M^\Lambda, \Delta^+ \models \neg\psi$. Then $M^\Lambda, \Sigma \not\models \Box_a \psi$, which is contradicted to supposition. Therefore $\Box_a \psi \in \Sigma$.

□

Based on our preparation above, now we can provide a proof of Kripke completeness of $H(\mathbf{K})$.

Theorem 19. For any formula φ , if φ is valid on all Kripke models then φ is a theorem of $H(\mathbf{K})$.

Proof. We show the contraposition. Let $\Lambda = H(\mathbf{K})$. Suppose $\not\vdash_\Lambda \varphi$. It follows that $\{\neg\varphi\}$ is Λ -consistent. For if we suppose $\{\neg\varphi\}$ is Λ -inconsistent, i.e. $\vdash_\Lambda \neg\varphi \rightarrow \perp$. Then $\vdash_\Lambda \varphi$, which is contradicted to $\not\vdash_\Lambda \varphi$. By Lemma 15, there exists a Λ -MCS Σ such that $\{\neg\varphi\} \subseteq \Sigma$ i.e. $\neg\varphi \in \Sigma$. Then by Lemma 18, $M^\Lambda, \Sigma \models \neg\varphi$, which implies $M^\Lambda, \Sigma \not\models \varphi$, i.e. φ is not valid on M^Λ , as desired. □

Next we establish the completeness of extensions of $H(\mathbf{K})$. First we establish the following lemmas.

Lemma 20. In the canonical model M^Λ for Λ , we have the following equivalence:

$$\Gamma R_a^\Lambda \Sigma \text{ iff } (\varphi \in \Sigma \text{ implies } \Diamond_a \varphi \in \Gamma) \text{ for all formulas } \varphi.$$

Proof. Fix any canonical model M^Λ , any $\Gamma, \Sigma \in W^\Lambda$ and any formula φ .

($\Rightarrow$) Suppose $\Gamma R_a^\Lambda \Sigma$. and $\varphi \in \Sigma$. We show $\Diamond_a \varphi \in \Gamma$. Since Γ is a MCS, by Proposition 13 (2), it suffices to show $\neg\Diamond_a \varphi \notin \Gamma$. Suppose for contradiction that $\neg\Diamond_a \varphi \in \Gamma$. Then by $\vdash_\Lambda \neg\Diamond_a \varphi \rightarrow \Box_a \neg\varphi$, we have $\Box_a \neg\varphi \in \Gamma$. By $\Gamma R_a^\Lambda \Sigma$, it follows that $\neg\varphi \in \Sigma$. Then $\Sigma \vdash_\Lambda \perp$, which is contradicted to the fact that Σ is a MCS. Therefore $\neg\Diamond_a \varphi \notin \Gamma$ i.e. $\Diamond_a \varphi \in \Gamma$ holds.

- ($\Leftarrow$) Suppose $\varphi \in \Sigma$ implies $\Diamond_a \varphi \in \Gamma$. We show $\Gamma R_a^\Lambda \Sigma$. It suffices to show for all formulas ψ ,

$$\Box_a \psi \in \Gamma \text{ implies } \psi \in \Sigma.$$

Suppose $\Box_a \psi \in \Gamma$ and $\psi \notin \Sigma$ for contradiction. Since Σ is a Λ -MCS, $\neg\psi \in \Sigma$. By our supposition, $\neg\psi \in \Sigma$ implies $\Diamond_a \neg\psi \in \Gamma$. By Lemma 18, we have

$M^\Lambda, \Gamma \models \Diamond_a \neg \psi$. Again by Lemma 18, $\Box_a \psi \in \Gamma$ implies $M^\Lambda, \Gamma \models \Box_a \psi$, i.e. for all Δ such that $\Gamma R_a^\Lambda \Delta$, $M^\Lambda, \Delta \models \psi$. But $M^\Lambda, \Gamma \models \Diamond_a \neg \psi$ implies that there exists some Π such that $\Gamma R_a^\Lambda \Pi$ and $M^\Lambda, \Pi \models \neg \psi$. Then we have a contradiction.

□

Lemma 20 implies that $\Gamma R_a^\Lambda \Sigma$ iff $\{\Diamond \varphi \mid \varphi \in \Sigma\} \subseteq \Gamma$. The following lemma shows the correspondence from axiom schemes to classes of frames.

Lemma 21. Let Λ be an axiomatic extension of $H(\mathbf{K})$ and $M^\Lambda = (W^\Lambda, R_a^\Lambda, V^\Lambda)$ be the canonical model for Λ .

- (1) If Λ contains an axiom scheme (T), then R_a^Λ is reflexive;
- (2) If Λ contains an axiom scheme (4), then R_a^Λ is transitive;
- (3) If Λ contains an axiom scheme (B), then R_a^Λ is symmetric;
- (4) If Λ contains an axiom scheme (5), then R_a^Λ is Euclidean;
- (5) If Λ contains an axiom scheme (D), then R_a^Λ is serial.

Proof. • (T) Suppose $\vdash_\Lambda \Box_a \varphi \rightarrow \varphi$. We show that R_a^Λ is reflexive. It suffices to show that

$$\Box_a \psi \in \Gamma \text{ implies } \psi \in \Gamma \text{ for all formulas } \psi.$$

Fix any $\Gamma \in W^\Lambda$. Suppose $\Box_a \psi \in \Gamma$. We show that $\psi \in \Gamma$, which follows by $\vdash_\Lambda \Box_a \varphi \rightarrow \varphi$ clearly.

- (4) Suppose $\vdash_\Lambda \Box_a \varphi \rightarrow \Box_a \Box_a \varphi$. We show that R_a^Λ is transitive. It suffices to show that

$$\forall \Gamma, \Sigma, \Delta \in W^\Lambda (\Gamma R_a^\Lambda \Sigma \text{ and } \Sigma R_a^\Lambda \Delta \text{ implies } \Gamma R_a^\Lambda \Delta).$$

Fix any $\Gamma, \Sigma, \Delta \in W^\Lambda$. Suppose $\Gamma R_a^\Lambda \Sigma$ and $\Sigma R_a^\Lambda \Delta$. We show that $\Gamma R_a^\Lambda \Delta$. It suffices to show $\Box_a \psi \in \Gamma$ implies $\psi \in \Delta$ for all formulas ψ . Fix any ψ . Suppose $\Box_a \psi \in \Gamma$. We show that $\psi \in \Delta$. By $\vdash_\Lambda \Box_a \varphi \rightarrow \Box_a \Box_a \varphi$, we have $\Box_a \Box_a \psi \in \Gamma$. Then $\Box_a \psi \in \Sigma$ since $\Gamma R_a^\Lambda \Sigma$. Hence $\psi \in \Delta$ by $\Sigma R_a^\Lambda \Delta$.

- (B) Suppose $\vdash_{\Lambda} \varphi \rightarrow \Box_a \Diamond_a \varphi$. We show that R_a^{Λ} is symmetric. It suffices to show that

$$\forall \Gamma, \Sigma \in W^{\Lambda} (\Gamma R_a^{\Lambda} \Sigma \text{ implies } \Sigma R_a^{\Lambda} \Gamma).$$

Fix any $\Gamma, \Sigma \in W^{\Lambda}$. Suppose $\Gamma R_a^{\Lambda} \Sigma$. We show that $\Sigma R_a^{\Lambda} \Gamma$. By Lemma 20, it suffices to show $\psi \in \Gamma$ implies $\Diamond_a \psi \in \Sigma$ for all formulas ψ . Fix any ψ . Suppose $\psi \in \Gamma$. We show that $\Diamond_a \psi \in \Sigma$. By $\vdash_{\Lambda} \varphi \rightarrow \Box_a \Diamond_a \varphi$, we have $\Box_a \Diamond_a \psi \in \Gamma$. Then by $\Gamma R_a^{\Lambda} \Sigma$, $\Diamond_a \psi \in \Sigma$ holds.

- (5) Suppose $\vdash_{\Lambda} \neg \Box_a \varphi \rightarrow \Box_a \neg \Box_a \varphi$. We show that R_a^{Λ} is Euclidean. It suffices to show that

$$\forall \Gamma, \Sigma, \Delta \in W^{\Lambda} (\Gamma R_a^{\Lambda} \Sigma \text{ and } \Gamma R_a^{\Lambda} \Delta \text{ implies } \Sigma R_a^{\Lambda} \Delta).$$

Fix any $\Gamma, \Sigma, \Delta \in W^{\Lambda}$. Suppose $\Gamma R_a^{\Lambda} \Sigma$ and $\Gamma R_a^{\Lambda} \Delta$. We show that $\Sigma R_a^{\Lambda} \Delta$. It suffices to show $\psi \in \Delta$ implies $\Diamond_a \psi \in \Sigma$ for all formulas ψ . Fix any ψ . Suppose $\psi \in \Delta$. We show that $\Diamond_a \psi \in \Sigma$. By $\psi \in \Delta$ and $\Gamma R_a^{\Lambda} \Delta$, we have $\Diamond_a \psi \in \Gamma$ i.e. $\neg \Box_a \neg \psi \in \Gamma$. Then by $\vdash_{\Lambda} \neg \Box_a \varphi \rightarrow \Box_a \neg \Box_a \varphi$, we have $\Box_a \neg \Box_a \neg \psi \in \Gamma$. Hence by $\Gamma R_a^{\Lambda} \Sigma$, $\neg \Box_a \neg \psi \in \Sigma$ i.e. $\Diamond_a \psi \in \Sigma$ holds.

- (D) Suppose $\vdash_{\Lambda} \Box_a \varphi \rightarrow \Diamond_a \varphi$. We show that R_a^{Λ} is serial. It suffices to show that

$$\forall \Gamma \exists \Sigma (\Gamma R_a^{\Lambda} \Sigma).$$

Let ψ be an axiom, say, $p \rightarrow p$. Then $\vdash_{\Lambda} \Box_a \psi$ holds. By supposition, we obtain $\vdash_{\Lambda} \Diamond_a \psi$. Fix any $\Gamma \in W^{\Lambda}$. It follows that $\Diamond_a \psi \in \Gamma$ i.e. $\neg \Box_a \neg \psi \in \Gamma$. Since Γ is a Λ -MCS, $\Box_a \neg \psi \notin \Gamma$. Hence by Lemma 17, $\{\neg \neg \psi\} \cup \{\gamma \mid \Box_a \gamma \in \Gamma\}$ is Λ -consistent. Then by Lemma 15 there exists an Λ -MCS Δ^+ such that $\Delta \subseteq \Delta^+$. Since $\{\gamma \mid \Box_a \gamma \in \Gamma\} \subseteq \Delta \subseteq \Delta^+$, $\Gamma R_a^{\Lambda} \Delta^+$ holds.

□

Now we are ready to prove Theorem 22.

Theorem 22 (Completeness). Let Σ be a possibly empty subset of $\{\mathbf{T}, \mathbf{B}, \mathbf{4}, \mathbf{5}, \mathbf{D}\}$ and $H(\mathbf{K}\Sigma)$ be the Hilbert system extending $H(\mathbf{K})$ with the axiom schemes corresponding to Σ . Then, for any formula φ , if φ is valid in *all* Kripke frames which satisfies the corresponding properties to Σ , then φ is a theorem in $H(\mathbf{K}\Sigma)$.

Proof. Similar to Theorem 19. Let $\Lambda = \mathbf{K}\Sigma$. Fix any formula φ . Suppose $\vDash_{\Lambda} \varphi$. Let M^{Λ} be the canonical model for Λ . By Lemma 21, it suffices to show

$$\exists \Gamma (\Gamma \in W^{\Lambda} \text{ and } M^{\Lambda}, \Gamma \not\models \varphi).$$

By $\not\models_{\Lambda} \varphi$, we have $\{\neg\varphi\}$ is Λ -consistent. Then by Lemma 15 there exists a Λ -MCS Δ such that $\{\neg\varphi\} \subseteq \Delta$ i.e. $\neg\varphi \in \Sigma$. Then by Lemma 18, $M^{\Lambda}, \Delta \models \neg\varphi$, which implies $M^{\Lambda}, \Delta \not\models \varphi$. $\square$

2.6.2 Completeness of H(K45PAL) and H(S5PAL) by translation

In this chapter we give a proof on completeness of H(K45PAL) and H(S5PAL). The proof is adapted from [7, Theorem 7.26].

When one looks closely at the axioms for **PAL**, one notices that what is the case after an announcement can be expressed by saying what is the case before the announcement [7, p. 186]. The following translation captures this observation.

Definition 23 (Translation). The translation $t: L^+ \rightarrow L$ is defined as follows:

$$\begin{aligned}
 t(p) &= p \\
 t(\neg\varphi) &= \neg t(\varphi) \\
 t(\varphi \wedge \psi) &= t(\varphi) \wedge t(\psi) \\
 t(\varphi \vee \psi) &= t(\varphi) \vee t(\psi) \\
 t(\varphi \rightarrow \psi) &= t(\varphi) \rightarrow t(\psi) \\
 t(\Box_a \varphi) &= \Box_a t(\varphi) \\
 t([\varphi]p) &= t(\varphi \rightarrow p) \\
 t([\varphi]\neg\psi) &= t(\varphi \rightarrow \neg[\varphi]\psi) \\
 t([\varphi](\psi \bullet \gamma)) &= t([\varphi]\psi \bullet [\varphi]\gamma) \quad (\bullet \in \{\wedge, \rightarrow\}) \\
 t([\varphi]\Box_a \psi) &= t(\varphi \rightarrow \Box_a [\varphi]\psi) \\
 t([\varphi][\psi]\gamma) &= t([\varphi \wedge [\varphi]\psi]\gamma)
 \end{aligned}$$

Recall in Definition 5, we have

$$c([\varphi]\psi) := (2 + c(\varphi)) \cdot c(\psi).$$

where the number 2 seems arbitrary, but it is necessary to obtain the following properties.

Lemma 24. For all formulas φ , ψ and γ :

- (1) $c(\psi) > c(\varphi)$ if $\varphi \in \text{Sub}(\psi)$ and φ is distinct from ψ as a formula,

- (2) $c([\varphi]p) > c(\varphi \rightarrow p)$,
- (3) $c([\varphi]\neg\psi) > c(\varphi \rightarrow \neg[\varphi]\psi)$,
- (4) $c([\varphi](\psi \bullet \gamma)) > c([\varphi]\psi \bullet [\varphi]\gamma)$ ($\bullet \in \{\wedge, \rightarrow\}$),
- (5) $c([\varphi]\Box_a\psi) > c(\varphi \rightarrow \Box_a[\varphi]\psi)$,
- (6) $c([\varphi][\psi]\gamma) > c([\varphi \wedge [\varphi]\psi]\gamma)$.

Proof. (1) Fix any φ and ψ such that $\varphi \in \text{Sub}(\psi)$ and φ is distinct from ψ as a formula. It follows that $\psi \neq p$. We divide the argument depending on ψ .

- Suppose $\psi \equiv \neg\gamma$. Then $c(\psi) = 1 + c(\gamma)$. Since $\varphi \neq \neg\gamma$, $c(\varphi) \leq c(\gamma)$, which implies $c(\psi) > c(\varphi)$.
- Suppose $\psi \equiv \gamma_1 \bullet \gamma_2$ ($\bullet \in \{\wedge, \rightarrow\}$). Then $c(\psi) = 1 + \max(c(\gamma_1), c(\gamma_2))$, while $\max(c(\varphi)) = \max(c(\gamma_1), c(\gamma_2))$. Therefore $c([\varphi](\psi_1 \bullet \psi_2)) > c([\varphi]\psi_i)$ holds.
- Suppose $\psi \equiv \Box_a\gamma$. Then $c(\psi) = 1 + c(\gamma)$, while $c(\varphi) \leq c(\gamma)$. Therefore $c([\varphi]\Box_a\psi) > c(\Box_a[\varphi]\psi)$.

- (2) Fix any φ and p . Then

$$c([\varphi]p) = (2 + c(\varphi)) \cdot c(p) = 2 + c(\varphi),$$

while

$$c(\varphi \rightarrow p) = 1 + \max(c(\varphi), c(p)).$$

Since $c(\varphi) \geq 1$,

$$c(\varphi \rightarrow p) = 1 + c(\varphi).$$

Therefore $c([\varphi]p) > c(\varphi \rightarrow p)$ holds.

- (3) Fix any φ and ψ . Then

$$c([\varphi]\neg\psi) = (2 + c(\varphi)) \cdot (1 + c(\psi)) = c(\varphi) \cdot c(\psi) + c(\varphi) + 2c(\psi) + 2,$$

while

$$\begin{aligned} c(\varphi \rightarrow \neg[\varphi]\psi) &= 1 + \max(c(\varphi), 1 + c([\varphi]\psi)) \\ &= 1 + \max(c(\varphi), 1 + (2 + c(\varphi)) \cdot c(\psi)) \\ &= c(\varphi) \cdot c(\psi) + 2c(\psi) + 2. \end{aligned}$$

Since $c(\varphi) \geq 1$, $c([\varphi]\neg\psi) > c(\varphi \rightarrow \neg[\varphi]\psi)$ holds.

(4) Fix any φ , ψ and γ . We divide the argument into the following three cases:

- Suppose $c(\psi) > c(\gamma)$. Then

$$c([\varphi](\psi \bullet \gamma)) = c(\varphi) \cdot c(\psi) + c(\varphi) + 2c(\psi) + 2,$$

while

$$c([\varphi]\psi \bullet [\varphi]\gamma) = c(\varphi) \cdot c(\psi) + 2c(\psi) + 1.$$

Therefore $c([\varphi](\psi \bullet \gamma)) > c([\varphi]\psi \bullet [\varphi]\gamma)$ ($\bullet \in \{\wedge, \rightarrow\}$) holds.

- Suppose $c(\psi) = c(\gamma)$. Then

$$c([\varphi](\psi \bullet \gamma)) = c(\varphi) \cdot c(\psi) + 3c(\psi) + 2,$$

while

$$c([\varphi]\psi \bullet [\varphi]\gamma) = c(\varphi) \cdot c(\psi) + 2c(\psi) + 1.$$

Therefore $c([\varphi](\psi \bullet \gamma)) > c([\varphi]\psi \bullet [\varphi]\gamma)$ ($\bullet \in \{\wedge, \rightarrow\}$) holds.

- Suppose $c(\psi) < c(\gamma)$. Then

$$c([\varphi](\psi \bullet \gamma)) = c(\varphi) \cdot c(\gamma) + c(\varphi) + 2c(\gamma) + 2,$$

while

$$c([\varphi]\psi \bullet [\varphi]\gamma) = c(\varphi) \cdot c(\gamma) + 2c(\gamma) + 1.$$

Therefore $c([\varphi](\psi \bullet \gamma)) > c([\varphi]\psi \bullet [\varphi]\gamma)$ ($\bullet \in \{\wedge, \rightarrow\}$) holds.

(5) Fix any φ and ψ . Then

$$c([\varphi]\Box_a\psi) = c(\varphi) \cdot c(\gamma) + c(\varphi) + 2c(\gamma) + 2,$$

while

$$c(\varphi \rightarrow \Box_a[\varphi]\psi) = c(\varphi) \cdot c(\gamma) + 2c(\gamma) + 2.$$

Since $c(\varphi) \geq 1$, $c([\varphi]\Box_a\psi) > c(\varphi \rightarrow \Box_a[\varphi]\psi)$ holds.

(6) Fix any φ , ψ and γ . Then

$$c([\varphi][\psi]\gamma) = (c(\varphi) \cdot c(\psi) + 2c(\varphi) + 2c(\psi) + 4) \cdot c(\gamma),$$

while

$$c([\varphi \wedge [\varphi]\psi]\gamma) = (c(\varphi) \cdot c(\psi) + 2c(\psi) + 3) \cdot c(\gamma).$$

Since $c(\varphi), c(\gamma) \geq 1$, $c([\varphi][\psi]\gamma) > c([\varphi \wedge [\varphi]\psi]\gamma)$ holds.

□

Lemma 25. Let $\Lambda \in \{\mathbf{K45PAL}, \mathbf{S5PAL}\}$. For all formulas $\varphi \in L^+$ it is the case that

$$\mathbf{H}(\Lambda) \vdash \varphi \leftrightarrow t(\varphi).$$

Proof. By induction on $c(\varphi)$. It suffices to show the following statement

$$\forall \varphi \in L^+ (c(\varphi) \leq n \text{ implies } \vdash \varphi \leftrightarrow t(\varphi)).$$

For base step, let $c(\varphi) = 1$. It follows that $\varphi \equiv p$. Then $\vdash p \leftrightarrow p$ is trivial.

For inductive step, we divide the argument into the following cases:

- $(\varphi \equiv \neg\psi)$ It suffices to show $\vdash \neg\psi \leftrightarrow \neg t(\psi)$. Since $\psi \in \text{Sub}(\neg\psi)$, by Lemma 24(1), $c(\psi) < c(\neg\psi)$. Then by induction hypothesis, $\vdash \psi \leftrightarrow t(\psi)$. Then implies the contraposition $\vdash \neg\psi \leftrightarrow \neg t(\psi)$.
- $(\varphi \equiv \psi \wedge \gamma)$ It suffices to show $\vdash \psi \wedge \gamma \leftrightarrow t(\psi) \wedge t(\gamma)$. By Lemma 24(1) and induction hypothesis, we obtain $\vdash \psi \leftrightarrow t(\psi)$ and $\vdash \gamma \leftrightarrow t(\gamma)$. Then it follows that $\vdash \psi \wedge \gamma \leftrightarrow t(\psi) \wedge t(\gamma)$.
- $(\varphi \equiv \psi \rightarrow \gamma)$ It suffices to show $\vdash (\psi \rightarrow \gamma) \leftrightarrow (t(\psi) \rightarrow t(\gamma))$. By Lemma 24(1) and induction hypothesis, we obtain $\vdash \psi \leftrightarrow t(\psi)$ and $\vdash \gamma \leftrightarrow t(\gamma)$. Then it follows that $\vdash (\psi \rightarrow \gamma) \leftrightarrow (t(\psi) \rightarrow t(\gamma))$.
- $(\varphi \equiv \Box_a \psi)$ It suffices to show $\vdash \Box_a \psi \leftrightarrow \Box_a t(\psi)$. By Lemma 24(1) and induction hypothesis, we obtain $\vdash \psi \leftrightarrow t(\psi)$. Then by (Nec), we obtain $\vdash \Box_a \psi \leftrightarrow \Box_a t(\psi)$.
- $(\varphi \equiv [\psi]p)$ It suffices to show $\vdash [\psi]p \leftrightarrow t(\psi \rightarrow p)$. By Lemma 24(2) and induction hypothesis, $\vdash (\psi \rightarrow p) \leftrightarrow t(\psi \rightarrow p)$. Moreover by ($[\Box]at$), we have $\vdash [\varphi]p \leftrightarrow (\varphi \rightarrow p)$. Therefore $\vdash [\varphi]p \leftrightarrow t(\varphi \rightarrow p)$ holds.
- $(\varphi \equiv [\psi]\neg\gamma)$ It suffices to show $\vdash [\psi]\neg\gamma \leftrightarrow t(\psi \rightarrow \neg[\psi]\gamma)$. By Lemma 24(3) and induction hypothesis, $\vdash (\psi \rightarrow \neg[\psi]\gamma) \leftrightarrow t(\psi \rightarrow \neg[\psi]\gamma)$. Moreover by ($[\Box]\neg$), we have $\vdash [\psi]\neg\gamma \leftrightarrow (\psi \rightarrow \neg[\psi]\gamma)$. Therefore $\vdash [\psi]\neg\gamma \leftrightarrow t(\varphi \rightarrow p)$ holds.
- $(\varphi \equiv [\psi](\gamma_1 \wedge \gamma_2))$ It suffices to show $\vdash [\psi](\gamma_1 \wedge \gamma_2) \leftrightarrow ([\psi]\gamma_1 \wedge [\psi]\gamma_2)$. By Lemma 24(4) and induction hypothesis, $\vdash [\psi]\gamma_1 \wedge [\psi]\gamma_2 \leftrightarrow t([\psi]\gamma_1 \wedge [\psi]\gamma_2)$. Moreover by ($[\Box]\wedge$), we have $\vdash [\psi](\gamma_1 \wedge \gamma_2) \leftrightarrow ([\psi]\gamma_1 \wedge [\psi]\gamma_2)$. Therefore $\vdash [\psi](\gamma_1 \wedge \gamma_2) \leftrightarrow t([\psi]\gamma_1 \wedge [\psi]\gamma_2)$ holds.

- $(\varphi \equiv [\psi](\gamma \rightarrow \chi))$ It suffices to show $\vdash [\psi](\gamma \rightarrow \chi) \leftrightarrow t([\psi]\gamma \rightarrow [\psi]\chi)$. By Lemma 24(4) and induction hypothesis, $\vdash ([\psi]\gamma \rightarrow [\psi]\chi) \leftrightarrow t([\psi]\gamma \rightarrow [\psi]\chi)$. Moreover by $([\] \rightarrow)$, we have $\vdash [\psi](\gamma \rightarrow \chi) \leftrightarrow ([\psi]\gamma \rightarrow [\psi]\chi)$. Therefore $\vdash [\psi](\gamma \rightarrow \chi) \leftrightarrow t([\psi]\gamma \rightarrow [\psi]\chi)$ holds.
- $(\varphi \equiv [\psi]\Box_a\gamma)$ It suffices to show $\vdash [\psi]\Box_a\gamma \leftrightarrow t(\varphi \rightarrow \Box_a[\psi]\gamma)$. By Lemma 24(5) and induction hypothesis, $\vdash \Box_a[\psi]\gamma \leftrightarrow t(\Box_a[\psi]\gamma)$, i.e. $\vdash \Box_a[\psi]\gamma \leftrightarrow t(\varphi \rightarrow \Box_a[\psi]\gamma)$. Moreover by $([\]\Box_a)$, we have $\vdash [\psi]\Box_a\gamma \leftrightarrow \Box_a[\psi]\gamma$. Therefore $\vdash [\psi]\Box_a\gamma \leftrightarrow t(\varphi \rightarrow \Box_a[\psi]\gamma)$ holds.
- $(\varphi \equiv [\psi][\gamma]\chi)$ It suffices to show $\vdash [\psi][\gamma]\chi \leftrightarrow t([\psi \wedge [\psi]\gamma]\chi)$. By Lemma 24(6) and induction hypothesis, $\vdash ([\psi \wedge [\psi]\gamma]\chi) \leftrightarrow t([\psi \wedge [\psi]\gamma]\chi)$. Moreover by $([\][\])$, we have $\vdash [\psi][\gamma]\chi \leftrightarrow [\psi \wedge [\psi]\gamma]\chi$. Therefore $\vdash [\psi][\gamma]\chi \leftrightarrow t([\psi \wedge [\psi]\gamma]\chi)$ holds.

□

Theorem 26 (Completeness of **PAL**). Let $\Lambda \in \{\mathbf{K45}, \mathbf{S5}\}$. For every $\varphi \in L^+$,

$$\mathbb{F}_{\mathbf{K45}} \models \varphi \text{ implies } \mathbf{H}(\mathbf{K45PAL}) \vdash \varphi,$$

$$\mathbb{F}_{\mathbf{S5}} \models \varphi \text{ implies } \mathbf{H}(\mathbf{S5PAL}) \vdash \varphi.$$

Proof. We comment on the case of **S5**. Fix any $\varphi \in L^+$. Suppose $\mathbb{F}_{\mathbf{S5}} \models \varphi$. By Lemma 25 and Theorem 11, we have $\mathbb{F}_{\mathbf{S5}} \models \varphi \leftrightarrow t(\varphi)$, where $t(\varphi)$ does not contain any public announcement operator. Hence $\mathbb{F}_{\mathbf{S5}} \models t(\varphi)$. Hence $\mathbf{H}(\mathbf{S5}) \vdash t(\varphi)$. Since $\mathbf{H}(\mathbf{S5})$ is a subsystem of $\mathbf{H}(\mathbf{S5PAL})$, we have $\mathbf{H}(\mathbf{S5PAL}) \vdash t(\varphi)$. Then by Lemma 25, $\mathbf{H}(\mathbf{S5PAL}) \vdash \varphi$. □

Chapter 3

Sequent Calculi for Modal Logic K45 and S5

3.1 Sequent calculi G(K45) and G(S5)

In this chapter we introduce sequent calculi G(K45) and G(S5).

Definition 27 (Multiset). A *multiset* is a set which allows for multiple instances for each of its elements.

For example, $\{\varphi, \varphi, \psi\}$ is a multi set where the element φ is repeated.

A *sequent* $\Gamma \Rightarrow \Delta$ is a pair of finite multisets Γ and Δ of formulas and it read as: if all formulas in Γ hold, then at least one formula in Δ holds. This section reviews the known sequent calculi G(K45) and G(S5) for multi-agent K45 and S5, respectively. Both calculi build on LK₀ (see [44]), which is the propositional fragment of a sequent calculus LK [10, 11] of the first-order classical logic.

Definition 28. A sequent calculus LK₀ consists of the following.

- Axioms:

$$\frac{}{\varphi \Rightarrow \varphi} \text{ (id)}.$$

- Structural Rules:

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \varphi} (\Rightarrow w) \quad \frac{\Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta} (w \Rightarrow)$$
$$\frac{\Gamma \Rightarrow \Delta, \varphi, \varphi}{\Gamma \Rightarrow \Delta, \varphi} (\Rightarrow c) \quad \frac{\varphi, \varphi, \Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta} (c \Rightarrow).$$

- Logical Rules:

$$\begin{array}{c}
 \frac{\varphi, \Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \neg \varphi} (\Rightarrow \neg) \quad \frac{\Gamma \Rightarrow \Delta, \varphi}{\neg \varphi, \Gamma \Rightarrow \Delta} (\neg \Rightarrow) \\
 \\
 \frac{\Gamma \Rightarrow \Delta, \varphi_1 \quad \Gamma \Rightarrow \Delta, \varphi_2}{\Gamma \Rightarrow \Delta, \varphi_1 \wedge \varphi_2} (\Rightarrow \wedge) \quad \frac{\varphi_i, \Gamma \Rightarrow \Delta}{\varphi_1 \wedge \varphi_2, \Gamma \Rightarrow \Delta} (\wedge \Rightarrow) (i \in \{1, 2\}) \\
 \\
 \frac{\varphi, \Gamma \Rightarrow \Delta, \psi}{\Gamma \Rightarrow \Delta, \varphi \rightarrow \psi} (\Rightarrow \rightarrow) \quad \frac{\Gamma \Rightarrow \Delta, \varphi \quad \psi, \Pi \Rightarrow \Sigma}{\varphi \rightarrow \psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (\rightarrow \Rightarrow).
 \end{array}$$

- Cut:

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut)_\varphi.$$

The sequent calculus **G(K45)** is **LK₀** expanded with the following rule:

$$\frac{\Gamma, \Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi} (\Box_{K45}).$$

The sequent calculus **G(S5)** expands **LK₀** with the following rules:

$$\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi} (\Box_{S5}) \quad \frac{\varphi, \Gamma \Rightarrow \Delta}{\Box_a \varphi, \Gamma \Rightarrow \Delta} (\Box_a \Rightarrow).$$

For each calculus, we define the notion of *derivability* of a sequent as a finite tree generated from axioms (id) by inference rules specific to the calculus. Moreover we use = as an abbreviation of finite applications of structural rules, such as:

$$\begin{array}{c}
 \vdots \mathcal{D}' \\
 \hline
 \varphi, \varphi \Rightarrow \psi \\
 \hline
 \varphi \Rightarrow \psi, \psi,
 \end{array}$$

where $(c \Rightarrow)$ and $(\Rightarrow w)$ is applied for once, respectively.

Proposition 29. Let $\Lambda \in \{\mathbf{K45}, \mathbf{S5}\}$.

- (1) If $\Gamma \Rightarrow \Delta$ is derivable in **G(Λ)** then $\bigwedge \Gamma \rightarrow \bigvee \Delta$ is a theorem of **H(Λ)**.
- (2) If φ is a theorem of **H(Λ)**, then $\Rightarrow \varphi$ is derivable in **G(Λ)**.

Proof. (1) By induction on height of the derivation $\mathcal{D}$ to obtain $\Gamma \Rightarrow \Delta$. We prove the following statement by induction on n :

$$\forall \Pi \Rightarrow \Sigma \text{ in } \mathcal{D} \text{ (if the height of } \Pi \Rightarrow \Sigma \text{ in } \mathcal{D} \text{ is } n, \text{ then } \vdash_{\Lambda} \bigwedge \Gamma \rightarrow \bigvee \Delta).$$

For base step, let $n = 0$. Then $\Gamma \Rightarrow \Delta$ is an axiom $\varphi \Rightarrow \varphi$. Hence $\vdash_{\Lambda} \varphi \rightarrow \varphi$ holds clearly.

For inductive step, let $n > 0$. Then, we divide our argument depending on the last applied rule to obtain $\Gamma \Rightarrow \Delta$. We comment on the cases of $(\Box_{\mathbf{K45}})$, $(\Box_{\mathbf{S5}})$ and $(\Box_a \Rightarrow)$.

- Let the last applied rule be $(\Box_{\mathbf{K45}})$:

$$\frac{\Gamma, \Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi} (\Box_{\mathbf{K45}}).$$

By induction hypothesis, we obtain $\vdash_{\mathbf{K45}} \bigwedge \Gamma \wedge \bigwedge \Box_a \Gamma \rightarrow \bigvee \Box_a \Delta \vee \varphi$. By (Nec) and $\vdash_{\mathbf{K45}} \Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a \varphi \rightarrow \Box_a \psi)$, we obtain $\vdash_{\mathbf{K45}} \Box_a(\bigwedge \Gamma \wedge \bigwedge \Box_a \Gamma) \rightarrow \Box_a(\bigvee \Box_a \Delta \vee \varphi)$. Then by $\vdash_{\mathbf{K45}} \Box_a \varphi \rightarrow \Box_a(\varphi \wedge \Box_a \varphi)$ and $\vdash_{\mathbf{K45}} \Box_a(\Box_a \varphi \vee \psi) \rightarrow \Box_a \varphi \vee \Box_a \psi$, we obtain our goal, which is $\vdash_{\mathbf{K45}} \bigwedge \Box_a \Gamma \rightarrow \bigvee \Box_a \Delta \vee \Box_a \varphi$.

- Let the last applied rule be $(\Box_{\mathbf{S5}})$:

$$\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi} (\Box_{\mathbf{S5}}).$$

By induction hypothesis, we obtain $\vdash_{\mathbf{S5}} \bigwedge \Box_a \Gamma \rightarrow \bigvee \Box_a \Delta \vee \varphi$. By (Nec) and $\vdash_{\mathbf{S5}} \Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a \varphi \rightarrow \Box_a \psi)$, we obtain $\vdash_{\mathbf{S5}} \Box_a \bigwedge \Box_a \Gamma \rightarrow \Box_a(\bigvee \Box_a \Delta \vee \varphi)$. Then by $\vdash_{\mathbf{S5}} \Box_a \varphi \rightarrow \varphi$ and $\vdash_{\mathbf{S5}} \Box_a(\Box_a \varphi \vee \psi) \rightarrow \Box_a \varphi \vee \Box_a \psi$, we obtain our goal, which is $\vdash_{\mathbf{S5}} \bigwedge \Box_a \Gamma \rightarrow \bigvee \Box_a \Delta \vee \Box_a \varphi$.

- Let the last applied rule be $(\Box_a \Rightarrow)$:

$$\frac{\varphi, \Gamma \Rightarrow \Delta}{\Box_a \varphi, \Gamma \Rightarrow \Delta} (\Box_a \Rightarrow).$$

By induction hypothesis, we obtain $\vdash_{\mathbf{S5}} \varphi \wedge \bigwedge \Gamma \rightarrow \bigvee \Delta$. By (Nec) and $\vdash_{\mathbf{S5}} \Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a \varphi \rightarrow \Box_a \psi)$, we obtain $\vdash_{\mathbf{S5}} \Box_a(\varphi \wedge \bigwedge \Gamma) \rightarrow \Box_a \bigvee \Delta$. Then by $\vdash_{\mathbf{S5}} \Box_a(\varphi \wedge \psi) \rightarrow \Box_a \varphi \wedge \Box_a \psi$ and $\vdash_{\mathbf{S5}} \Box_a \varphi \rightarrow \varphi$, we obtain our goal, which is $\vdash_{\mathbf{S5}} \Box_a \varphi \wedge \bigwedge \Gamma \rightarrow \bigvee \Delta$.

(2) We divide the argument into cases for **G(K45)** and **G(S5)**:

• (**G(K45)**):

– (K):

$$\begin{array}{c}
 \frac{\varphi \Rightarrow \varphi \quad \psi \Rightarrow \psi}{\varphi \rightarrow \psi, \varphi \Rightarrow \psi} (\rightarrow \Rightarrow) \\
 \frac{\varphi \rightarrow \psi, \varphi \Rightarrow \psi}{\varphi \rightarrow \psi, \varphi, \Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \psi} (w) \\
 \frac{\varphi \rightarrow \psi, \varphi, \Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \psi}{\Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \Box_a \psi} (\Box_{K45}) \\
 \frac{\Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \Box_a \psi}{\Box_a(\varphi \rightarrow \psi) \Rightarrow \Box_a \varphi \rightarrow \Box_a \psi} (\rightarrow \Rightarrow) \\
 \frac{\Box_a(\varphi \rightarrow \psi) \Rightarrow \Box_a \varphi \rightarrow \Box_a \psi}{\Rightarrow \Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a \varphi \rightarrow \Box_a \psi)} (\rightarrow \Rightarrow).
 \end{array}$$

– (4):

$$\begin{array}{c}
 \frac{\Box_a \varphi \Rightarrow \Box_a \varphi}{\varphi, \Box_a \varphi \Rightarrow \Box_a \varphi} (w \Rightarrow) \\
 \frac{\varphi, \Box_a \varphi \Rightarrow \Box_a \varphi}{\Box_a \varphi \Rightarrow \Box_a \Box_a \varphi} (\Box_{K45}) \\
 \frac{\Box_a \varphi \Rightarrow \Box_a \Box_a \varphi}{\Rightarrow \Box_a \varphi \rightarrow \Box_a \Box_a \varphi} (\rightarrow \Rightarrow).
 \end{array}$$

– (5):

$$\begin{array}{c}
 \frac{\Box_a \varphi \Rightarrow \Box_a \varphi}{\Rightarrow \Box_a \varphi, \neg \Box_a \varphi} (\Rightarrow \neg) \\
 \frac{\Rightarrow \Box_a \varphi, \neg \Box_a \varphi}{\Rightarrow \Box_a \varphi, \Box_a \neg \Box_a \varphi} (\Box_{K45}) \\
 \frac{\Rightarrow \Box_a \varphi, \Box_a \neg \Box_a \varphi}{\neg \Box_a \varphi \Rightarrow \Box_a \neg \Box_a \varphi} (\neg \Rightarrow) \\
 \frac{\neg \Box_a \varphi \Rightarrow \Box_a \neg \Box_a \varphi}{\Rightarrow \neg \Box_a \varphi \rightarrow \Box_a \neg \Box_a \varphi} (\rightarrow \Rightarrow).
 \end{array}$$

– (MP):

$$\frac{\frac{\Rightarrow \varphi \quad \psi \Rightarrow \psi}{\varphi \rightarrow \psi, \varphi \Rightarrow \psi} (\rightarrow \Rightarrow) \quad \frac{\varphi \rightarrow \psi, \varphi \Rightarrow \psi}{\Rightarrow \psi} (Cut)_{\varphi \rightarrow \psi}.$$

– (Nec):

$$\frac{\Rightarrow \varphi}{\Rightarrow \Box_a \varphi} (\Box_{K45}).$$

• (**G(S5)**)

– (K):

$$\begin{array}{c}
 \frac{\varphi \Rightarrow \varphi \quad \psi \Rightarrow \psi}{\varphi \rightarrow \psi, \varphi \Rightarrow \psi} (\rightarrow \Rightarrow) \\
 \frac{\varphi \rightarrow \psi, \varphi \Rightarrow \psi}{\varphi \rightarrow \psi, \Box_a \varphi \Rightarrow \psi} (\Box_a \Rightarrow) \\
 \frac{\varphi \rightarrow \psi, \Box_a \varphi \Rightarrow \psi}{\Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \psi} (\Box_a \Rightarrow) \\
 \frac{\Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \Box_a \psi}{\Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \Box_a \psi} (\Box_{S5}) \\
 \frac{\Box_a(\varphi \rightarrow \psi), \Box_a \varphi \Rightarrow \Box_a \psi}{\Box_a(\varphi \rightarrow \psi) \Rightarrow \Box_a \varphi \rightarrow \Box_a \psi} (\rightarrow \Rightarrow) \\
 \frac{\Box_a(\varphi \rightarrow \psi) \Rightarrow \Box_a \varphi \rightarrow \Box_a \psi}{\Rightarrow \Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a \varphi \rightarrow \Box_a \psi)} (\rightarrow \Rightarrow).
 \end{array}$$

– (T):

$$\frac{\frac{\varphi \Rightarrow \varphi}{\Box_a \varphi \Rightarrow \varphi} (\Box_a \Rightarrow)}{\Rightarrow \Box_a \varphi \rightarrow \varphi} (\rightarrow \Rightarrow).$$

– (4):

$$\frac{\frac{\Box_a \varphi \Rightarrow \Box_a \varphi}{\Box_a \varphi \Rightarrow \Box_a \Box_a \varphi} (\Box_{S5})}{\Rightarrow \Box_a \varphi \rightarrow \Box_a \Box_a \varphi} (\rightarrow \Rightarrow).$$

– (5):

$$\frac{\frac{\frac{\Box_a \varphi \Rightarrow \Box_a \varphi}{\Rightarrow \Box_a \varphi, \neg \Box_a \varphi} (\Rightarrow \neg)}{\Rightarrow \Box_a \varphi, \Box_a \neg \Box_a \varphi} (\Box_{S5})}{\neg \Box_a \varphi \Rightarrow \Box_a \neg \Box_a \varphi} (\neg \Rightarrow)}{\Rightarrow \neg \Box_a \varphi \rightarrow \Box_a \neg \Box_a \varphi} (\rightarrow \Rightarrow).$$

– (MP):

$$\frac{\frac{\Rightarrow \varphi \quad \psi \Rightarrow \psi}{\varphi \rightarrow \psi \quad \psi \Rightarrow \psi} (\rightarrow \Rightarrow)}{\Rightarrow \psi} (Cut)_{\varphi \rightarrow \psi}.$$

– (Nec):

$$\frac{\Rightarrow \varphi}{\Rightarrow \Box_a \varphi} (\Box_{S5}).$$

□

3.2 Cut-elimination Theorem of G(**K45**)

In this section we establish the cut-elimination for **G(K45)** adapted from [42, Theorem 2]. The following is an extended version of the cut rule from [20], which plays the similar role essentially as Mix rule by Gentzen [10] to handle the difficulty caused by contraction rules.

Definition 30. (*Ecut*) is the following rule:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho,$$

where $m, n \in \mathbb{N}$, i.e., m and n are possibly zero.

It is noted that when $m = n = 1$ (*Ecut*) is the same as (*Cut*).

Definition 31. We define $G^-(\mathbf{K45})$ as the same calculus as $G(\mathbf{K45})$ except that the cut rule is excluded. The sequent calculus $G^*(\mathbf{K45})$ is the same calculus as $G(\mathbf{K45})$ except that the cut rule is replaced with $(Ecut)$.

Definition 32. The formulas that do not change in an inference rule except for $(\Box_{\mathbf{K45}})$ in Definition 28 and $(\Box_{\mathbf{S5}})$ in Definition 28 are called *parameters*. For the other formulas, those occurring in the lower sequent of an inference rule are called *principal formulas*, and those occurring in the upper sequent are called *active formulas*. For $(\Box_{\mathbf{K45}})$ and $(\Box_{\mathbf{S5}})$, the principal formulas are defined to be $\Box_a\Gamma$, $\Box_a\Delta$ and $\Box_a\varphi$ hence there are no parameters in these rules.

For example, the principal formula of $(\Rightarrow \vee)$ is $\varphi \vee \psi$.

Theorem 33. If $\Gamma \Rightarrow \Delta$ is derivable in $G^*(\mathbf{K45})$, then it is also derivable in $G^-(\mathbf{K45})$. Therefore, if $\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{K45})$ then it is also derivable in $G^-(\mathbf{K45})$.

Proof. It suffices to prove the first statement. For our purpose, we establish the following (*Hauptsatz*): If in the following derivation $\mathcal{D}$ of $G^*(\mathbf{K45})$:

$$\mathcal{D} \equiv \frac{\frac{\vdots \mathcal{L}}{\Gamma \Rightarrow \Delta, \varphi^m} \text{rule}(\mathcal{L}) \quad \frac{\frac{\vdots \mathcal{R}}{\varphi^n, \Pi \Rightarrow \Sigma} \text{rule}(\mathcal{R})}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\varphi,$$

there is no other $(Ecut)$ in derivations of $\Gamma \Rightarrow \Delta, \varphi^m$ and $\varphi^n, \Pi \Rightarrow \Sigma$, then there is a derivation in $G^-(\mathbf{K45PAL})$ of the sequent $\Gamma, \Pi \Rightarrow \Delta, \Sigma$. We prove this statement on a derivation $\mathcal{D}$ by double induction on the *complexity* $c(\mathcal{D})$ and *weight* $w(\mathcal{D})$ of the derivation $\mathcal{D}$ (or the lexicographic order of the complexity and the weight), which are defined as follows (see [20, pp.135-136]):

- The *complexity* $c(\mathcal{D})$ of a derivation $\mathcal{D}$ is $c(\varphi)$ of the cut formula φ of $(Ecut)$, as defined in Definition 5.
- The *weight* $w(\mathcal{D})$ of a derivation $\mathcal{D}$ is the number of all sequents in derivations $\mathcal{L}$ and $\mathcal{R}$.

First we consider the case where $m = 0$ or $n = 0$.

- Suppose $m = 0$. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\Gamma \Rightarrow \Delta \quad \varphi^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut).$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (w).$$

- Suppose $n = 0$. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\Gamma \Rightarrow \Delta \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut).$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (w).$$

Since in both cases, our goal can be obtained by weakening rules, we suppose $m > 0$ and $n > 0$ below. We divide the argument into the following cases.

- (1) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be (id).
- (2) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be structural rules.
- (3) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be logical rules or ($\Box_{\mathbf{K45}}$) where the cut formula is not principal.
- (4) Let $\text{rule}(\mathcal{L})$ and $\text{rule}(\mathcal{R})$ be rules of the same connective and the cut formula is principal in both rules.

We only comment on a typical case of (4). Let $\text{rule}(\mathcal{L})$ and $\text{rule}(\mathcal{R})$ be ($\Box_{\mathbf{K45}}$). Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma, \Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi, \Box_a \rho^{m-1}}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, \Box_a \rho^{m-1}} (\Box_{\mathbf{K45}}) \quad \frac{\rho^n, \Box_a \rho^n, \Pi, \Box_a \Pi \Rightarrow \Box_a \Sigma, \psi}{\Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \Box_a \psi} (\Box_{\mathbf{K45}})}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \varphi, \Box_a \psi} (Ecut)_\rho.$$

Here we divide the argument into the following two cases: $\varphi \equiv \rho$ and $\varphi \not\equiv \rho$.

- Suppose $\varphi \equiv \rho$. Then our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma, \Box_a \Gamma \Rightarrow \Box_a \Delta, \rho, \Box_a \rho^{m-1}}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \rho^m} (\Box_{\mathbf{K45}}) \quad \frac{\rho^n, \Box_a \rho^n, \Pi, \Box_a \Pi \Rightarrow \Box_a \Sigma, \psi}{\Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \Box_a \psi} (\Box_{\mathbf{K45}})}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi} (Ecut)_{\Box_a \rho}.$$

We transform this derivation into:

$$\begin{array}{c}
 \mathcal{L} = \frac{\Gamma, \Box_a \Gamma \Rightarrow \Box_a \Delta, \rho, \Box_a \rho^{m-1} \quad \Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \Box_a \psi}{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \rho, \Box_a \psi} (Ecut)_{\Box_a \rho}^1 \\
 \vdots \\
 \mathcal{R} = \frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \rho^m \quad \rho^n, \Box_a \rho^n, \Pi, \Box_a \Pi \Rightarrow \Box_a \Sigma, \psi}{\rho^n, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \psi} (Ecut)_{\Box_a \rho}^2 \\
 \vdots \\
 \frac{\begin{array}{c} \vdots \mathcal{L} \\ \Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \rho, \Box_a \psi \end{array} \quad \begin{array}{c} \vdots \mathcal{R} \\ \rho^n, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \psi \end{array}}{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \rho, \Box_a \psi \quad \rho^n, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \psi} (Ecut)_{\rho}^3 \\
 \frac{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \rho, \Box_a \psi \quad \rho^n, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \psi}{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \rho, \Box_a \psi} (c) \\
 \frac{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \rho, \Box_a \psi}{\Gamma, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \psi} (\Box)_{\mathbf{K45}} \\
 \frac{\Gamma, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \psi}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \psi} (\Rightarrow c) \\
 \frac{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \psi}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi} .
 \end{array}$$

By induction hypotheses, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced, and $(Ecut)^3$ can be reduced since its complexity is reduced since $c(\rho) < c(\Box_a \rho)$.

- Suppose $\varphi \neq \rho$. We transform this derivation into:

$$\begin{array}{c}
 \frac{\Gamma, \Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi, \Box_a \rho^{m-1} \quad \Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \Box_a \psi}{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \varphi} (Ecut)_{\Box_a \rho} \\
 \frac{\Gamma, \Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \varphi}{\Gamma, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \varphi} (w \Rightarrow) \\
 \frac{\Gamma, \Box_a \Gamma, \Pi, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \psi, \varphi}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \varphi, \Box_a \psi} (\Box)_{\mathbf{K45}} .
 \end{array}$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

□

3.3 Analytic Cut Property of **G(S5)**

A formula $p \Rightarrow \Box_a \neg \Box_a \neg p$ is derivable in **G(S5)** with the cut rule as follows:

$$\frac{\frac{\Box_a \neg p \Rightarrow \Box_a \neg p}{\Rightarrow \neg \Box_a \neg p, \Box_a \neg p} (\Rightarrow \neg) \quad \frac{p \Rightarrow p}{\neg p, p \Rightarrow} (\neg \Rightarrow)}{\Rightarrow \Box_a \neg \Box_a \neg p, \Box_a \neg p} (\Box_{SS}) \quad \frac{p \Rightarrow p}{\Box_a \neg p, p \Rightarrow} (\Box_a \Rightarrow)}{p \Rightarrow \Box_a \neg \Box_a \neg p} (Cut)_{\neg p} .$$

Definition 34. The sequent calculus $G^-(S5)$ is the same calculus as $G(S5)$ except that (Cut) is eliminated.

Proposition 35. The sequent $p \Rightarrow \Box_a \neg \Box_a \neg p$ is not derivable in $G^-(S5)$.

Proof. See Section A.1. □

As shown in Proposition 35, we cannot eliminate the above application of the cut rule. Instead, Takano [44] established the analytic cut property of $G(S5)$ syntactically (a semantic argument can be found in [43]).

Definition 36. We say that an $(Ecut)_\rho$ of Definition 30 is *admissible* in $G^a(S5)$ if derivability in $G^a(S5PAL)$ of both upper sequents $\Gamma \Rightarrow \Delta, \rho$ and $\rho, \Pi \Rightarrow \Sigma$ of the $(Ecut)_\rho$ implies derivability in $G^a(S5)$ of the lower sequent $\Gamma, \Pi \Rightarrow \Delta, \Sigma$.

Definition 37. We say that an inference rule (r) :

$$\frac{\Gamma_1 \Rightarrow \Delta_1 \quad \cdots \quad \Gamma_n \Rightarrow \Delta_n}{\Gamma_0 \Rightarrow \Delta_0} (r)$$

has the *subformula property* if every formula occurring in the upper sequent(s) $\Gamma_i \Rightarrow \Delta_i$ of the rule (r) constitutes a subformula of some formulas in its lower sequent $\Gamma_0 \Rightarrow \Delta_0$.

Proposition 38. Every inference rule of $G(S5)$ except for (Cut) has the subformula property.

Definition 39. A cut

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut)_\varphi$$

is *analytic* if $\varphi \in CL(\Gamma, \Pi, \Delta, \Sigma)$, otherwise it is *non-analytic*. It follows that if $\varphi \in Sub(\Gamma, \Pi, \Delta, \Sigma)$, the (Cut) is analytic. A derivation without non-analytic cuts is said to be an *analytic derivation*, or an *a-derivation* for short. The sequent calculus $G^a(S5)$ is the same calculus as $G(S5)$ except that (Cut) is always analytic.

The following argument of Takano [44, Lemma 3.2] plays a crucial role in the proof of analytic cut property of $G(S5)$.

Lemma 40 (Takano [44]). Suppose that $(Ecut)_\psi$ is admissible in $G^a(S5)$ for all subformulas ψ of φ . If the sequent $\Box_a \Gamma \Rightarrow \Box_a \Delta$ is derivable in $G^a(S5)$ but with an application of $(\Box_{S5})$ for its lowest inference, then the sequent $\Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi$ is derivable in $G^a(S5)$.

Proof. See Section A.2. □

Definition 41. We define $G^*(S5)$ as the same calculus as $G(S5)$ except that the cut rule is replaced with $(Ecut)$.

The following theorem is a simplified version of the main theorem in [44].

Theorem 42 (Takano [44]). If $\Gamma \Rightarrow \Delta$ is derivable in $G^*(S5)$ then it is also derivable in $G^a(S5)$. Therefore, if $\Gamma \Rightarrow \Delta$ is derivable in $G(S5)$ then it is also derivable in $G^a(S5)$.

Proof. The following argument is an improvement of Takano's argument in [44], in that we use $(Ecut)$ instead of Mix rule by Gentzen [10] to simplify the outline of the argument. It suffices for our purpose to show the following:

Any derivation $\mathcal{D}$ with an unacceptable $Ecut$ for its lowest inference, and not containing other unacceptable $Ecuts$, can be transformed into another derivation in $G^a(S5)$ without changing the endsequent, where no unacceptable $Ecut$ occurs.

In this way, we can transform all the unacceptable $(Ecut)$ s in a derivation from top to bottom and finally turn it into a derivation in $G^a(S5)$.

We prove the statement by double induction on complexity and weight of $\mathcal{D}$ in $G(S5)$: if in

$$\mathcal{D} \equiv \frac{\frac{\vdots \mathcal{L}}{\Gamma \Rightarrow \Delta, \varphi^m} \text{rule}(\mathcal{L}) \quad \frac{\vdots \mathcal{R}}{\varphi^n, \Pi \Rightarrow \Sigma} \text{rule}(\mathcal{R})}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\varphi,$$

there is no $(Ecut)$ in derivations in $G^a(S5)$ of $\Gamma \Rightarrow \Delta, \varphi^m$ and $\varphi^n, \Pi \Rightarrow \Sigma$, then there is a derivation in $G^a(S5)$ with the same endsequent. We use the following two induction hypotheses:

- For all derivation in $G(S5)$ whose complexity is smaller than c , there is another derivation with the same endsequent in $G^a(S5)$.
- For all derivation in $G(S5)$ whose complexity is c and the weight is smaller than w , there is another derivation with the same endsequence in $G^a(S5)$.

Besides, if $m = 0$ or $n = 0$, it follows our goal can be obtained by weakening rules, hence we suppose $m > 0$ and $n > 0$ below. We divide the argument into the following cases determined by the rules applied last above the $Ecut$.

1. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be (id) .
2. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be structural rules.
3. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be an a-cut.
4. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be logical rules where the cut formula is not principal.
5. Let $\text{rule}(\mathcal{L})$ and $\text{rule}(\mathcal{R})$ be rules of the same logical connective and the cut formula is principal in both rules.

Since cases (1), (2) and (4) are handled similarly to the proof of Theorem 33, we discuss only (3) and (5) for the cut formula of the form $\Box_a\varphi$. Lemma 40 is needed in case (5).

- (3-1) Suppose the left-upper sequent is the lower sequent of an a-cut. Our $\mathcal{D}$ looks like as follows:

$$\frac{\frac{\Gamma' \Rightarrow \Delta', \psi, \rho^a \quad \psi, \Gamma'' \Rightarrow \Delta'', \rho^b}{\Gamma', \Gamma'' \Rightarrow \Delta', \Delta'', \rho^{a+b}} (Cut)_\psi^a \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma', \Gamma'', \Pi \Rightarrow \Delta', \Delta'', \Sigma} (Ecut)_\rho.$$

where $\psi \in \text{Sub}(\Gamma', \Gamma'', \Delta', \Delta'', \rho^{a+b})$. Here we divide the argument into the following two cases: $\psi \equiv \rho$ and $\psi \not\equiv \rho$.

- Suppose $\psi \equiv \rho$. We transform $\mathcal{D}$ into:

$$\frac{\frac{\Gamma' \Rightarrow \Delta', \psi, \rho^a \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma', \Pi \Rightarrow \Delta', \Sigma} (Ecut)_\rho}{\frac{\Gamma', \Pi \Rightarrow \Delta', \Sigma}{\Gamma', \Gamma'', \Pi \Rightarrow \Delta', \Delta'', \Sigma} (w)}.$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Suppose $\psi \not\equiv \rho$. We transform $\mathcal{D}$ into:

$$\frac{\frac{\Gamma' \Rightarrow \Delta', \psi, \rho^a \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma', \Pi \Rightarrow \Delta', \psi, \Sigma} (Ecut)_\rho^1 \quad \frac{\psi, \Gamma'' \Rightarrow \Delta, \rho^b \quad \rho^n, \Pi \Rightarrow \Sigma}{\psi, \Gamma'', \Pi \Rightarrow \Delta'', \Sigma} (Ecut)_\rho^2}{\frac{\Gamma', \Gamma'', \Pi, \Pi \Rightarrow \Delta', \Delta'', \Sigma, \Sigma}{\Gamma', \Gamma'', \Pi \Rightarrow \Delta', \Delta'', \Sigma} (c)} (Cut)_\psi^3.$$

In this derivation $(Ecut)^1$ and $(Ecut)^2$ can be eliminated by induction hypothesis since their weights are reduced. For $(Cut)^3$, we divide the argument into the following two cases: $\psi \in \text{Sub}(\rho)$ and $\psi \notin \text{Sub}(\rho)$.

- * Suppose $\psi \in \text{Sub}(\rho)$. Since $\psi \neq \rho$, complexity of $(Cut)^3$ is reduced. Hence it can be eliminated by induction hypothesis.
 - * Suppose $\psi \notin \text{Sub}(\rho)$. By $\psi \in \text{Sub}(\Gamma', \Gamma'', \Delta', \Delta'', \rho^{a+b})$, we have $\psi \in \text{Sub}(\Gamma', \Gamma'', \Delta', \Delta'')$. Then $(Cut)^3$ is an a-cut since $\psi \in \text{Sub}(\Gamma', \Gamma'', \Delta', \Delta'') \subseteq \text{Sub}(\Gamma', \Gamma'', \Pi, \Pi, \Delta', \Delta'', \Sigma, \Sigma)$.
- (3-2) Suppose the right-upper sequent is the lower sequent of an a-cut. The argument is similar to the left case. Our $\mathcal{D}$ looks like as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^a, \Pi' \Rightarrow \Sigma', \psi \quad \psi, \rho^b, \Pi'' \Rightarrow \Sigma''}{\rho^{a+b}, \Pi', \Pi'' \Rightarrow \Sigma', \Sigma''} (Cut_\psi^a)}{\Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Sigma', \Sigma''} (Ecut)_\rho .$$

where $\psi \in \text{Sub}(\Pi', \Pi'', \Sigma', \Sigma'', \rho^{a+b})$. Again we divide the argument into the following two cases: $\psi \equiv \rho$ and $\psi \neq \rho$.

- Suppose $\psi \equiv \rho$. We transform $\mathcal{D}$ into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \psi, \rho^b, \Pi'' \Rightarrow \Sigma''}{\Gamma, \Pi'' \Rightarrow \Delta, \Sigma''} (Ecut)_\rho}{\Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Sigma', \Sigma''} (w) .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Suppose $\psi \neq \rho$. We transform $\mathcal{D}$ into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^a, \Pi' \Rightarrow \Sigma', \psi}{\Gamma, \Pi' \Rightarrow \Delta, \Sigma', \psi} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \psi, \rho^b, \Pi'' \Rightarrow \Sigma''}{\psi, \Gamma, \Pi'' \Rightarrow \Delta, \Sigma''} (Ecut)_\rho^2}{\frac{\Gamma, \Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Delta \Sigma', \Sigma''}{\Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Sigma', \Sigma''} (c)} (Cut)_\psi^3 .$$

In this derivation $(Ecut)^1$ and $(Ecut)^2$ can be eliminated by induction hypothesis since their weights are reduced. For $(Cut)^3$, we divide the argument into the following two cases: $\psi \in \text{Sub}(\rho)$ and $\psi \notin \text{Sub}(\rho)$.

- * Suppose $\psi \in \text{Sub}(\rho)$. Since $\psi \neq \rho$, complexity of $(Cut)^3$ is reduced. Hence it can be eliminated by induction hypothesis.
 - * Suppose $\psi \notin \text{Sub}(\rho)$. By $\psi \in \text{Sub}(\Pi', \Pi'', \Sigma', \Sigma'', \rho^{a+b})$, we have $\psi \in \text{Sub}(\Pi', \Pi'', \Sigma', \Sigma'')$. Then $(Cut)^3$ is an a-cut since $\psi \in \text{Sub}(\Pi', \Pi'', \Sigma', \Sigma'') \subseteq \text{Sub}(\Gamma, \Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Delta\Sigma', \Sigma'')$.
- (5-1) Suppose the left upper sequent is $(\Box_{\mathbf{S5}})$ and the right upper sequent is $(\Box_a \Rightarrow)$ where the cut formula is the principal formula in both rules. In this case, the derivation $\mathcal{D}$ runs as follows:

$$\frac{\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi, \Box_a \rho^m}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, \Box_a \rho^m} (\Box_{\mathbf{S5}}) \quad \frac{\rho, \Box_a \rho^{n-1}, \Pi \Rightarrow \Sigma}{\Box_a \rho^n, \Pi \Rightarrow \Sigma} (\Box_a \Rightarrow)}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma} (Ecut) .$$

Here we divide the argument into the following two cases: $\varphi \equiv \rho$ and $\varphi \neq \rho$.

- Suppose $\varphi \equiv \rho$. Then the derivation $\mathcal{D}$ runs as follows:

$$\frac{\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi^{m-1}, \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi^m} (\Box_{\mathbf{S5}}) \quad \frac{\varphi, \Box_a \varphi^{n-1} \Rightarrow \Sigma}{\Box_a \varphi^n, \Pi \Rightarrow \Sigma} (\Box_a \Rightarrow)}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Sigma} (Ecut) .$$

We transform this derivation into:

$$\begin{aligned} \mathcal{L} &= \frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi^{m-1}, \varphi}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \varphi, \Sigma} (Ecut)^1 \\ \mathcal{R} &= \frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi^m \quad \varphi, \Box_a \varphi^{n-1} \Rightarrow \Sigma}{\varphi, \Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Sigma} (Ecut)^2 \\ &\quad \vdots \mathcal{L} \quad \quad \quad \vdots \mathcal{R} \\ &= \frac{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \varphi, \Sigma \quad \varphi, \Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Sigma}{\Box_a \Gamma, \Box_a \Gamma, \Pi, \Pi \Rightarrow \Box_a \Delta, \Box_a \Delta, \Sigma, \Sigma} (Cut)^3 \\ &\quad \frac{\Box_a \Gamma, \Box_a \Gamma, \Pi, \Pi \Rightarrow \Box_a \Delta, \Box_a \Delta, \Sigma, \Sigma}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Sigma} (c) . \end{aligned}$$

By induction hypotheses $(Ecut)^1$ and $(Ecut)^2$ can be eliminated because their complexity are the same and their weight are reduced. $(Cut)^3$ can also be eliminated since its complexity is reduced.

- Suppose $\varphi \neq \rho$. By Lemma 40, if $\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, \Box_a \rho^m$ is derivable in $\mathbf{G}^a(\mathbf{S5})$, then $\Box_a \Gamma \Rightarrow \Box_a(\Delta_\rho), \Box_a \varphi, \rho$ is also derivable in $\mathbf{G}^a(\mathbf{S5})$. Hence we transform the derivation into:

$$\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, \Box_a \rho^m \quad \frac{\rho, \Box_a \rho^{n-1}, \Pi \Rightarrow \Sigma}{\Box_a \Gamma, \Pi, \rho \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma} (Ecut)^1}{\frac{\Box_a \Gamma \Rightarrow \Box_a(\Delta_\rho), \Box_a \varphi, \rho \quad \Box_a \Gamma, \Pi, \rho \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma}{\Box_a \Gamma, \Box_a \Gamma, \Pi \Rightarrow \Box_a(\Delta_\rho), \Box_a \Delta, \Box_a \varphi, \Box_a \varphi, \Sigma} (Cut)^2}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma}.$$

By induction hypotheses, $(Ecut)^1$ can be eliminated since its weight is reduced, and $(Cut)^2$ can be reduced since its complexity is reduced.

- (5-2) Suppose both the upper sequents are lower sequents of $(\Box_{S5})$ s and the cut formula is the principal formula. In this case, the derivation $\mathcal{D}$ runs as follows:

$$\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \rho^m \quad \frac{\Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \varphi}{\Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \Box_a \varphi} (\Box_{S5})}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \varphi} (Ecut).$$

We transform the derivation into:

$$\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \rho^m \quad \Box_a \rho^n, \Box_a \Pi \Rightarrow \Box_a \Sigma, \varphi}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \varphi} (Ecut) \quad \frac{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \varphi}{\Box_a \Gamma, \Box_a \Pi \Rightarrow \Box_a \Delta, \Box_a \Sigma, \Box_a \varphi} (\Box_{S5}).$$

By induction hypotheses the $(Ecut)$ can be eliminated since the weight is reduced.

We have finished establishing Theorem 42. $\square$

3.4 Decidability of $\mathbf{G(K45)}$ and $\mathbf{G(S5)}$

In this section we discuss the decision problem of $\mathbf{G(K45)}$ and $\mathbf{G(S5)}$ based on [33, Section 4]. A given logic $\mathbf{L}$ is said to be *decidable* if there exists an algorithm with which for any formula φ , we can decide whether φ is provable in $\mathbf{L}$ or not. Otherwise it is *undecidable*. The *decision problem* of a given logic $\mathbf{L}$ is the problem

as to whether **L** is decidable or not. As a consequence of the cut-elimination theorem, $G(\mathbf{K45})$ is decidable. $G(\mathbf{S5})$ is decidable as a subsequence of Theorem 42.

First we explain a method of checking the provability of a given sequent due to Gentzen. Let $\Lambda \in \{\mathbf{K45}, \mathbf{S5}\}$. We start by defining the following notions:

Definition 43. • A sequent $\Sigma \Rightarrow \Pi$ is *reduced* if every formula in it occurs at most three times in the antecedent and also at most three times in the succedent.

- A sequent $\Sigma \Rightarrow \Pi$ is *1-reduced* if every formula in it occurs exactly one time in the antecedent and exactly one time in the succedent.
- A sequent $\Gamma^\dagger \Rightarrow \Delta^\dagger$ is called a *contraction* of a sequent $\Gamma \Rightarrow \Delta$ if it is obtained from $\Gamma \Rightarrow \Delta$ by applying the contraction rules repeatedly.

We can show easily that for any given sequent $\Gamma \Rightarrow \Delta$ there exists a 1-reduced sequent $\Gamma^* \Rightarrow \Delta^*$, which is a contraction of $\Gamma \Rightarrow \Delta$, such that $\Gamma \Rightarrow \Delta$ is provable in $G(\Lambda)$ if and only if $\Gamma^* \Rightarrow \Delta^*$ is provable in $G(\Lambda)$ by applications of contraction rules and weakening rules, respectively. Thus there exists an *effective* procedure of getting the 1-reduced sequent $\mathcal{S}'$ for a given sequent $\mathcal{S}$, whose provability is the same as that of $\mathcal{S}$. The following Lemma 44 shows that every derivable sequent can be reduced without changing its derivability.

Lemma 44. Suppose that $\Gamma \Rightarrow \Delta$ is a sequent which is derivable in $G(\Lambda)$ and that $\Gamma^* \Rightarrow \Delta^*$ is the 1-reduced contraction of $\Gamma \Rightarrow \Delta$. Then there exists a cut-free derivation of $\Gamma^* \Rightarrow \Delta^*$ in $G(\Lambda)$ such that every sequent appearing in it is reduced.

Proof. Let $\mathcal{D}$ be a cut-free derivation of $\Gamma \Rightarrow \Delta$. We prove Lemma 44 by induction on *height* of $\mathcal{D}$. Let the height of $\mathcal{D}$ be n . We prove the following statement by induction on $k \leq n$:

$\forall \Sigma \Rightarrow \Pi$ in $\mathcal{D}$ (if the height of $\Sigma \Rightarrow \Pi$ in $\mathcal{D}$ is k ,
then there exists a cut-free proof of $\Sigma^* \Rightarrow \Pi^*$).

1. For basic step, let $\Gamma \Rightarrow \Delta$ be an axiom. Our goal is trivial.
2. For inductive step, suppose that $\Gamma \Rightarrow \Delta$ is the lower sequent of an application of rule I . We divide our argument by the number of upper sequents of I :

- Suppose I is $(c \Rightarrow)$ or $(\Rightarrow c)$. Let the upper sequent of I be the form of $\Sigma \Rightarrow \Pi$. Then by induction hypothesis there exists a cut-free proof of $\Sigma^* \Rightarrow \Pi^*$. It follows that $\Sigma^* \Rightarrow \Pi^*$ is a reduced contraction of $\Gamma \Rightarrow \Delta$. Then we can obtain $\Gamma^* \Rightarrow \Delta^*$ by applying contraction rules.
- Suppose I has a single upper sequent. By induction hypothesis we obtain $\Sigma^* \Rightarrow \Pi^*$. Let $\Gamma' \Rightarrow \Delta'$ be the sequent obtained by applying the rule I to $\Sigma^* \Rightarrow \Pi^*$. By checking the form of every rule of $\mathbf{G}(\Lambda)$, we can see that $\Gamma' \Rightarrow \Delta'$ is reduced. Hence we can obtain $\Sigma^* \Rightarrow \Pi^*$.
- Suppose I has two upper sequents. The same argument holds.

Here we shall explain why we define reduced sequents with the number “three”. Let I in the proof be $(\rightarrow \Rightarrow)$:

$$\frac{\Gamma \Rightarrow \varphi, \Theta \quad \psi, \Pi, \Sigma \Rightarrow \Delta}{\Pi, \varphi \rightarrow \psi, \Gamma, \Sigma \Rightarrow \Delta, \Theta} (\rightarrow \Rightarrow).$$

Suppose $\varphi \rightarrow \psi$ occurs once in Γ and once in Π, Σ . Then the antecedent of the lower sequent will contain *three* occurrences of $\varphi \rightarrow \psi$. To make the lower sequent *reduced*, we have to admit three occurrences of the same formula. $\square$

Given any sequent $\Gamma \Rightarrow \Delta$ in $\mathbf{G}(\Lambda)$, we will try to search for a proof of this sequent. Since there may be infinitely many possible proofs of it, it is necessary to make some restrictions on proofs. We say a derivation $\mathcal{D}$ has *redundancies* if the same sequent appears twice in a branche of $\mathcal{D}$.

Lemma 45. Any derivations $\mathcal{D}$ in $\mathbf{G}(\Lambda)$ can be transformed to a derivation $\mathcal{D}'$ which satisfies the following properties, with the same endsequence.

- (1) $\mathcal{D}'$ has the *subformula property*.
- (2) $\mathcal{D}'$ does not consist *redundancies*. As shown in the following proof, a redundancy of $\Sigma \Rightarrow \Pi$

$$\begin{array}{c} \Sigma \Rightarrow \Pi \\ \vdots \\ (\mathcal{P}_0) \\ \vdots \\ \vdots \\ \Sigma \Rightarrow \Pi \\ \vdots \\ (\mathcal{P}_1) \\ \vdots \\ \Gamma \Rightarrow \Delta \end{array}$$

can be eliminated as follow:

$$\begin{array}{c} \vdots (\mathcal{P}_0) \\ \Sigma \Rightarrow \Pi \\ \vdots (\mathcal{P}_1) \\ \Gamma \Rightarrow \Delta \end{array}.$$

(3) $\mathcal{D}'$ consists only of *reduced sequents*.

Proof. (1) Follows from Theorems 33 and 52.

(2) As shown in the following derivation, a redundancy of $\Sigma \Rightarrow \Pi$

$$\begin{array}{c} \Sigma \Rightarrow \Pi \\ \vdots (\mathcal{P}_0) \\ \vdots \\ \Sigma \Rightarrow \Pi \\ \vdots (\mathcal{P}_1) \\ \Gamma \Rightarrow \Delta \end{array}$$

can be eliminated as follow:

$$\begin{array}{c} \vdots (\mathcal{P}_0) \\ \Sigma \Rightarrow \Pi \\ \vdots (\mathcal{P}_1) \\ \Gamma \Rightarrow \Delta \end{array}.$$

(3) Follows from Lemma 44.

□

All three properties are necessary to reduce the number of all possible proofs to finitely many. To see this, Suppose that a given sequent $\varphi, \Pi \Rightarrow \Sigma$ satisfies only

(1) and (2) of Lemma 45. Then, any sequent of the form $\overbrace{\varphi, \dots, \varphi}^n, \Lambda \Rightarrow \Theta$ also satisfies these for any natural number n .

Now we are ready for showing the decidability of $\mathbf{G}(\Lambda)$, which is a known fact. For instance, the decidability of $\mathbf{G}(\mathbf{S5})$ is shown by Indrzejczak [19].

Theorem 46. $\mathbf{G}(\mathbf{K45})$ and $\mathbf{G}(\mathbf{S5})$ are decidable.

Proof. Let $\Gamma \Rightarrow \Delta$ be any sequent in $\mathbf{G}(\mathbf{K45})$ or $\mathbf{G}(\mathbf{S5})$. Suppose it is reduced. We will check whether there is a derivation of it satisfying all of these three restrictions or not. By Lemma 45(1) and (3), if there exists a derivation $\mathcal{D}$, we can transform that to a derivation $\mathcal{D}'$ which consists only of reduced sequents such that every

formula in $\mathcal{D}'$ is a subformula of some formulas in $\Gamma \Rightarrow \Delta$. Clearly, the number of such sequents is finite. Moreover, by Lemma 45(2), the height of the derivation of $\Gamma \Rightarrow \Delta$ must be finite. Thus, our decision algorithm is to generate all possible derivations one by one and to check whether it is a correct derivation of $\Gamma \Rightarrow \Delta$ or not. If we can find one correct derivation, $\Gamma \Rightarrow \Delta$ is derivable in $\mathbf{G}(\Lambda)$. If not, then by Lemma 45, $\Gamma \Rightarrow \Delta$ is not derivable.

□

Chapter 4

Sequent Calculi for PAL

4.1 Sequent Calculi G(K45PAL) and G(S5PAL)

Definition 47. The sequent calculi G(K45PAL) and G(S5PAL) are the ones which are obtained from G(K45) and G(S5), respectively, by supplementing the following rules on public announcements:

$$\begin{array}{c}
 \frac{\varphi, \Gamma \Rightarrow \Delta, p}{\Gamma \Rightarrow \Delta, [\varphi]p} (\Rightarrow [] \text{ at}) \quad \frac{\Gamma \Rightarrow \Delta, \varphi \quad p, \Pi \Rightarrow \Sigma}{[\varphi]p, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([] \text{ at } \Rightarrow) \\
 \\
 \frac{\varphi, [\varphi]\psi, \Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, [\varphi]\neg\psi} (\Rightarrow [] \neg) \quad \frac{\Gamma \Rightarrow \varphi, \Delta \quad \Pi \Rightarrow \Sigma, [\varphi]\psi}{[\varphi]\neg\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([] \neg \Rightarrow) \\
 \\
 \frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_1 \quad \Gamma \Rightarrow \Delta, [\varphi]\psi_2}{\Gamma \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2)} (\Rightarrow [] \wedge) \quad \frac{[\varphi]\psi_i, \Gamma \Rightarrow \Delta}{[\varphi](\psi_1 \wedge \psi_2), \Gamma \Rightarrow \Delta} ([] \wedge \Rightarrow) \\
 \\
 \frac{[\varphi]\psi, \Gamma \Rightarrow \Delta, [\varphi]\gamma}{\Gamma \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma)} (\Rightarrow [] \rightarrow) \quad \frac{\Gamma \Rightarrow \Delta, [\varphi]\psi \quad [\varphi]\gamma, \Pi \Rightarrow \Sigma}{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([] \rightarrow \Rightarrow) \\
 \\
 \frac{\varphi, \Gamma \Rightarrow \Delta, \Box_a[\varphi]\psi}{\Gamma \Rightarrow \Delta, [\varphi]\Box_a\psi} (\Rightarrow [] \Box) \quad \frac{\Gamma \Rightarrow \Delta, \varphi \quad \Box_a[\varphi]\psi, \Pi \Rightarrow \Sigma}{[\varphi]\Box_a\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([] \Box \Rightarrow) \\
 \\
 \frac{\Gamma \Rightarrow \Delta, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma \Rightarrow \Delta, [\varphi][\psi]\gamma} (\Rightarrow [] []) \quad \frac{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma \Rightarrow \Delta}{[\varphi][\psi]\gamma, \Gamma \Rightarrow \Delta} ([] [] \Rightarrow).
 \end{array}$$

As the reader may see, these new rules are naturally obtained from recursion axioms for public announcements in Table 2.3. For example, rules $(\Rightarrow [] [])$ and $([] [] \Rightarrow)$ are obtained from axiom $([] [])$:

$$([] []) \quad [\varphi][\psi]\gamma \leftrightarrow [\varphi \wedge [\varphi]\psi]\gamma.$$

Moreover, a formula $\Box_a[\varphi]\psi$ in the premise of the rule $(\Rightarrow \Box_a)$ is *not* a subformula of the conclusion of the rule, but it is in the closure of the conclusion. For each of the new rules above, every formula in premises of the rule is an element of the closure of the conclusion of the rule. In this sense, our calculi enjoy the extended subformula property in terms of the notion of closure (Definition 6).

The following lemmas play an important role in Section 4.2 and Section 3.3.

Lemma 48. The following inequalities hold in L^+ :

- (1) $c(\psi) > c(\varphi)$ if $\varphi \in \text{Sub}(\psi)$ and φ is distinct from ψ as a formula,
- (2) $c([\varphi]\neg\psi) > c([\varphi]\psi)$,
- (3) $c([\varphi](\psi_1 \bullet \psi_2)) > c([\varphi]\psi_i)$ ($\bullet \in \{\wedge, \rightarrow\}$),
- (4) $c([\varphi]\Box_a\psi) > c(\Box_a[\varphi]\psi)$,
- (5) $c([\varphi][\psi]\gamma) > c([\varphi \wedge [\varphi]\psi]\gamma)$.

Proof. Follows directly from Lemma 24. □

By Lemma 48, we obtain the following.

Lemma 49. $c(\psi) > c(\varphi)$ if $\varphi \in \text{CL}(\psi)$ and φ is distinct from ψ as a formula.

Proof. Follows directly from Lemma 48 and Definition 6. For instance, let $\psi \equiv [\gamma]p$. Then $\varphi \equiv \gamma \rightarrow p$. Hence by Lemma 48(2), $c(\psi) > c(\varphi)$. □

Proposition 50. Let $\Lambda \in \{\mathbf{K45PAL}, \mathbf{S5PAL}\}$.

- (1) If $\Gamma \Rightarrow \Delta$ is derivable in $\mathbf{G}(\Lambda)$ then $\bigwedge \Gamma \rightarrow \bigvee \Delta$ is a theorem of $\mathbf{H}(\Lambda)$.
- (2) If φ is a theorem of $\mathbf{H}(\Lambda)$, then $\Rightarrow \varphi$ is derivable in $\mathbf{G}(\Lambda)$.

Proof. (1) By induction on height of the derivation $\mathcal{D}$ to obtain $\Gamma \Rightarrow \Delta$. We prove the following statement by induction on n :

$$\forall \Pi \Rightarrow \Sigma \text{ in } \mathcal{D} \text{ (if the height of } \Pi \Rightarrow \Sigma \text{ in } \mathcal{D} \text{ is } n, \text{ then } \vdash_{\Lambda} \bigwedge \Gamma \rightarrow \bigvee \Delta).$$

For base step, let $n = 0$. Then $\Gamma \Rightarrow \Delta$ is an axiom $\varphi \Rightarrow \varphi$. Hence $\vdash_{\Lambda} \varphi \rightarrow \varphi$ holds clearly.

For inductive step, let $n > 0$. Then, we divide our argument depending on the last applied rule to obtain $\Gamma \Rightarrow \Delta$. Since Proposition 29 holds, we should only consider additional rules for $\mathbf{G}(\mathbf{K45PAL})$ and $\mathbf{G}(\mathbf{S5PAL})$. Since the new rules are naturally obtained from recursion axioms for public announcements in Table 2.3, (1) holds clearly. Here we show one typical case of $(\Box_a \Box_a \Rightarrow)$:

- Let the last applied rule be $([] [] \Rightarrow)$:

$$\frac{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma \Rightarrow \Delta}{[\varphi][\psi]\gamma, \Gamma \Rightarrow \Delta} ([] [] \Rightarrow).$$

By induction hypothesis, we obtain $\vdash_{\Lambda} [\varphi \wedge [\varphi]\psi]\gamma \wedge \bigwedge \Gamma \rightarrow \bigvee \Delta$. By $\vdash_{\Lambda} [\varphi][\psi]\gamma \leftrightarrow [\varphi \wedge [\varphi]\psi]\gamma$, we obtain our goal, which is $\vdash_{\Lambda} [\varphi][\psi]\gamma \wedge \bigwedge \Gamma \rightarrow \bigvee \Delta$.

- (2) Since Proposition 29 holds, we should only consider additional axioms for $G(\mathbf{K45PAL})$ and $G(\mathbf{S5PAL})$. We divide the argument into the following cases:

- $([] at)$

- $(\Rightarrow)$

$$\frac{\frac{\frac{\varphi \Rightarrow \varphi \quad p \Rightarrow p}{\varphi, [\varphi]p \Rightarrow p} ([] at \Rightarrow)}{[\varphi]p \Rightarrow \varphi \rightarrow p} (\Rightarrow \rightarrow)}{\Rightarrow [\varphi]p \rightarrow (\varphi \rightarrow p)} (\Rightarrow \rightarrow).$$

- $(\Leftarrow)$

$$\frac{\frac{\frac{\varphi \Rightarrow \varphi \quad p \Rightarrow p}{\varphi, \varphi \rightarrow p \Rightarrow p} (\rightarrow \Rightarrow)}{\varphi \rightarrow p \Rightarrow [\varphi]p} (\Rightarrow [] at)}{\Rightarrow (\varphi \rightarrow p) \rightarrow [\varphi]p} (\Rightarrow \rightarrow).$$

- $([] \neg)$

- $(\Rightarrow)$

$$\frac{\frac{\frac{\varphi \Rightarrow \varphi \quad [\varphi]\psi \Rightarrow [\varphi]\psi}{\varphi, [\varphi]\psi, [\varphi]\neg\psi \Rightarrow} ([] \neg \Rightarrow)}{\varphi, [\varphi]\neg\psi \Rightarrow \neg[\varphi]\psi} (\Rightarrow \neg)}{[\varphi]\neg\psi \Rightarrow \varphi \rightarrow \neg[\varphi]\psi} (\Rightarrow \rightarrow)}{\Rightarrow [\varphi]\neg\psi \rightarrow (\varphi \rightarrow \neg[\varphi]\psi)} (\Rightarrow \rightarrow).$$

- $(\Leftarrow)$

$$\frac{\frac{\frac{\varphi \Rightarrow \varphi \quad [\varphi]\psi \Rightarrow [\varphi]\psi}{\varphi, [\varphi]\psi, \varphi \rightarrow \neg[\varphi]\psi \Rightarrow} (\rightarrow \Rightarrow)}{\varphi \rightarrow \neg[\varphi]\psi \Rightarrow [\varphi]\neg\psi} (\Rightarrow [] \neg)}{\Rightarrow (\varphi \rightarrow \neg[\varphi]\psi) \rightarrow [\varphi]\neg\psi} (\Rightarrow \rightarrow).$$

- $([] \wedge)$

– ($\Rightarrow$)

$$\frac{\frac{[\varphi]\psi \Rightarrow [\varphi]\psi}{[\varphi](\psi \wedge \gamma) \Rightarrow [\varphi]\psi} (\Box \wedge \Rightarrow) \quad \frac{[\varphi]\gamma \Rightarrow [\varphi]\gamma}{[\varphi](\psi \wedge \gamma) \Rightarrow [\varphi]\gamma} (\Box \wedge \Rightarrow)}{[\varphi](\psi \wedge \gamma) \Rightarrow [\varphi]\psi \wedge [\varphi]\gamma} (\Rightarrow \wedge)$$

$$\frac{[\varphi](\psi \wedge \gamma) \Rightarrow [\varphi]\psi \wedge [\varphi]\gamma}{\Rightarrow [\varphi](\psi \wedge \gamma) \rightarrow ([\varphi]\psi \wedge [\varphi]\gamma)} (\Rightarrow \rightarrow)$$

– ($\Leftarrow$)

$$\frac{\frac{[\varphi]\psi \Rightarrow [\varphi]\psi}{[\varphi]\psi \wedge [\varphi]\gamma \Rightarrow [\varphi]\psi} (\wedge \Rightarrow) \quad \frac{[\varphi]\gamma \Rightarrow [\varphi]\gamma}{[\varphi]\psi \wedge [\varphi]\gamma \Rightarrow [\varphi]\gamma} (\wedge \Rightarrow)}{[\varphi]\psi \wedge [\varphi]\gamma \Rightarrow [\varphi](\psi \wedge \gamma)} (\Rightarrow \wedge)$$

$$\frac{[\varphi]\psi \wedge [\varphi]\gamma \Rightarrow [\varphi](\psi \wedge \gamma)}{\Rightarrow ([\varphi]\psi \wedge [\varphi]\gamma) \rightarrow [\varphi](\psi \wedge \gamma)} (\Rightarrow \rightarrow)$$

• ($\Box \rightarrow$)

– ($\Rightarrow$)

$$\frac{\frac{[\varphi]\psi \Rightarrow [\varphi]\psi \quad [\varphi]\gamma \Rightarrow [\varphi]\gamma}{[\varphi]\psi, [\varphi](\psi \rightarrow \gamma) \Rightarrow [\varphi]\gamma} (\Box \rightarrow \Rightarrow)}{[\varphi](\psi \rightarrow \gamma) \Rightarrow [\varphi]\psi \rightarrow [\varphi]\gamma} (\Rightarrow \rightarrow)$$

$$\frac{[\varphi](\psi \rightarrow \gamma) \Rightarrow [\varphi]\psi \rightarrow [\varphi]\gamma}{\Rightarrow [\varphi](\psi \rightarrow \gamma) \rightarrow ([\varphi]\psi \rightarrow [\varphi]\gamma)} (\Rightarrow \rightarrow)$$

– ($\Leftarrow$)

$$\frac{\frac{[\varphi]\psi \Rightarrow [\varphi]\psi \quad [\varphi]\gamma \Rightarrow [\varphi]\gamma}{[\varphi]\psi, [\varphi]\psi \rightarrow [\varphi]\gamma \Rightarrow [\varphi]\gamma} (\rightarrow \Rightarrow)}{[\varphi]\psi \rightarrow [\varphi]\gamma \Rightarrow [\varphi](\psi \rightarrow \gamma)} (\Rightarrow \Box \rightarrow)$$

$$\frac{[\varphi]\psi \rightarrow [\varphi]\gamma \Rightarrow [\varphi](\psi \rightarrow \gamma)}{\Rightarrow ([\varphi]\psi \rightarrow [\varphi]\gamma) \rightarrow [\varphi](\psi \rightarrow \gamma)} (\Rightarrow \rightarrow)$$

• ($\Box \Box$)

– ($\Rightarrow$)

$$\frac{\frac{\varphi \Rightarrow \varphi \quad \Box_a[\varphi]\psi \Rightarrow \Box_a[\varphi]\psi}{\varphi, [\varphi]\Box_a\psi \Rightarrow \Box_a[\varphi]\psi} (\Box \Box \Rightarrow)}{[\varphi]\Box_a\psi \Rightarrow \varphi \rightarrow \Box_a[\varphi]\psi} (\Rightarrow \rightarrow)$$

$$\frac{[\varphi]\Box_a\psi \Rightarrow \varphi \rightarrow \Box_a[\varphi]\psi}{\Rightarrow [\varphi]\Box_a\psi \rightarrow (\varphi \rightarrow \Box_a[\varphi]\psi)} (\Rightarrow \rightarrow)$$

– ($\Leftarrow$)

$$\frac{\frac{\varphi \Rightarrow \varphi \quad \Box_a[\varphi]\psi \Rightarrow \Box_a[\varphi]\psi}{\varphi, \varphi \rightarrow \Box_a[\varphi]\psi \Rightarrow \Box_a[\varphi]\psi} (\rightarrow \Rightarrow)}{\varphi \rightarrow \Box_a[\varphi]\psi \Rightarrow [\varphi]\Box_a\psi} (\Box \Box \Rightarrow)$$

$$\frac{\varphi \rightarrow \Box_a[\varphi]\psi \Rightarrow [\varphi]\Box_a\psi}{(\varphi \rightarrow \Box_a[\varphi]\psi) \rightarrow [\varphi]\Box_a\psi} (\Rightarrow \rightarrow)$$

• ($\Box \Box \Box$)

– ($\Rightarrow$)

$$\frac{\frac{[\varphi \wedge [\varphi]\psi]\gamma \Rightarrow [\varphi \wedge [\varphi]\psi]\gamma}{[\varphi][\psi]\gamma \Rightarrow [\varphi \wedge [\varphi]\psi]\gamma} \quad \square \square \Rightarrow}{\Rightarrow [\varphi][\psi]\gamma \rightarrow [\varphi \wedge [\varphi]\psi]\gamma} (\Rightarrow \rightarrow).$$

– ($\Leftarrow$)

$$\frac{\frac{[\varphi \wedge [\varphi]\psi]\gamma \Rightarrow [\varphi \wedge [\varphi]\psi]\gamma}{[\varphi \wedge [\varphi]\psi]\gamma \Rightarrow [\varphi][\psi]\gamma} (\Rightarrow \square \square)}{\Rightarrow [\varphi \wedge [\varphi]\psi]\gamma \rightarrow [\varphi][\psi]\gamma} (\rightarrow \Rightarrow).$$

□

4.2 Cut-elimination Theorem of $G(\mathbf{K45PAL})$

This section establishes the cut-elimination theorem for $G(\mathbf{K45PAL})$.

Definition 51. We define $G^-(\mathbf{K45PAL})$ as the same calculus as $G(\mathbf{K45PAL})$ except that the cut rule is excluded. The sequent calculus $G^*(\mathbf{K45PAL})$ is the same calculus as $G(\mathbf{K45PAL})$ except that the cut rule is replaced with (*Ecut*).

Theorem 52. If $\Gamma \Rightarrow \Delta$ is derivable in $G^*(\mathbf{K45PAL})$, then it is also derivable in $G^-(\mathbf{K45PAL})$. Therefore, if $\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{K45PAL})$ then it is also derivable in $G^-(\mathbf{K45PAL})$.

Proof. It suffices to prove the first statement. For our purpose, we establish the following (*Hauptsatz*): If in the following derivation $\mathcal{D}$ of $G^*(\mathbf{K45PAL})$:

$$\mathcal{D} \equiv \frac{\frac{\vdots \mathcal{L}}{\Gamma \Rightarrow \Delta, \varphi^m} \text{rule}(\mathcal{L}) \quad \frac{\vdots \mathcal{R}}{\varphi^n, \Pi \Rightarrow \Sigma} \text{rule}(\mathcal{R})}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_{\varphi},$$

there is no other (*Ecut*) in derivations of $\Gamma \Rightarrow \Delta, \varphi^m$ and $\varphi^n, \Pi \Rightarrow \Sigma$, then there is a derivation in $G^-(\mathbf{K45PAL})$ of the sequent $\Gamma, \Pi \Rightarrow \Delta, \Sigma$. We prove this statement on a derivation $\mathcal{D}$ by double induction on the *complexity* $c(\mathcal{D})$ and *weight* $w(\mathcal{D})$ of the derivation $\mathcal{D}$.

If $m = 0$ or $n = 0$, it follows our goal can be obtained by weakening rules. Hence we suppose $m > 0$ and $n > 0$ below. We divide the argument into the following cases.

1. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be (*id*).

2. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be structural rules.
3. Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be logical rules or rules for $\Box_a$ and public announcements where the cut formula is not principal.
4. Let $\text{rule}(\mathcal{L})$ and $\text{rule}(\mathcal{R})$ be rules of the same connective and the cut formula is principal in both rules.

Since cases (1) and (2) are handled similarly to the proof of Theorem 33, we comment on the cases where our new public announcement rules are concerned with.

- (3) Cases for $\text{rule}(\mathcal{L})$:
 - Let $\text{rule}(\mathcal{L})$ be $([] \text{ at } \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \varphi, \rho^a \quad p, \Pi \Rightarrow \Sigma, \rho^b}{[\varphi]p, \Gamma, \Pi \Rightarrow \Delta, \Sigma, \rho^{a+b}} ([]) \text{ at } \Rightarrow}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} \rho^n, \Theta \Rightarrow \Omega (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \varphi, \rho^a \quad \rho^n, \Theta \Rightarrow \Omega}{\Gamma, \Theta \Rightarrow \Delta, \Omega, \varphi} (Ecut)_\rho^1 \quad \frac{p, \Pi \Rightarrow \Sigma, \rho^b \quad \rho^n, \Theta \Rightarrow \Omega}{p, \Pi, \Theta \Rightarrow \Sigma, \Omega} (Ecut)_\rho^2}{\frac{[\varphi]p, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Sigma, \Omega, \Omega}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([]) \text{ at } \Rightarrow.$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{L})$ be $(\Rightarrow [] \text{ at})$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\varphi, \Gamma \Rightarrow \Delta, p, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi]p, \rho^m} (\Rightarrow []) \text{ at} \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, [\varphi]p, \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\varphi, \Gamma \Rightarrow \Delta, p, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\varphi, \Gamma \Rightarrow \Delta, p, \Pi \Sigma} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, [\varphi]p, \Sigma} (\Rightarrow []) \text{ at}.$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{L})$ be $([] \neg \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \varphi, \Delta, \rho^a \quad \Pi \Rightarrow \Sigma, [\varphi]\psi, \rho^b}{[\varphi]\neg\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma, \rho^{a+b}} ([]) \neg \Rightarrow}{\frac{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega}{\rho^n, \Theta \Rightarrow \Omega} (Ecut)_\rho}.$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \varphi, \Delta, \rho^a \quad \rho^n, \Theta \Rightarrow \Omega}{\Gamma, \Theta \Rightarrow \Delta, \Omega, \varphi} (Ecut)_\rho^1 \quad \frac{\Pi \Rightarrow \Sigma, [\varphi]\psi, \rho^b \quad \rho^n, \Theta \Rightarrow \Omega}{\Pi, \Theta \Rightarrow \Sigma, \Omega, [\varphi]\psi} (Ecut)_\rho^2}{\frac{[\varphi]p, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Sigma, \Omega, \Omega}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([]) \neg \Rightarrow.$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{L})$ be $(\Rightarrow [] \neg)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\varphi, [\varphi]\psi, \Gamma \Rightarrow \Delta, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi]\neg\psi, \rho^m} (\Rightarrow []) \neg \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, [\varphi]\neg\psi, \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\varphi, [\varphi]\psi, \Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\varphi, [\varphi]\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, [\varphi]\neg\psi, \Sigma} (\Rightarrow []) \neg.$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{L})$ be $([] \wedge \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{[\varphi]\psi_i, \Gamma \Rightarrow \Delta, \rho^m}{[\varphi](\psi_1 \wedge \psi_2), \Gamma \Rightarrow \Delta, \rho^m} ([]) \wedge \Rightarrow \quad \rho^n, \Pi \Rightarrow \Sigma}{[\varphi](\psi_1 \wedge \psi_2), \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{[\varphi]\psi_i, \Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{[\varphi]\psi_i, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho}{[\varphi](\psi_1 \wedge \psi_2), \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([]) \wedge \Rightarrow.$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{L})$ be $(\Rightarrow [] \wedge)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_1, \rho^m \quad \Gamma \Rightarrow \Delta, [\varphi]\psi_2, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2), \rho^m} (\Rightarrow [] \wedge) \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2), \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_1, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi_1} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_2, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi_2} (Ecut)_\rho^2}{\Gamma, \Pi \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2), \Sigma} ([\wedge] \rightarrow \Rightarrow).$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{L})$ be $([] \rightarrow \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, [\varphi]\psi, \rho^a \quad [\varphi]\gamma, \Pi \Rightarrow \Sigma, \rho^b}{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi \Rightarrow \Delta, \Sigma, \rho^{a+b}} ([\rightarrow] \rightarrow \Rightarrow) \quad \rho^n, \Theta \Rightarrow \Omega}{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, [\varphi]\psi, \rho^a \quad \rho^n, \Theta \Rightarrow \Omega}{\Gamma, \Theta \Rightarrow \Delta, \Omega, [\varphi]\psi} (Ecut)_\rho^1 \quad \frac{[\varphi]\gamma, \Pi \Rightarrow \Sigma, \rho^b \quad \rho^n, \Theta \Rightarrow \Omega}{[\varphi]\gamma, \Pi, \Theta \Rightarrow \Sigma, \Omega} (Ecut)_\rho^2}{\frac{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Sigma, \Omega, \Omega}{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([\rightarrow] \rightarrow \Rightarrow).$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{L})$ be $(\Rightarrow [] \rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{[\varphi]\psi, \Gamma \Rightarrow \Delta, [\varphi]\gamma, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma), \rho^m} (\Rightarrow [] \rightarrow) \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma), \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{[\varphi]\psi, \Gamma \Rightarrow \Delta, [\varphi]\gamma, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{[\varphi]\psi, \Gamma, \Pi \Rightarrow \Delta, [\varphi]\psi, \Sigma} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma), \Sigma} (\Rightarrow [] \rightarrow).$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{L})$ be $([] \sqsupset \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \varphi, \rho^a \quad \Box_a[\varphi]\psi, \Pi \Rightarrow \Sigma, \rho^b}{[\varphi]\Box_a\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([] \sqsupset \Rightarrow) \quad \rho^n, \Theta \Rightarrow \Omega}{[\varphi]\Box_a\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \varphi, \rho^a \quad \rho^n, \Theta \Rightarrow \Omega}{\Gamma, \Theta \Rightarrow \Delta, \Omega, \varphi} (Ecut)_\rho^1 \quad \frac{\Box_a[\varphi]\psi, \Pi \Rightarrow \Sigma, \rho^b \quad \rho^n, \Theta \Rightarrow \Omega}{[\varphi]\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega} (Ecut)_\rho^2}{\frac{[\varphi]\Box_a\psi, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Sigma, \Omega, \Omega}{[\varphi]\Box_a\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([] \sqsupset \Rightarrow).$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{L})$ be $(\Rightarrow [] \sqsupset)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\varphi, \Gamma \Rightarrow \Delta, \Box_a[\varphi]\psi, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi]\Box_a\psi, \rho^m} (\Rightarrow [] \sqsupset) \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Box_a[\varphi]\psi, \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\varphi, \Gamma \Rightarrow \Delta, \Box_a[\varphi]\psi, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\varphi, \Gamma, \Pi \Rightarrow \Delta, \Sigma, \Box_a[\varphi]\psi} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \Box_a[\varphi]\psi} (\Rightarrow [] \sqsupset).$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{L})$ be $([] [] \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma \Rightarrow \Delta, \rho^m}{[\varphi][\psi]\gamma, \Gamma \Rightarrow \Delta, \rho^m} ([] [] \Rightarrow) \quad \rho^n, \Pi \Rightarrow \Sigma}{[\varphi][\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho}{[\varphi][\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([] [] \Rightarrow).$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{L})$ be $(\Rightarrow [] [])$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, [\varphi \wedge [\varphi]\psi]\gamma, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi][\psi]\gamma, \rho^m} (\Rightarrow [] []) \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi][\psi]\gamma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, [\varphi \wedge [\varphi]\psi]\gamma, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi \wedge [\varphi]\psi]\gamma} (Ecut)_\rho$$

$$\frac{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi][\psi]\gamma} (\Rightarrow [] []).$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- (3) Cases for $\text{rule}(\mathcal{R})$:

- Let $\text{rule}(\mathcal{R})$ be $([] \text{ at } \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\rho^a, \Pi \Rightarrow \Sigma, \varphi \quad \rho^b, p, \Theta \Rightarrow \Omega}{\rho^{a+b}, [\varphi]p, \Pi, \Theta \Rightarrow \Sigma, \Omega} ([] \text{ at } \Rightarrow) \quad \Gamma \Rightarrow \Delta, \rho^m}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^a, \Pi \Rightarrow \Sigma, \varphi}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \varphi} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^b, p, \Theta \Rightarrow \Omega}{\Gamma, p, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho^2}{\frac{[\varphi]p, \Gamma, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([] \text{ at } \Rightarrow).$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{R})$ be $(\Rightarrow [] \text{ at})$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\rho^n, \varphi, \Pi \Rightarrow \Sigma, p}{\rho^n, \Pi \Rightarrow \Sigma, [\varphi]p} (\Rightarrow [] \text{ at}) \quad \Gamma \Rightarrow \Delta, \rho^m}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \varphi, \Pi \Rightarrow \Sigma, p}{\varphi, \Gamma, \Pi \Rightarrow \Delta, \Sigma, \rho} (Ecut)_\rho$$

$$\frac{\varphi, \Gamma, \Pi \Rightarrow \Delta, \Sigma, \rho}{[\varphi]p, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} ([] \text{ at } \Rightarrow).$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{R})$ be $([] \neg \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^a, \Pi \Rightarrow \varphi, \Sigma \quad \rho^b, \Theta \Rightarrow \Omega, [\varphi]\psi}{\rho^{a+b}, [\varphi]\neg\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega} ([]) \neg \Rightarrow}{[\varphi]\neg\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^a, \Pi \Rightarrow \varphi, \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \varphi} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^b, \Theta \Rightarrow \Omega, [\varphi]\psi}{\Gamma, \Theta \Rightarrow \Delta, \Omega, [\varphi]\psi} (Ecut)_\rho^2}{\frac{[\varphi]\neg\psi, \Gamma, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega}{[\varphi]\neg\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([]) \neg \Rightarrow .$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{R})$ be $(\Rightarrow [] \neg)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, \varphi, [\varphi]\psi, \Pi \Rightarrow \Sigma}{\rho^n, \Pi \Rightarrow \Sigma, [\varphi]\neg\psi} (\Rightarrow []) \neg}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\neg\psi} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \varphi, [\varphi]\psi, \Pi \Rightarrow \Sigma}{\varphi, [\varphi]\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\neg\psi} (\Rightarrow []) \neg .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{R})$ be $([] \wedge \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, [\varphi]\psi_i, \Pi \Rightarrow \Sigma}{\rho^n, [\varphi](\psi_1 \wedge \psi_2), \Pi \Rightarrow \Sigma} ([]) \wedge \Rightarrow}{[\varphi](\psi_1 \wedge \psi_2), \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, [\varphi]\psi_i, \Pi \Rightarrow \Sigma}{[\varphi]\psi_i, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho}{[\varphi](\psi_1 \wedge \psi_2), \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([]) \wedge \Rightarrow .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{R})$ be $(\Rightarrow [] \wedge)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, \Pi \Rightarrow \Sigma, [\varphi]\psi_1 \quad \rho^n, \Pi \Rightarrow \Sigma, [\varphi]\psi_2}{\rho^n, \Pi \Rightarrow \Sigma, [\varphi](\psi_1 \wedge \psi_2)} (\Rightarrow [] \wedge)}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi](\psi_1 \wedge \psi_2)} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma, [\varphi]\psi_1}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi_1} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma, [\varphi]\psi_2}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi_2} (Ecut)_\rho^2}{\Gamma, \Gamma, \Pi \Rightarrow \Delta, \Delta, \Sigma, [\varphi](\psi_1 \wedge \psi_2)} (\Rightarrow [] \wedge) .$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{R})$ be $([] \rightarrow \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^a, \Pi \Rightarrow \Sigma, [\varphi]\psi \quad \rho^b, [\varphi]\gamma, \Theta \Rightarrow \Omega}{\rho^{a+b}, [\varphi](\psi \rightarrow \gamma), \Pi, \Theta \Rightarrow \Sigma, \Omega} ([] \rightarrow \Rightarrow)}{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^a, \Pi \Rightarrow \Sigma, [\varphi]\psi}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^b, [\varphi]\gamma, \Theta \Rightarrow \Omega}{[\varphi]\gamma, \Gamma, \Theta \Rightarrow \Delta, \Omega} (Ecut)_\rho^2}{\frac{[\varphi](\psi \rightarrow \gamma), \Gamma, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega}{[\varphi](\psi \rightarrow \gamma), \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([] \rightarrow \Rightarrow) .$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{R})$ be $(\Rightarrow [] \rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, [\varphi]\psi, \Pi \Rightarrow \Sigma, [\varphi]\gamma}{\rho^n, \Pi \Rightarrow \Sigma, [\varphi](\psi \rightarrow \gamma)} (\Rightarrow [] \rightarrow)}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi](\psi \rightarrow \gamma)} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, [\varphi]\psi, \Pi \Rightarrow \Sigma, [\varphi]\gamma}{[\varphi]\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\gamma} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi](\psi \rightarrow \gamma)} (\Rightarrow [] \rightarrow) .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{R})$ be $([] \square \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^a, \Pi \Rightarrow \Sigma, \varphi \quad \rho^b, \square_a[\varphi]\psi, \Theta \Rightarrow \Omega}{\rho^{a+b}, [\varphi]\square_a\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega} ([]) \square \Rightarrow}{[\varphi]\square_a\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^a, \Pi \Rightarrow \Sigma, \varphi}{\Gamma \Pi \Rightarrow \Delta, \Sigma, \varphi} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^b, \square_a[\varphi]\psi, \Theta \Rightarrow \Omega}{[\varphi]\psi, \Gamma, \Theta \Rightarrow \Delta, \Omega} (Ecut)_\rho^2}{\frac{[\varphi]\square_a\psi, \Gamma, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega}{[\varphi]\square_a\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} ([]) \square \Rightarrow .$$

By induction hypothesis, $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their weights are reduced.

- Let $\text{rule}(\mathcal{R})$ be $(\Rightarrow [] \square)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, \varphi, \Pi \Rightarrow \Sigma, \square_a[\varphi]\psi}{\rho^n, \Pi \Rightarrow \Sigma, [\varphi]\square_a\psi} (\Rightarrow [] \square)}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\square_a\psi} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \varphi, \Pi \Rightarrow \Sigma, \square_a[\varphi]\psi}{\varphi, \Gamma, \Pi \Rightarrow \Delta, \square_a[\varphi]\psi} (Ecut)_\rho}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\square_a\psi} (\Rightarrow [] \square) .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{R})$ be $([] [] \Rightarrow)$. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, [\varphi \wedge [\varphi]\psi]\gamma, \Pi \Rightarrow \Sigma}{\rho^n, [\varphi][\psi]\gamma, \Pi \Rightarrow \Sigma} ([]) [] \Rightarrow}{[\varphi][\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho .$$

We transform this derivation into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, [\varphi \wedge [\varphi]\psi]\gamma, \Pi \Rightarrow \Sigma}{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\rho}{[\varphi][\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([]) [] \Rightarrow .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- Let $\text{rule}(\mathcal{R})$ be $(\Rightarrow [] [])$. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^n, \Pi \Rightarrow \Sigma, [\varphi \wedge [\varphi]\psi]\gamma \quad \rho^n, \Pi \Rightarrow \Sigma, [\varphi][\psi]\gamma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi][\psi]\gamma} (\Rightarrow [] [])}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi][\psi]\gamma} (Ecut)_\rho.$$

We transform this derivation into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^n, \Pi \Rightarrow \Sigma, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi \wedge [\varphi]\psi]\gamma} (Ecut)_\rho \quad \frac{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi][\psi]\gamma} (\Rightarrow [] []).$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

- (4)

- Suppose $\text{rule}(\mathcal{L})$ is $(\Rightarrow [] \text{at})$ and $\text{rule}(\mathcal{R})$ is $([] \text{at} \Rightarrow)$ where the cut formula is principal in both rules. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\frac{\varphi, \Gamma \Rightarrow \Delta, \rho^{m-1}, p}{\Gamma \Rightarrow \Delta, \rho^{m-1}, [\varphi]p} (\Rightarrow [] \text{at}) \quad \frac{\rho^a, \Pi \Rightarrow \Sigma, \varphi \quad \rho^b, p, \Theta \Rightarrow \Omega}{\rho^{a+b}, [\varphi]p, \Pi, \Theta \Rightarrow \Sigma, \Omega} ([] \text{at} \Rightarrow)}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_{[\varphi]p},$$

where $\rho \equiv [\varphi]p$. We transform this derivation into the following:

$$\begin{aligned} \mathcal{L} &= \frac{\Gamma \Rightarrow \Delta, \rho^{m-1}, [\varphi]p \quad \rho^a, \Pi \Rightarrow \Sigma, \varphi}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \varphi} (Ecut)_{[\varphi]p}^1 \\ \mathcal{R} &= \frac{\varphi, \Gamma \Rightarrow \Delta, \rho^{m-1}, p \quad \rho^{a+b}, [\varphi]p, \Pi, \Theta \Rightarrow \Sigma, \Omega}{\varphi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, p} (Ecut)_{[\varphi]p}^2 \\ \mathcal{D}' &= \frac{\begin{array}{c} \vdots \mathcal{L} \\ \vdots \mathcal{R} \end{array} \quad \frac{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \varphi \quad \varphi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, p}{\Gamma, \Gamma, \Pi, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Delta, \Sigma, \Omega, p} (Ecut)_\varphi^3}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, p} (c) \\ &= \frac{\begin{array}{c} \vdots \mathcal{D}' \\ \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, p \end{array} \quad \frac{\Gamma \Rightarrow \Delta, \rho^{m-1}, [\varphi]p \quad \rho^b, p, \Theta \Rightarrow \Omega}{p, \Gamma, \Theta \Rightarrow \Delta, \Omega} (Ecut)_p^5}{\Gamma, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega, \Omega} (c) \quad \frac{\Gamma, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega, \Omega}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c). \end{aligned}$$

By induction hypothesis $(Ecut)^1$, $(Ecut)^2$ and $(Ecut)^4$ can be eliminated since their weights are reduced. Moreover $(Ecut)^3$ and $(Ecut)^5$ can be eliminated since their complexities are reduced by Lemma 49 and $p, \varphi \in CL([\varphi]p)$.

- Suppose $\text{rule}(\mathcal{L})$ is $(\Rightarrow [] \neg)$ and $\text{rule}(\mathcal{R})$ is $([] \neg \Rightarrow)$ where the cut formula is principal in both rules. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\frac{\varphi, [\varphi]\psi, \Gamma \Rightarrow \Delta, \rho^{m-1}}{\Gamma \Rightarrow \Delta, [\varphi]\neg\psi, \rho^{m-1}} (\Rightarrow [] \neg) \quad \frac{\rho^a, \Pi \Rightarrow \varphi, \Sigma \quad \rho^b, \Theta \Rightarrow \Omega, [\varphi]\psi}{\rho^{a+b}, [\varphi]\neg\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega} ([] \neg \Rightarrow)}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_{[\varphi]\neg\psi},$$

where $\rho \equiv [\varphi]\neg\psi$. We transform this derivation into:

$$\mathcal{R} = \frac{\varphi, [\varphi]\psi, \Gamma \Rightarrow \Delta, \rho^{m-1} \quad \rho^{a+b}, [\varphi]\neg\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega}{\varphi, [\varphi]\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_{[\varphi]\neg\psi}^1$$

$$\mathcal{L}' = \frac{\Gamma \Rightarrow \Delta, [\varphi]\neg\psi, \rho^{m-1} \quad \rho^b, \Theta \Rightarrow \Omega, [\varphi]\psi}{\Gamma, \Theta \Rightarrow \Delta, \Omega, [\varphi]\psi} (Ecut)_{[\varphi]\neg\psi}^3$$

$$\mathcal{L}'' = \frac{\Gamma \Rightarrow \Delta, [\varphi]\neg\psi, \rho^{m-1} \quad \rho^a, \Pi \Rightarrow \varphi, \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \varphi} (Ecut)_{[\varphi]\neg\psi}^2.$$

By induction hypothesis $(Ecut)^1$, $(Ecut)^2$ and $(Ecut)^3$ can be eliminated since their complexities are the same and their weights are reduced. Then, we obtain the following:

$$\frac{\frac{\frac{\vdots \mathcal{L}'}{\Gamma, \Theta \Rightarrow \Delta, \Omega, [\varphi]\psi} \quad \frac{\frac{\vdots \mathcal{L}''}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, \varphi} \quad \frac{\frac{\vdots \mathcal{R}}{\varphi, [\varphi]\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega}}{[\varphi]\psi, \Gamma, \Gamma, \Pi, \Pi, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Sigma, \Omega} (Ecut)_{\varphi}^4}{\Gamma, \Gamma, \Gamma, \Pi, \Pi, \Theta, \Theta \Rightarrow \Delta, \Delta, \Delta, \Sigma, \Sigma, \Omega, \Omega} (Ecut)_{[\varphi]\psi}^5}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c).$$

By induction hypothesis $(Ecut)^4$ can be eliminated since its complexity is reduced by Lemma 49 and $\varphi \in CL([\varphi]\neg\psi)$. Moreover $(Ecut)^5$ can be eliminated since its complexity is reduced by Lemma 48 (2): $c([\varphi]\neg\psi) > c([\varphi]\psi)$ and $\varphi \in CL([\varphi]\neg\psi)$.

- Suppose $\text{rule}(\mathcal{L})$ is $(\Rightarrow [] \wedge)$ and $\text{rule}(\mathcal{R})$ is $([] \wedge \Rightarrow)$ where the cut formula is principal in both rules. In this case, the derivation $\mathcal{D}$ is of

the following form:

$$\frac{\frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_1, \rho^m \quad \Gamma \Rightarrow \Delta, [\varphi]\psi_2, \rho^m}{\Gamma \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2), \rho^m} (\Rightarrow [\wedge]) \quad \frac{\rho^n, [\varphi]\psi_i, \Pi \Rightarrow \Sigma}{\rho^n, [\varphi](\psi_1 \wedge \psi_2), \Pi \Rightarrow \Sigma} ([\wedge] \Rightarrow)}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_{[\varphi](\psi_1 \wedge \psi_2)},$$

where $\rho \equiv [\varphi](\psi_1 \wedge \psi_2)$. We transform this derivation into:

$$\mathcal{L} = \frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_1, \rho^m \quad \rho^n, [\varphi](\psi_1 \wedge \psi_2), \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi_i} (Ecut)_{[\varphi](\psi_1 \wedge \psi_2)}^1$$

$$\mathcal{R} = \frac{\Gamma \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2), \rho^m \quad \rho^n, [\varphi]\psi_i, \Pi \Rightarrow \Sigma}{[\varphi]\psi_i, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_{[\varphi](\psi_1 \wedge \psi_2)}^2.$$

By induction hypothesis $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their complexities are the same and their weights are reduced. Then, we obtain the following:

$$\frac{\frac{\vdots \mathcal{L}}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi_i} \quad \frac{\vdots \mathcal{R}}{[\varphi]\psi_i, \Gamma, \Pi \Rightarrow \Delta, \Sigma}}{\frac{\Gamma, \Pi, \Gamma, \Pi \Rightarrow \Delta, \Sigma, \Delta, \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (c)} (Ecut)_{[\varphi]\psi_i}^3,$$

where $(Ecut)^3$ can be eliminated since its complexity is reduced by Lemma 48 (3): $c([\varphi](\psi_1 \wedge \psi_2)) > c([\varphi]\psi_i)$.

- Suppose $\text{rule}(\mathcal{L})$ is $(\Rightarrow [\rightarrow])$ and $\text{rule}(\mathcal{R})$ is $([\rightarrow] \Rightarrow)$ where the cut formula is principal in both rules. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\frac{[\varphi]\psi, \Gamma \Rightarrow \Delta, [\varphi]\gamma, \rho^{m-1}}{\Gamma \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma), \rho^{m-1}} (\Rightarrow [\rightarrow]) \quad \frac{\rho^a, \Pi \Rightarrow \Sigma, [\varphi]\psi \quad \rho^b, [\varphi]\gamma, \Theta \Rightarrow \Omega}{\rho^{a+b}, [\varphi](\psi \rightarrow \gamma), \Pi, \Theta \Rightarrow \Sigma, \Omega} ([\rightarrow] \Rightarrow)}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_{[\varphi](\psi \rightarrow \gamma)}.$$

where $\rho \equiv [\varphi](\psi \rightarrow \gamma)$. We transform this derivation into:

$$\mathcal{L}' = \frac{\Gamma \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma), \rho^{m-1} \quad \rho^a, \Pi \Rightarrow \Sigma, [\varphi]\psi}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi} (Ecut)_{[\varphi](\psi \rightarrow \gamma)}^2$$

$$\mathcal{L}'' = \frac{[\varphi]\psi, \Gamma \Rightarrow \Delta, [\varphi]\gamma, \rho^{m-1} \quad \rho^{a+b}, [\varphi](\psi \rightarrow \gamma), \Pi, \Theta \Rightarrow \Sigma, \Omega}{[\varphi]\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, [\varphi]\gamma} (Ecut)_{[\varphi](\psi \rightarrow \gamma)}^1$$

$$\mathcal{R} = \frac{\Gamma \Rightarrow \Delta, [\varphi](\psi \rightarrow \gamma), \rho^{m-1} \quad \rho^b, [\varphi]\gamma, \Theta \Rightarrow \Omega}{[\varphi]\gamma, \Gamma, \Theta \Rightarrow \Delta, \Omega} (Ecut)_{[\varphi](\psi \rightarrow \gamma)}^3.$$

By induction hypothesis $(Ecut)^1$, $(Ecut)^2$ and $(Ecut)^3$ can be eliminated since their complexities are the same and their weights are reduced. Then, we obtain the following:

$$\frac{\frac{\frac{\vdots \mathcal{L}'}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi]\psi} \quad \frac{\frac{\vdots \mathcal{L}''}{[\varphi]\psi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, [\varphi]\gamma} \quad \frac{\vdots \mathcal{R}}{[\varphi]\gamma, \Gamma, \Theta \Rightarrow \Delta, \Omega}}{[\varphi]\psi, \Gamma, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega, \Omega} (Ecut)_{[\varphi]\psi}^5}{\frac{\Gamma, \Gamma, \Gamma, \Pi, \Pi, \Theta, \Theta, \Delta, \Delta, \Delta, \Sigma, \Sigma, \Omega, \Omega}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} (Ecut)_{[\varphi]\psi}^4.$$

By induction hypothesis $(Ecut)^4$ and $(Ecut)^5$ can be eliminated since their complexities are reduced by Lemma 48 (3): $c([\varphi](\psi_1 \rightarrow \psi_2)) > c([\varphi]\psi_i)$.

- Suppose $\text{rule}(\mathcal{L})$ is $(\Rightarrow [] \Box)$ and $\text{rule}(\mathcal{R})$ is $([] \Box \Rightarrow)$ where the cut formula is principal in both rules. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\frac{\varphi, \Gamma \Rightarrow \Delta, \Box_a[\varphi]\psi, \rho^{m-1}}{\Gamma \Rightarrow \Delta, [\varphi]\Box_a\psi, \rho^{m-1}} (\Rightarrow [] \Box) \quad \frac{\rho^a, \Pi \Rightarrow \Sigma, \varphi \quad \rho^b, \Box_a[\varphi]\psi, \Theta \Rightarrow \Omega}{\rho^{a+b}, [\varphi]\Box_a\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega} ([] \Box \Rightarrow)}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (Ecut)_{[\varphi]\Box_a\psi},$$

where $\rho \equiv [\varphi]\Box_a\psi$. We transform this derivation into:

$$\begin{aligned} \mathcal{L}' &= \frac{\Gamma \Rightarrow \Delta, [\varphi]\Box_a\psi, \rho^{m-1} \quad \rho^a, \Pi \Rightarrow \Sigma, \varphi}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \varphi} (Ecut)_{[\varphi]\Box_a\psi}^2 \\ \mathcal{L}'' &= \frac{\varphi, \Gamma \Rightarrow \Delta, \Box_a[\varphi]\psi, \rho^{m-1} \quad \rho^{a+b}, [\varphi]\Box_a\psi, \Pi, \Theta \Rightarrow \Sigma, \Omega}{\varphi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, \Box_a[\varphi]\psi} (Ecut)_{[\varphi]\Box_a\psi}^1 \\ \mathcal{R} &= \frac{\Gamma \Rightarrow \Delta, [\varphi]\Box_a\psi, \rho^{m-1} \quad \rho^b, \Box_a[\varphi]\psi, \Theta \Rightarrow \Omega}{\Box_a[\varphi]\psi, \Gamma, \Theta \Rightarrow \Delta, \Omega} (Ecut)_{[\varphi]\Box_a\psi}^3. \end{aligned}$$

By induction hypothesis $(Ecut)^1$, $(Ecut)^2$ and $(Ecut)^3$ can be eliminated since their complexities are the same and their weights are reduced. Then, we obtain the following:

$$\frac{\frac{\frac{\vdots \mathcal{L}'}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \varphi} \quad \frac{\frac{\vdots \mathcal{L}''}{\varphi, \Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega, \Box_a[\varphi]\psi} \quad \frac{\vdots \mathcal{R}}{\Box_a[\varphi]\psi, \Gamma, \Theta \Rightarrow \Delta, \Omega}}{\varphi, \Gamma, \Gamma, \Pi, \Theta, \Theta \Rightarrow \Delta, \Delta, \Sigma, \Omega, \Omega} (Ecut)_{\Box_a[\varphi]\psi}^4}{\frac{\Gamma, \Gamma, \Gamma, \Pi, \Pi, \Theta, \Theta, \Delta, \Delta, \Delta, \Sigma, \Sigma, \Omega, \Omega}{\Gamma, \Pi, \Theta \Rightarrow \Delta, \Sigma, \Omega} (c)} (Ecut)_{\varphi}^5.$$

By induction hypothesis $(Ecut)^4$ can be eliminated since its complexity is reduced by Lemma 48 (4): $c([\varphi]\Box_a\psi) > c(\Box_a[\varphi]\psi)$. Moreover $(Ecut)^5$ can be eliminated since its complexity is reduced by Lemma 48 (2): $c([\varphi]\neg\psi) > c([\varphi]\psi)$ and $\varphi \in CL([\varphi]\Box_a\psi)$.

- Suppose $\text{rule}(\mathcal{L})$ is $(\Rightarrow \Box \Box)$ and $\text{rule}(\mathcal{R})$ is $(\Box \Box \Rightarrow)$ where the cut formula is principal in both rules. In this case, the derivation $\mathcal{D}$ is of the following form:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^{m-1}, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma \Rightarrow \Delta, \rho^{m-1}, [\varphi][\psi]\gamma} (\Rightarrow \Box \Box) \quad \frac{[\varphi \wedge [\varphi]\psi]\gamma, \rho^{n-1}, \Pi \Rightarrow \Sigma}{\rho^{n-1}, [\varphi][\psi]\gamma, \Pi \Rightarrow \Sigma} (\Box \Box \Rightarrow)}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_{[\varphi][\psi]\gamma},$$

where $\rho \equiv [\varphi][\psi]\gamma$. We transform this derivation into:

$$\mathcal{L} = \frac{\Gamma \Rightarrow \Delta, \rho^{m-1}, [\varphi \wedge [\varphi]\psi]\gamma \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi \wedge [\varphi]\psi]\gamma} (Ecut)_{[\varphi][\psi]\gamma}^1$$

$$\mathcal{R} = \frac{\Gamma \Rightarrow \Delta, \rho^m \quad [\varphi \wedge [\varphi]\psi]\gamma, \rho^{n-1}, \Pi \Rightarrow \Sigma}{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_{[\varphi][\psi]\gamma}^2.$$

By induction hypothesis $(Ecut)^1$ and $(Ecut)^2$ can be eliminated since their complexities are the same and their weights are reduced. Then, we obtain the following:

$$\frac{\frac{\vdots \mathcal{L}}{\Gamma, \Pi \Rightarrow \Delta, \Sigma, [\varphi \wedge [\varphi]\psi]\gamma} \quad \frac{\vdots \mathcal{R}}{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma, \Pi \Rightarrow \Delta, \Sigma}}{\frac{\Gamma, \Gamma, \Pi, \Pi \Rightarrow \Delta, \Delta, \Sigma, \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (c)} (Ecut)_{[\varphi \wedge [\varphi]\psi]\gamma}^3,$$

where $(Ecut)^3$ can be eliminated because its complexity is reduced by Lemma 48 (5): $c([\varphi][\psi]\gamma) > c([\varphi \wedge [\varphi]\psi]\gamma)$.

□

As a corollary of Theorem 52, we can show that $\mathbf{G(K45PAL)}$ is a conservative extension of $\mathbf{G(K45)}$ as follows.

Corollary 53. If a sequent $\Gamma \Rightarrow \Delta$ in L is derivable in $\mathbf{G(K45PAL)}$ then it is also derivable in $\mathbf{G(K45)}$.

4.3 Extended Analytic Cut Property of $G(\mathbf{S5PAL})$

This section establishes that every application of the cut rule in $G(\mathbf{S5PAL})$ can be replaced with an application of the cut rule such that the cut formula is in the *closure* of the conclusion of the cut.

4.3.1 Formulation

Definition 54. We say that an $(Ecut)_\rho$ of Definition 30 is *admissible* in $G^a(\mathbf{S5PAL})$ if derivability in $G^a(\mathbf{S5PAL})$ of both upper sequents $\Gamma \Rightarrow \Delta, \rho$ and $\rho, \Pi \Rightarrow \Sigma$ of the $(Ecut)_\rho$ implies derivability in $G^a(\mathbf{S5PAL})$ of the lower sequent $\Gamma, \Pi \Rightarrow \Delta, \Sigma$.

In order to establish the following main lemma regarding the extended analytic cut property of $G(\mathbf{S5PAL})$, we employ Takano's argument for [44, Lemma 3.2] with necessary modifications to account for our public announcement operators.

Definition 55. The sequent calculus $G^a(\mathbf{S5PAL})$ is the same calculus as $G(\mathbf{S5PAL})$ except that (Cut) is always analytic.

In what follows, when $\Gamma \Rightarrow \Delta$ is derivable in $G^a(\mathbf{S5PAL})$, we denote $G^a(\mathbf{S5PAL}) \vdash \Gamma \Rightarrow \Delta$. We can establish the following in the same way as in the proof of [44, Lemma 3.1].

Lemma 56. Suppose $\Xi \subseteq CL(\Gamma, \Delta)$. If the sequent $\Phi, \Gamma \Rightarrow \Delta, \Psi$ is derivable in $G^a(\mathbf{S5PAL})$ for every partition $\langle \Phi; \Psi \rangle$ of Ξ , then $\Gamma \Rightarrow \Delta$ has a derivation in $G^a(\mathbf{S5PAL})$.

Proof. We prove this by induction on the number n of formulas in Ξ denoted as $|\Xi|$ (see [44, Lemma 3.1]).

$\forall \Xi \forall \Gamma \forall \Delta$ (if $|\Xi| = n$ and $\Xi \subseteq CL(\Gamma, \Delta)$ then

$\forall \langle \Phi; \Psi \rangle : \text{partition of } \Xi (G^a(\mathbf{S5PAL}) \vdash \Phi, \Gamma \Rightarrow \Delta, \Psi \text{ implies } G^a(\mathbf{S5PAL}) \vdash \Gamma \Rightarrow \Delta)$).

1. Let $n = 0$. For $\langle ; \rangle$ is the only partition, Lemma 56 holds clearly.
2. Let $n > 0$. Fix any Ξ' such that $|\Xi'| = n + 1$ and $\Xi' = \Xi \cup \{\varphi\} \subseteq CL(\Gamma, \Delta)$ (hence $|\Xi| = n$), and fix any Γ and Δ such that $\Xi' \subseteq CL(\Gamma, \Delta)$. Suppose $G^a(\mathbf{S5PAL}) \vdash \Phi', \Gamma \Rightarrow \Delta, \Psi'$ for all partitions $\langle \Phi'; \Psi' \rangle$ of Ξ' . We show that $G^a(\mathbf{S5PAL}) \vdash \Gamma \Rightarrow \Delta$. By induction hypothesis, it suffices to show that $G^a(\mathbf{S5PAL}) \vdash \Phi, \Gamma \Rightarrow \Delta, \Psi$ for all partitions $\langle \Phi; \Psi \rangle$ of Ξ . Fix any

partition $\langle \Phi; \Psi \rangle$ of Ξ . Consider the following partitions of Ξ' : $\langle \Phi, \varphi; \Psi \rangle$ and $\langle \Phi; \Psi, \varphi \rangle$. By supposition, $\Phi, \Gamma \Rightarrow \Delta, \Psi, \varphi$ and $\varphi, \Phi, \Gamma \Rightarrow \Delta, \Psi$ are derivable in $G^a(\mathbf{S5PAL})$. We have the following derivation:

$$\frac{\frac{\Phi, \Gamma \Rightarrow \Delta, \Psi, \varphi \quad \varphi, \Phi, \Gamma \Rightarrow \Delta, \Psi}{\Phi, \Phi, \Gamma, \Gamma \Rightarrow \Delta, \Delta, \Psi, \Psi} (Cut)_\varphi}{\Phi, \Gamma \Rightarrow \Delta, \Psi} (c),$$

where $(Cut)_\varphi$ is analytic since $\varphi \in \Xi' \subseteq CL(\Gamma, \Delta)$. We have shown that $G^a(\mathbf{S5PAL}) \vdash \Phi, \Gamma \Rightarrow \Delta, \Psi$, as required. $\square$

Next we provide a proof of Lemma 57, whose statement is the following.

Lemma 57. Suppose that $(Ecut)_\psi$ is admissible in $G^a(\mathbf{S5PAL})$ for all $\psi \in CL(\varphi)$. If the sequent $\Box_a \Gamma \Rightarrow \Box_a \Delta$ is derivable in $G^a(\mathbf{S5PAL})$, but with an application of $(\Box_5)$ for its lowest inference, then the sequent $\Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi$ is derivable in $G^a(\mathbf{S5PAL})$, where Δ_φ denotes the set which is obtained from Δ by deleting all the occurrences of φ .

Proof. Let $\mathcal{D}$ be such derivation of $\Box_a \Gamma \Rightarrow \Box_a \Delta$ in $G^a(\mathbf{S5PAL})$ with an application of $(\Box_5)$ for its lowest inference. We divide our argument into the following two cases: $\Box_a \varphi \in CL(\Box_a \Gamma, \Box_a(\Delta_\varphi))$ and $\Box_a \varphi \notin CL(\Box_a \Gamma, \Box_a(\Delta_\varphi))$. When $\Box_a \varphi \in CL(\Box_a \Gamma, \Box_a(\Delta_\varphi))$ we have the following derivation:

$$\frac{\frac{\vdots \mathcal{D} \quad \varphi \Rightarrow \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta \quad \Box_a \varphi \Rightarrow \varphi} (\Box_a \Rightarrow)}{\Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi} (Cut)_{\Box_a \varphi}^a,$$

where $(Cut)_{\Box_a \varphi}^a$ is analytic by assumption. So, in what follows, we always assume

(a) $\Box_a \varphi \notin CL(\Box_a \Gamma, \Box_a(\Delta_\varphi))$.

Moreover, we replace any application of (Cut) in $\mathcal{D}$

$$\frac{\Theta \Rightarrow \Omega, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Theta, \Pi \Rightarrow \Omega, \Sigma} (Cut)_\varphi$$

such that $\varphi \in \Theta \cup \Pi \cup \Omega \cup \Sigma$ with applications of weakening rules. By (a) $\Box_a \varphi \notin CL(\Box_a \Gamma, \Box_a(\Delta_\varphi))$, we can establish the following claim (as did in the proof of [44, Lemma 3.2]).

Claim 1. None of the sequents of $\mathcal{D}$ contain $\Box_a\varphi$ in its antecedent.

(Proof of Claim) We show that $\Box_a\varphi \notin \Pi$ for all sequents $\Pi \Rightarrow \Sigma$ in $\mathcal{D}$. Let the height of $\mathcal{D}$ be n . We prove the following statement by induction on $k \leq n$:

$\forall \Pi \Rightarrow \Sigma$ in $\mathcal{D}$ (if the height of $\Pi \Rightarrow \Sigma$ in $\mathcal{D}$ is $n - k$, then $\Box_a\varphi \notin \Pi$).

For the base step, let $k = 0$. Our derivation $\mathcal{D}$ is of the following form:

$$\frac{\Box_a\Gamma \Rightarrow \Box_a\Delta', \psi}{\Box_a\Gamma \Rightarrow \Box_a\Delta} (\Box_{ss}).$$

By (a) $\Box_a\varphi \notin \text{CL}(\Box_a\Gamma, \Box_a(\Delta_\varphi))$, $\Box_a\varphi \notin \Box_a\Gamma$ holds.

For inductive step, let $k > 0$ and fix any sequent $\Pi \Rightarrow \Sigma$ in $\mathcal{D}$ whose height is $n - k$. Then, we divide our argument depending on the last applied rule to obtain $\Pi \Rightarrow \Sigma$. First, we discuss the inference rules except (Cut) . For all right rules except (Cut) , our goal is immediate. So, we can focus on the remaining left rules of $\mathbf{G^a(S5PAL)}$. We only comment on rules with public announcement operators, since the rest are handled similarly to Lemma 42.

- Let the last applied rule be $([] \text{ at } \Rightarrow)$:

$$\frac{\Pi' \Rightarrow \Sigma', \chi \quad p, \Pi'' \Rightarrow \Sigma''}{[\chi]p, \Pi', \Pi'' \Rightarrow \Sigma', \Sigma''} ([] \text{ at } \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\varphi \notin \{[\chi]p\} \cup \Pi'$, which implies $\Box_a\varphi \notin \Pi'$ and $\Box_a\varphi \notin \Pi''$.

- Let the last applied rule be $([] \neg \Rightarrow)$:

$$\frac{\Pi' \Rightarrow \chi, \Sigma \quad \Pi' \Rightarrow [\chi]\psi, \Sigma}{[\chi]\neg\psi, \Pi' \Rightarrow \Sigma} ([] \neg \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\varphi \notin \{[\chi]\neg\psi\} \cup \Pi'$, which implies $\Box_a\varphi \notin \Pi'$.

- Let the last applied rule be $([] \wedge \Rightarrow)$:

$$\frac{[\chi]\psi_i, \Pi' \Rightarrow \Sigma}{[\chi](\psi_1 \wedge \psi_2), \Pi' \Rightarrow \Sigma} ([] \wedge \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\varphi \notin \{[\chi](\psi_1 \wedge \psi_2)\} \cup \Pi'$. Since $\Box_a\varphi \neq [\chi]\psi_i$, it follows that $\Box_a\varphi \notin \{[\chi]\psi_i\} \cup \Pi'$.

- Let the last applied rule be $(\Box \rightarrow \Rightarrow)$:

$$\frac{\Pi' \Rightarrow \Sigma, [\chi]\psi \quad [\chi]\gamma, \Pi' \Rightarrow \Sigma}{[\chi](\psi \rightarrow \gamma), \Pi' \Rightarrow \Sigma} (\Box \rightarrow \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{[\chi](\psi \rightarrow \gamma)\} \cup \Pi'$. Since $\Box_a \varphi \neq [\chi]\gamma$, it follows that $\Box_a \varphi \notin \{[\chi]\gamma\} \cup \Pi'$.

- Let the last applied rule be $(\Box \Box \Rightarrow)$:

$$\frac{\Pi' \Rightarrow \chi \quad \Box_a [\chi]\psi \Rightarrow \Sigma}{[\chi]\Box_a \psi, \Pi' \Rightarrow \Sigma} (\Box \Box \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{[\chi]\Box_a \psi\} \cup \Pi'$. Again we divide our argument into the following two cases: $\Box_a \varphi \equiv \Box_a [\chi]\psi$ and $\Box_a \varphi \neq \Box_a [\chi]\psi$:

- Suppose $\Box_a \varphi \equiv \Box_a [\chi]\psi$. It follows that $\Box_a \varphi \in \text{CL}([\chi]\Box_a \psi)$. By the closure property, $[\chi]\Box_a \psi \in \text{CL}(\Box_a \Gamma, \Box_a \Delta)$. Since $[\chi]\Box_a \psi \neq \Box_a [\chi]\psi$, we have $[\chi]\Box_a \psi \in \text{CL}(\Box_a \Gamma, \Box_a (\Delta_\varphi))$. Then $\Box_a \varphi \in \text{CL}([\chi]\Box_a \psi) \subseteq \text{CL}(\Box_a \Gamma, \Box_a (\Delta_\varphi))$, which is contradicted to (a).
 - Suppose $\Box_a \varphi \neq \Box_a [\chi]\psi$. It follows that $\Box_a \varphi \notin \{\Box_a [\chi]\psi\} \cup \Pi'$, which suffices our goal.
- Let the last applied rule be $(\Box \Box \Rightarrow)$:

$$\frac{[\chi \wedge [\chi]\psi]\gamma, \Pi' \Rightarrow \Delta}{[\chi][\psi]\gamma, \Pi' \Rightarrow \Delta} (\Box \Box \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{[\chi][\psi]\gamma\} \cup \Pi'$. Since $\Box_a \varphi \neq [\chi \wedge [\chi]\psi]\gamma$, it follows that $\Box_a \varphi \notin \{[\chi \wedge [\chi]\psi]\gamma\} \cup \Pi'$.

Second, let the last applied rule to obtain $\Pi \Rightarrow \Sigma$ be $(Cut)^a$.

$$\frac{\Pi_1 \Rightarrow \Sigma_1, \psi \quad \psi, \Pi_2 \Rightarrow \Sigma_2}{\Pi_1, \Pi_2 \Rightarrow \Sigma_1, \Sigma_2} (Cut)^a_\psi$$

By induction hypothesis, we have $\Box_a \varphi \notin \Pi_1 \cup \Pi_2$. We divide our argument into the following two cases: $\Box_a \varphi \equiv \psi$ and $\Box_a \varphi \neq \psi$. Suppose $\Box_a \varphi \neq \psi$. Then $\Box_a \varphi \notin \Pi_1$ and $\Box_a \varphi \notin \{\psi\} \cup \Pi_2$ hold clearly. So, let us suppose $\Box_a \varphi \equiv \psi$. We show that this

case is impossible. Then, the derivation under consideration is of the following form:

$$\frac{\Pi_1 \Rightarrow \Sigma_1, \Box_a \varphi \quad \Box_a \varphi, \Pi_2 \Rightarrow \Sigma_2}{\Pi_1, \Pi_2 \Rightarrow \Sigma_1, \Sigma_2} (Cut)_{\Box_a \varphi}^a.$$

Since $(Cut)_{\Box_a \varphi}^a$ is an a -cut, $\Box_a \varphi \in \text{CL}(\Pi_1, \Pi_2, \Sigma_1, \Sigma_2)$. By the extended subformula property, $\Pi_1 \cup \Pi_2 \cup \Sigma_1 \cup \Sigma_2 \subseteq \text{CL}(\Box_a \Gamma, \Box_a(\Delta_\varphi), \varphi)$ hence $\text{CL}(\Pi_1, \Pi_2, \Sigma_1, \Sigma_2) \subseteq \text{CL}(\Box_a \Gamma, \Box_a(\Delta_\varphi), \varphi)$. Thus, $\Box_a \varphi \in \text{CL}(\Box_a \Gamma, \Box_a(\Delta_\varphi), \varphi)$. Since (a) $\Box_a \varphi \notin \text{CL}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$, we obtain $\Box_a \varphi \in \text{CL}(\varphi)$, which is impossible. **(QED of Claim)**

Define Ξ to be a sequence of all formulas except $\Box_a \varphi$ which occur in the lower sequents of some applications of $(\Box_{ss})$ in $\mathcal{D}$. It follows that Ξ only consist $\Box_a$ -formulas. By the extended subformula property, we have $\Xi \subseteq \text{CL}(\Box_a \Gamma, \Box_a(\Delta_\varphi), \varphi)$. Our goal is to show that $\Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi$ is derivable in $G^a(\mathbf{S5PAL})$. By Lemma 56, it suffices to show the following.

(2.1) For every partition $\langle \Box_a \Phi; \Box_a \Psi \rangle$ of Ξ , $\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$.

Fix any partition $\langle \Box_a \Phi; \Box_a \Psi \rangle$ of Ξ . Then, we divide our argument into the following two cases:

- (i) $(\Box_a \Phi \cap \Box_a(\Delta_\varphi) \neq \emptyset \text{ or } \Box_a \Gamma \cap \Box_a \Psi \neq \emptyset)$ or
- (ii) $(\Box_a \Phi \cap \Box_a(\Delta_\varphi) = \emptyset \text{ and } \Box_a \Gamma \cap \Box_a \Psi = \emptyset)$.

Suppose that (i) $\Box_a \Phi \cap \Box_a(\Delta_\varphi) \neq \emptyset$ or $\Box_a \Gamma \cap \Box_a \Psi \neq \emptyset$. Then, either $\Box_a \Phi \Rightarrow \Box_a(\Delta_\varphi)$ or $\Box_a \Gamma \Rightarrow \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$ by weakening rules. Hence $\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$ by weakening rules.

Therefore, we always suppose that (ii) $\Box_a \Phi \cap \Box_a(\Delta_\varphi) = \emptyset$ and $\Box_a \Gamma \cap \Box_a \Psi = \emptyset$ in what follows. By definition of Ξ , we have $\Box_a \Gamma \subseteq \Xi$ and $\Box_a(\Delta_\varphi) \subseteq \Xi$. Since $\langle \Box_a \Phi; \Box_a \Psi \rangle$ is a partition of Ξ , it follows from (ii) that (c) $\Box_a \Gamma \subseteq \Box_a \Phi$ and $\Box_a(\Delta_\varphi) \subseteq \Box_a \Psi$.

Let us consider the uppermost application $\text{rule}(\mathcal{D}')$ of $(\Box_{ss})$ in $\mathcal{D}$:

$$\frac{\mathcal{D}' \quad \Box_a \Pi \Rightarrow \Box_a \Sigma, \psi}{\Box_a \Pi \Rightarrow \Box_a \Sigma, \Box_a \psi} (\Box_{ss}) = \text{rule}(\mathcal{D}')$$

such that (d) $\Box_a \Pi \subseteq \Box_a \Phi$ and $\Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$. Such an application exists certainly, because, in the lowermost application $(\Box_{ss})$ in the derivation $\mathcal{D}$ (i.e., $\text{rule}(\mathcal{D})$), by (c) $\Box_a \Gamma \subseteq \Box_a \Phi$ and $\Box_a(\Delta_\varphi) \subseteq \Box_a \Psi$ hence $\Box_a \Delta \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$. Moreover we have the following claim:

Claim 2. $\psi \not\equiv \Box_a \varphi$.

Now our goal is to show the following item:

(2.1') $\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$.

It suffices to show the following.

(2.2) For every sequent $\Theta \Rightarrow \Omega$ which lies above the lower sequent of $\text{rule}(\mathcal{D}')$, (i.e., in $\mathcal{D}'$), the sequent $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$.

Suppose (2.2) holds. By applying (2.2) to the upper sequent $\Box_a \Pi \Rightarrow \Box_a \Sigma, \psi$ of $\text{rule}(\mathcal{D}')$, it follows that $\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \psi, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$. To show (2.1'), we divide the argument into the following two cases: $\psi \equiv \varphi$ and $\psi \not\equiv \varphi$.

First, let $\psi \equiv \varphi$. Then, by (d) $\Box_a \Pi \subseteq \Box_a \Phi$ and $\Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$, we have $\Box_a(\Sigma_\varphi) \subseteq \Box_a \Psi$. Hence (2.1') holds by structural rules:

$$\frac{\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \varphi, \Box_a \Psi}{\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi}.$$

Second, let $\psi \not\equiv \varphi$. Then by (d) $\Box_a \Pi \subseteq \Box_a \Phi$ and $\Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$, we have $\Box_a(\Sigma_\varphi) \cup \{\Box_a \psi\} \subseteq \Box_a \Psi$. Hence (2.1') holds by structural rules:

$$\frac{\frac{\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \psi, \Box_a \Psi}{\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \Box_a \psi, \Box_a \Psi} (\Box_{ss})}{\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi}.$$

Therefore we are going to establish (2.2). Fix any sequent $\Theta \Rightarrow \Omega$ in $\mathcal{D}'$. Let n be the height of derivation $\mathcal{D}'$. To show (2.2), it suffices to prove the following statement by induction on $k \leq n$:

$\forall \Theta \Rightarrow \Omega$ in $\mathcal{D}'$ (if the height of $\Theta \Rightarrow \Omega$ is k ,
then $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5PAL})$).

For the base step, let $k = 0$. Then $\Theta \Rightarrow \Omega \equiv \rho \Rightarrow \rho$ and $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi \equiv \Box_a \Phi, \rho \Rightarrow \rho, \Box_a \Psi$ by Claim 1 of $\rho \not\equiv \Box_a \varphi$. Then $\Box_a \Phi, \rho \Rightarrow \rho, \Box_a \Psi$ can be derived by structural rules.

For inductive step, let $k > 0$ and fix any sequent $\Theta \Rightarrow \Omega$ in $\mathcal{D}'$ whose height is k . Then, we divide our argument depending on the last applied rule to obtain $\Theta \Rightarrow \Omega$. We only comment on the cases where $\Theta \Rightarrow \Omega$ is the lower sequent of rules with public announcement operators.

- Let $\text{rule}(\mathcal{D}')$ be $(\Rightarrow [] \text{at})$. We have the derivation as follows:

$$\frac{\rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, p}{\Theta \Rightarrow \Omega', [\rho]p} (\Rightarrow [] \text{at}).$$

By induction hypothesis, we obtain $\Box_a \Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, p, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, p, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, [\rho]p, \Box_a \Psi} (\Rightarrow [] \text{at}).$$

- Let $\text{rule}(\mathcal{D}')$ be $([] \text{at} \Rightarrow)$. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \Omega', \rho \quad p, \Theta'' \Rightarrow \Omega''}{[\rho]p, \Theta', \Theta'' \Rightarrow \Omega', \Omega''} ([] \text{at} \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \rho, \Box_a \Psi$ and $\Box_a \Phi, p, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \rho, \Box_a \Psi \quad \Box_a \Phi, p, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, [\rho] \neg \gamma, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} (\Box \text{at} \Rightarrow)}{\Box_a \Phi, [\rho] \neg \gamma, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi} (c).$$

- Let $\text{rule}(\mathcal{D}')$ be $(\Rightarrow [] \neg)$. We have the derivation as follows:

$$\frac{\rho, [\rho]\gamma, \Theta \Rightarrow \Omega'}{\Theta \Rightarrow [\rho] \neg \gamma, \Omega'} (\Rightarrow [] \neg).$$

By induction hypothesis, we obtain $\Box_a \Phi, \rho, [\rho]\gamma, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \rho, [\rho]\gamma, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow [\rho] \neg \gamma, \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow [] \neg).$$

- Let $\text{rule}(\mathcal{D}')$ be $([] \neg \Rightarrow)$. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \rho, \Omega \quad \Theta' \Rightarrow [\rho]\gamma, \Omega}{[\rho] \neg \gamma, \Theta' \Rightarrow \Omega} ([] \neg \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \rho, \Omega_{\Box_a \varphi}, \Box_a \Psi$ and $\Box_a \Phi, \Theta' \Rightarrow [\rho]\gamma, \Omega_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \rho, \Omega_{\Box_a \varphi}, \Box_a \Psi \quad \Box_a \Phi, \Theta' \Rightarrow [\rho]\gamma, \Omega_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, [\rho]\neg\gamma, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} (\Box \neg \Rightarrow)}{\Box_a \Phi, [\rho]\neg\gamma, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi} (c).$$

- Let $\text{rule}(\mathcal{D}')$ be $(\Rightarrow \Box \wedge)$. We have the derivation as follows:

$$\frac{\Theta \Rightarrow [\rho]\gamma_1, \Omega' \quad \Theta \Rightarrow [\rho]\gamma_2, \Omega'}{\Theta \Rightarrow [\rho](\gamma_1 \wedge \gamma_2), \Omega'} (\Rightarrow \Box \wedge).$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta \Rightarrow [\rho]\gamma_1, \Omega'_{\Box_a \varphi}, \Box_a \Psi$ and $\Box_a \Phi, \Theta \Rightarrow [\rho]\gamma_2, \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \Theta \Rightarrow [\rho]\gamma_1, \Omega'_{\Box_a \varphi}, \Box_a \Psi \quad \Box_a \Phi, \Theta \Rightarrow [\rho]\gamma_2, \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow [\rho](\gamma_1 \wedge \gamma_2), \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow \Box \wedge).$$

- Let $\text{rule}(\mathcal{D}')$ be $(\Box \wedge \Rightarrow)$. We have the derivation as follows:

$$\frac{[\rho]\gamma_i, \Theta' \Rightarrow \Omega}{[\rho](\gamma_1 \wedge \gamma_2), \Theta' \Rightarrow \Omega} (\Box \wedge \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \Phi, [\rho]\gamma_i, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, [\rho]\gamma_i, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, [\rho](\gamma_1 \wedge \gamma_2), \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi} (\Box \wedge \Rightarrow).$$

- Let $\text{rule}(\mathcal{D}')$ be $(\Rightarrow \Box \rightarrow)$. We have the derivation as follows:

$$\frac{[\rho]\gamma, \Theta \Rightarrow [\rho]\chi, \Omega'}{\Theta \Rightarrow [\rho](\gamma \rightarrow \chi), \Omega'} (\Rightarrow \Box \rightarrow).$$

By induction hypothesis, we obtain $\Box_a \Phi, [\rho]\gamma, \Theta \Rightarrow [\rho]\chi, \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, [\rho]\gamma, \Theta \Rightarrow [\rho]\chi, \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow [\rho](\gamma \rightarrow \chi), \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow \Box \rightarrow).$$

- Let $\text{rule}(\mathcal{D}')$ be $([] \rightarrow \Rightarrow)$. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \Omega, [\rho]\gamma \quad [\rho]\chi, \Theta' \Rightarrow \Omega}{[\rho](\gamma \rightarrow \chi), \Theta' \Rightarrow \Omega} ([] \rightarrow \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, [\rho]\gamma, \Box_a \Psi$ and $\Box_a \Phi, [\rho]\chi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, [\rho]\gamma, \Box_a \Psi \quad \Box_a \Phi, [\rho]\chi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, [\rho](\gamma \rightarrow \chi), \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} ([]) \rightarrow \Rightarrow}{\Box_a \Phi, [\rho](\gamma \rightarrow \chi), \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi} (c).$$

- Let $\text{rule}(\mathcal{D}')$ be $(\Rightarrow [] \Box)$. We have the derivation as follows:

$$\frac{\rho, \Theta \Rightarrow \Box_a[\rho]\gamma, \Omega'}{\Theta \Rightarrow [\rho]\Box_a\gamma, \Omega'} (\Rightarrow [] \Box_a).$$

Again we divide the argument into the following two cases: $\Box_a[\rho]\gamma \equiv \Box_a\varphi$ and $\Box_a[\rho]\gamma \not\equiv \Box_a\varphi$.

- Suppose $\Box_a[\rho]\gamma \equiv \Box_a\varphi$. By induction hypothesis, we obtain $\Box_a \Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \rho, \Theta \Rightarrow \Box_a[\rho]\gamma, \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow w)}{\Box_a \Phi, \Theta \Rightarrow [\rho]\Box_a\gamma, \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow [] \Box_a).$$

- Suppose $\Box_a[\rho]\gamma \not\equiv \Box_a\varphi$. By induction hypothesis, we obtain $\Box_a \Phi, \rho, \Theta \Rightarrow \Box_a[\rho]\gamma, \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \rho, \Theta \Rightarrow \Box_a[\rho]\gamma, \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow [\rho]\Box_a\gamma, \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow [] \Box_a).$$

- Let $\text{rule}(\mathcal{D}')$ be $([] \Box \Rightarrow)$. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \rho, \Omega' \quad \Box_a[\rho]\gamma, \Theta'' \Rightarrow \Omega''}{[\rho]\Box_a\gamma, \Theta', \Theta'' \Rightarrow \Omega', \Omega''} ([] \Box \Rightarrow).$$

Again we divide the argument into the following two cases: $\rho \equiv \Box_a\varphi$ and $\rho \not\equiv \Box_a\varphi$.

- Suppose $\rho \equiv \Box_a \varphi$. By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi$ and $\Box_a \Phi, \Box_a [\rho] \gamma, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta' \Rightarrow \rho, \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow w) \quad \Box_a \Phi, \Box_a [\rho] \gamma, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\frac{\Box_a \Phi, \Box_a \Phi, [\rho] \Box_a \gamma, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi}{\Box_a \Phi, [\rho] \Box_a \gamma, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi}} (\Box \Box \Rightarrow) \quad (c).$$

- Suppose $\rho \not\equiv \Box_a \varphi$. By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \rho, \Omega'_{\Box_a \varphi}, \Box_a \Psi$ and $\Box_a \Phi, \Box_a [\rho] \gamma, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \rho, \Omega'_{\Box_a \varphi}, \Box_a \Psi \quad \Box_a \Phi, \Box_a [\rho] \gamma, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, [\rho] \Box_a \gamma, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} (\Box \Box \Rightarrow)}{\Box_a \Phi, [\rho] \Box_a \gamma, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi} (c).$$

- Let rule($\mathcal{D}'$) be $(\Rightarrow \Box \Box)$. We have the derivation as follows:

$$\frac{\Theta \Rightarrow [\rho \wedge [\rho] \gamma] \chi, \Omega'}{\Theta \Rightarrow [\rho] [\gamma] \chi, \Omega'} (\Rightarrow \Box \Box).$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta \Rightarrow [\rho \wedge [\rho] \gamma] \chi, \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \Theta \Rightarrow [\rho \wedge [\rho] \gamma] \chi, \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow [\rho] [\gamma] \chi, \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Rightarrow \Box \Box).$$

- Let rule($\mathcal{D}'$) be $(\Box \Box \Rightarrow)$. We have the derivation as follows:

$$\frac{[\rho \wedge [\rho] \gamma] \chi, \Theta' \Rightarrow \Omega'}{[\rho] [\gamma] \chi, \Theta' \Rightarrow \Omega'} (\Box \Box \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \Phi, [\rho \wedge [\rho] \gamma] \chi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, [\rho \wedge [\rho] \gamma] \chi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, [\rho] [\gamma] \chi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi} (\Box \Box \Rightarrow).$$

Therefore, we have finished establishing Lemma 57. $\square$

4.3.2 Proof of Extended Analytic Cut Property for $G(S5)$

Definition 58. We define $G^*(S5PAL)$ as the same calculus as $G(S5PAL)$ except that the cut rule is replaced with $(Ecut)$.

Theorem 59. If $\Gamma \Rightarrow \Delta$ is derivable in $G^*(S5PAL)$ then it is also derivable in $G^a(S5PAL)$. Therefore, if $\Gamma \Rightarrow \Delta$ is derivable in $G(S5PAL)$ then it is also derivable in $G^a(S5PAL)$.

Proof. The following argument is also an improvement of Takano's argument in [44], in that we use $(Ecut)$ instead of Mix rule by Gentzen [10] to simplify the outline of the argument. It suffices to prove the first statement. For our purpose, we establish the following: If in the following derivation $\mathcal{D}$ of $G^*(S5PAL)$:

$$\mathcal{D} \equiv \frac{\frac{\vdots \mathcal{L}}{\Gamma \Rightarrow \Delta, \varphi^m} \text{rule}(\mathcal{L}) \quad \frac{\vdots \mathcal{R}}{\varphi^n, \Pi \Rightarrow \Sigma} \text{rule}(\mathcal{R})}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Ecut)_\varphi$$

there is no other $(Ecut)$ in derivations of $\Gamma \Rightarrow \Delta, \varphi^m$ and $\varphi^n, \Pi \Rightarrow \Sigma$, then there is a derivation in $G^a(S5PAL)$ (i.e., all cuts are analytic) of the sequent $\Gamma, \Pi \Rightarrow \Delta, \Sigma$. That is, we can assume without loss of generality that the last $(Ecut)$ is non-analytic.

We prove this statement on a derivation $\mathcal{D}$ by double induction on the complexity and weight (or the lexicographic order of the complexity and the weight) of a derivation $\mathcal{D}$, which are defined as the same as in the proof of Theorem 52. If $m = 0$ or $n = 0$, it follows that our goal can be obtained by weakening rules, hence we suppose $m > 0$ and $n > 0$ below. We divide the argument into the following cases determined by the rules applied last above the $(Ecut)$.

- (1) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be (id).
- (2) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be a structural rule.
- (3) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be an analytic cut.
- (4) Let $\text{rule}(\mathcal{L})$ or $\text{rule}(\mathcal{R})$ be a logical rule or a rule for $\Box_a$ and public announcements where the cut formula is not principal.

- (5) Let $\text{rule}(\mathcal{L})$ and $\text{rule}(\mathcal{R})$ be rules of the same connective and the cut formula is principal in both rules.

All the cases are handled similarly as in the proof of Theorems 42 and 52. We comment on only (3) and (5) for the cut formula of the form $\Box_a \varphi$. Lemma 57 is needed in case (5).

• (3)

- Let $\text{rule}(\mathcal{L})$ be an analytic cut. Our $\mathcal{D}$ looks as follows:

$$\frac{\frac{\Gamma' \Rightarrow \Delta', \psi, \rho^a \quad \psi, \Gamma'' \Rightarrow \Delta'', \rho^b}{\Gamma', \Gamma'' \Rightarrow \Delta', \Delta'', \rho^{a+b}} (Cut)_\psi^a \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma', \Gamma'', \Pi \Rightarrow \Delta', \Delta'', \Sigma} (Ecut)_\rho$$

where $\psi \in \text{CL}(\Gamma', \Gamma'', \Delta', \Delta'', \rho^{a+b})$ since (Cut) is analytic. Here we divide the argument into the following two cases: $\psi \equiv \rho$ and $\psi \not\equiv \rho$. Suppose $\psi \equiv \rho$. We transform $\mathcal{D}$ into:

$$\frac{\frac{\Gamma' \Rightarrow \Delta', \psi, \rho^a \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma', \Pi \Rightarrow \Delta', \Sigma} (Ecut)_\rho}{\Gamma', \Gamma'', \Pi \Rightarrow \Delta', \Delta'', \Sigma} (w) .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

Suppose $\psi \not\equiv \rho$. We transform $\mathcal{D}$ into:

$$\frac{\frac{\Gamma' \Rightarrow \Delta', \psi, \rho^a \quad \rho^n, \Pi \Rightarrow \Sigma}{\Gamma', \Pi \Rightarrow \Delta', \psi, \Sigma} (Ecut)_\rho^1 \quad \frac{\psi, \Gamma'' \Rightarrow \Delta'', \rho^b \quad \rho^n, \Pi \Rightarrow \Sigma}{\psi, \Gamma'', \Pi \Rightarrow \Delta'', \Sigma} (Ecut)_\rho^2}{\frac{\Gamma', \Gamma'', \Pi, \Pi \Rightarrow \Delta', \Delta'', \Sigma, \Sigma}{\Gamma', \Gamma'', \Pi \Rightarrow \Delta', \Delta'', \Sigma} (c)} (Ecut)_\psi^3 .$$

In this derivation $(Ecut)^1$ and $(Ecut)^2$ can be eliminated by induction hypothesis since their weights are reduced. For $(Ecut)^3$, we divide the argument into the following two cases: $\psi \in \text{CL}(\rho)$ and $\psi \notin \text{CL}(\rho)$.

- * Suppose $\psi \in \text{CL}(\rho)$. Since $\psi \not\equiv \rho$, the complexity of $(Ecut)^3$ is reduced by Lemma 49: $c(\rho) > c(\psi)$. Hence it can be eliminated by induction hypothesis.

- * Suppose $\psi \notin \text{CL}(\rho)$. Since the original (Cut) is analytic, $\psi \in \text{CL}(\Gamma', \Gamma'', \Delta', \Delta'', \rho^{a+b})$. Then we have $\psi \in \text{CL}(\Gamma', \Gamma'', \Delta', \Delta'')$. Thus $(Ecut)^3$ becomes an analytic cut since $\psi \in \text{CL}(\Gamma', \Gamma'', \Delta', \Delta'') \subseteq \text{CL}(\Gamma', \Gamma'', \Pi, \Pi, \Delta', \Delta'', \Sigma, \Sigma)$.
- Let $\text{rule}(\mathcal{R})$ be an analytic cut. Our $\mathcal{D}$ looks as follows:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \frac{\rho^a, \Pi' \Rightarrow \Sigma', \psi \quad \psi, \rho^b, \Pi'' \Rightarrow \Sigma''}{\rho^{a+b}, \Pi', \Pi'' \Rightarrow \Sigma', \Sigma''} (Cut)_\psi}{\Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Sigma', \Sigma''} (Ecut)_\rho$$

where $\psi \in \text{CL}(\rho^{a+b}, \Pi', \Pi'', \Sigma', \Sigma'')$ since (Cut) is analytic. Again we divide the argument into the following two cases: $\psi \equiv \rho$ and $\psi \neq \rho$. Suppose $\psi \equiv \rho$. We transform $\mathcal{D}$ into:

$$\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \psi, \rho^b, \Pi'' \Rightarrow \Sigma''}{\Gamma, \Pi'' \Rightarrow \Delta, \Sigma''} (Ecut)_\rho}{\Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Sigma', \Sigma''} (w) .$$

By induction hypothesis, this $(Ecut)$ can be eliminated since its weight is reduced.

Suppose $\psi \neq \rho$. We transform $\mathcal{D}$ into:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \rho^m \quad \rho^a, \Pi' \Rightarrow \Sigma', \psi}{\Gamma, \Pi' \Rightarrow \Delta, \Sigma', \psi} (Ecut)_\rho^1 \quad \frac{\Gamma \Rightarrow \Delta, \rho^m \quad \psi, \rho^b, \Pi'' \Rightarrow \Sigma''}{\psi, \Gamma, \Pi'' \Rightarrow \Delta, \Sigma''} (Ecut)_\rho^2}{\frac{\Gamma, \Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Delta, \Sigma', \Sigma''}{\Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Sigma', \Sigma''} (Ecut)_\psi^3} (c) .$$

In this derivation $(Ecut)^1$ and $(Ecut)^2$ can be eliminated by induction hypothesis since their weights are reduced. For $(Ecut)^3$, we divide the argument into the following two cases: $\psi \in \text{CL}(\rho)$ and $\psi \notin \text{CL}(\rho)$.

- * Suppose $\psi \in \text{CL}(\rho)$. Since $\psi \neq \rho$, the complexity of $(Ecut)^3$ is reduced by Lemma 49: $c(\rho) > c(\psi)$. Hence it can be eliminated by induction hypothesis.
- * Suppose $\psi \notin \text{CL}(\rho)$. Since the original (Cut) is analytic, $\psi \in \text{CL}(\rho^{a+b}, \Pi', \Pi'', \Sigma', \Sigma'')$. Then we have $\psi \in \text{CL}(\Pi', \Pi'', \Sigma', \Sigma'')$. Thus $(Ecut)^3$ becomes an analytic cut since $\psi \in \text{CL}(\Pi', \Pi'', \Sigma', \Sigma'') \subseteq \text{CL}(\Gamma, \Gamma, \Pi', \Pi'' \Rightarrow \Delta, \Delta, \Sigma', \Sigma'')$.

- (5) Suppose $\text{rule}(\mathcal{L})$ is $(\Box_{S5})$ and $\text{rule}(\mathcal{R})$ is $(\Box_a \Rightarrow)$ where the cut formula is the principal formula in both rules. In this case, the derivation $\mathcal{D}$ runs as follows:

$$\frac{\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi, (\Box_a \rho)^m}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, (\Box_a \rho)^m} (\Box_{S5}) \quad \frac{\rho, (\Box_a \rho)^{n-1}, \Pi \Rightarrow \Sigma}{(\Box_a \rho)^n, \Pi \Rightarrow \Sigma} (\Box_a \Rightarrow)}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma} (Ecut)_{\Box_a \rho}.$$

Here we divide the argument into the following two cases: $\varphi \equiv \rho$ and $\varphi \not\equiv \rho$. Since the standard argument can be applied to the case of $\varphi \equiv \rho$, we focus on the case of $\varphi \not\equiv \rho$. Suppose $\varphi \not\equiv \rho$. By Lemma 57, if $\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, (\Box_a \rho)^m$ is derivable in $G^a(\mathbf{S5PAL})$, then $\Box_a \Gamma \Rightarrow \Box_a(\Delta_\rho), \Box_a \varphi, \rho$ is also derivable in $G^a(\mathbf{S5PAL})$. Hence we transform the derivation into:

$$\frac{\frac{\Box_a \Gamma \Rightarrow \Box_a(\Delta_\rho), \Box_a \varphi, \rho}{\Box_a \Gamma, \Box_a \Gamma, \Pi \Rightarrow \Box_a(\Delta_\rho), \Box_a \Delta, \Box_a \varphi, \Box_a \varphi, \Sigma} \quad \frac{\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi, (\Box_a \rho)^m \quad \rho, (\Box_a \rho)^{n-1}, \Pi \Rightarrow \Sigma}{\Box_a \Gamma, \Pi, \rho \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma} (Ecut)_{\Box_a \rho}^1}{\Box_a \Gamma, \Pi \Rightarrow \Box_a \Delta, \Box_a \varphi, \Sigma} (Ecut)_\rho^2 (c).$$

By induction hypotheses, $(Ecut)^1$ can be eliminated since its weight is reduced, and $(Ecut)^2$ can be eliminated since its complexity is reduced by Lemma 49 and $\rho \in \text{Sub}(\Box_a \rho) \subseteq \text{CL}(\Box_a \rho)$.

□

As a corollary of Theorem 59, we can show that $G(\mathbf{S5PAL})$ is a conservative extension of $G(\mathbf{S5})$ as follows.

Corollary 60. If a sequent from L is derivable in $G(\mathbf{S5PAL})$ then it is also derivable in $G(\mathbf{S5})$.

Proof. Suppose a sequent from L is derivable in $G(\mathbf{S5PAL})$. By Theorem 59, it follows that the sequent is derivable in $G^a(\mathbf{S5PAL})$ by a derivation $\mathcal{D}$. To regard $\mathcal{D}$ as a derivation in $G^a(\mathbf{S5})$ it suffices to show that each application of analytic cut in $G(\mathbf{S5PAL})$ preserves the subformula property (not the extended subformula property in terms of CL) when the conclusion of the cut is in $\mathcal{L}$. Let the cut formula ρ be in the closure of the conclusion $\Gamma, \Pi \Rightarrow \Delta, \Sigma$ of the cut, i.e., $\rho \in \text{CL}(\Gamma, \Pi, \Delta, \Sigma)$. Since the conclusion is from L , we get $\text{CL}(\Gamma, \Pi, \Delta, \Sigma) = \text{Sub}(\Gamma, \Pi, \Delta, \Sigma)$. So, $\rho \in \text{Sub}(\Gamma, \Pi, \Delta, \Sigma)$ holds, as desired. □

4.4 Decidability of $G(\mathbf{K45PAL})$ and $G(\mathbf{S5PAL})$

In this section we establish the decidabilities of $G(\mathbf{K45PAL})$ and $G(\mathbf{S5PAL})$, which is similar to Theorem 46. Let $\Lambda \in \{\mathbf{K45PAL}, \mathbf{S5PAL}\}$. The following lemmas hold similarly to 45, respectively.

Lemma 61. Suppose that $\Gamma \Rightarrow \Delta$ is a sequent which is derivable in $G(\Lambda)$ and that $\Gamma^* \Rightarrow \Delta^*$ is the 1-reduced contraction of $\Gamma \Rightarrow \Delta$. Then there exists a derivation $\mathcal{D}$ of $\Gamma^* \Rightarrow \Delta^*$ in $G(\Lambda)$ such that

- (1) Every formula in $\mathcal{D}$ consists of a closure of some formulas in $\Gamma^* \Rightarrow \Delta^*$.
- (2) Every sequent appearing in $\mathcal{D}$ is reduced.

Proof. Follows from Theorems 42 and 59 and Lemma 44. $\square$

Lemma 62. Any derivations $\mathcal{D}$ in $G(\Lambda)$ can be transformed to a derivation $\mathcal{D}'$ which satisfies the following properties, with the same endsequence.

- (1) $\mathcal{D}'$ has the *extended subformula property*.
- (2) $\mathcal{D}'$ does not consist *redundancies*. As shown in the following proof, a redundancy of $\Sigma \Rightarrow \Pi$

$$\begin{array}{c} \Sigma \Rightarrow \Pi \\ \vdots \\ (\mathcal{P}_0) \\ \vdots \\ \Sigma \Rightarrow \Pi \\ \vdots \\ (\mathcal{P}_1) \\ \vdots \\ \Gamma \Rightarrow \Delta \end{array}$$

can be eliminated as follow:

$$\begin{array}{c} \vdots \\ (\mathcal{P}_0) \\ \vdots \\ \Sigma \Rightarrow \Pi \\ \vdots \\ (\mathcal{P}_1) \\ \vdots \\ \Gamma \Rightarrow \Delta \end{array}.$$

- (3) $\mathcal{D}'$ consists only of *reduced sequents*.

Proof. Follows from Theorems 42 and 59 and Lemma 45. $\square$

Hence we obtain Theorem 63.

Theorem 63. $G(\mathbf{K45PAL})$ and $G(\mathbf{S5PAL})$ are decidable.

Proof. Follows from Lemmas 61 and 62, similar to Theorem 46. Given a sequent $\Gamma \Rightarrow \Delta$ in $\mathbf{G}(\mathbf{K45PAL})$ or $\mathbf{G}(\mathbf{S5PAL})$, we check whether there is a derivation of it satisfying all of these three restrictions or not. By Lemma 62(1) and (3), if there exists a derivation $\mathcal{D}$, we can transform that to a derivation $\mathcal{D}'$ which consists only of reduced sequents such that every formula in $\mathcal{D}'$ is a subformula of some formulas in $\Gamma \Rightarrow \Delta$. Clearly, the number of such sequents is finite. Moreover, by Lemma 62(2), the height of the derivation of $\Gamma \Rightarrow \Delta$ must be finite. Thus, our decision algorithm is to generate all possible derivations one by one and to check whether it is a correct derivation of $\Gamma \Rightarrow \Delta$ or not. If we can find one correct derivation, $\Gamma \Rightarrow \Delta$ is derivable in $\mathbf{G}(\Lambda)$. If not, then by Lemma 62, $\Gamma \Rightarrow \Delta$ is not derivable. $\square$

Chapter 5

Conclusion

5.1 Summary

In this thesis, we explore various aspects of Public Announcement Logic (**PAL**) through both semantic and syntactic analyses. In Chapter 2, We give an overview to the Hilbert Systems $H(\mathbf{K45})$, $H(\mathbf{S5})$, $H(\mathbf{K45PAL})$, and $H(\mathbf{S5PAL})$, highlighting their foundational axioms and proving their soundness. For completeness, both canonical models and translation approaches are employed. Chapter 3 introduces the sequent calculi $G(\mathbf{K45})$ and $G(\mathbf{S5})$. It also details the cut-elimination theorem for $G(\mathbf{K45})$ and explores the analytic cut property of $G(\mathbf{S5})$, drawing on Takano's 1992 methodology.

As a key contribution of [22], non-labeled sequent calculi $G(\mathbf{K45PAL})$, and $G(\mathbf{S5PAL})$ were introduced in Chapter 4. We give a detailed proof of the cut-elimination theorem for $G(\mathbf{K45PAL})$ and analytic cut property for $G(\mathbf{S5PAL})$ by extending the notion of subformulas to closures. In both proofs, the restricted complexity measure $c(\varphi)$ adapted from [7] plays a crucial role from a proof-theoretic viewpoint. This aspect was not emphasized in [50] due to the fact that the cut-elimination theorem of the labelled sequent calculus for public announcement expansion in [50] was *reduced* to that of multi-agent **S5** (similarly to semantic completeness, as seen in the proof of Proposition 22).

Also the decision problem for these calculi is discussed in detail in the end of Chapters 3 and 4, respectively. The decidability argument for our non-labelled sequent calculi can be regarded as a merit compared to the existing labelled sequent calculi [25, 36, 30, 50] for public announcement logic, because the notion of label makes [50]'s argument for the decidability more involved than ours ([30, 36] did

not discuss the decidability of the calculi).

5.2 Further Directions

There remains a number of future directions of further research. To begin with, there are other extensions of logics with public announcements that also incorporate recursion axioms, such as Intuitionistic **PAL**, as documented in the work of Ma et al. [23], where they gave a Hilbert-style axiomatization of Intuitionistic **PAL**; Nomura et al. [29], in which a labelled sequent calculi for Intuitionistic **PAL** is introduced; and Balbiani and Galmiche [35], where they gave a sound and complete axiomatization of Intuitionistic **PAL**. Also there is Bilattice **PAL** [40] and Relevant **PAL** [39]. We can endeavor to develop sequent calculi for those logics with the similar approach as in this paper.

Also, in light of the cut-free non-labelled sequent calculus in [49] for action model logic that is based on **S4**, it would be worthwhile to explore the development of a comparable calculus for that based on **S5** (for a cut-free labelled sequent calculus for action model logic, the reader can refer to [28]).

Moreover, although cut-elimination theorem does not hold in $G(\mathbf{S5})$, there are attempts for obtaining cut-free sequent calculi such as hypersequent calculus by [1], the structured system of [41], and bi-sequent calculus in [18]. Based on these calculi, it would be interesting to see if we can obtain a cut-free sequent calculus for **PAL** and how the complexity of Definition 5 will be used.

Finally, as a significant result of proof theory, it is worth examining if the interpolation theorem introduced by William Craig [6] holds in $G(\mathbf{K45PAL})$ and $G(\mathbf{S5PAL})$. This attempt could help in understanding the expressive power of our sequent calculi.

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Appendix A

Omitted Proofs

A.1 Proof of Proposition 35

In this section, we provide a proof of Proposition 35, whose statement is the following.

The sequent $p \Rightarrow \Box_a \neg \Box_a \neg p$ is not derivable in $G^-(S5)$.

Proof. We prove the following statements instead:

1. $\forall m, n \in \mathbb{N} (G^-(S5) \not\vdash p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n)$.
2. $\forall a, b, c, d, e \in \mathbb{N} (G^-(S5) \not\vdash (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e)$.

Then Proposition 35 becomes a special case of (1) where $m = n = 1$. Fix any derivation $\mathcal{D}$. Let $\text{root}(\mathcal{D}) = \Gamma \Rightarrow \Delta$ represent the endsequent of $\mathcal{D}$ is $\Gamma \Rightarrow \Delta$. We prove the following statements by simultaneous induction on $\mathcal{D}$.

1. $\forall m, n \in \mathbb{N} (\text{root}(\mathcal{D}) \neq p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n)$ and
2. $\forall a, b, c, d, e \in \mathbb{N} (\text{root}(\mathcal{D}) \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e)$.

For the base step, let $\mathcal{D}$ be an axiom of the form $\varphi \Rightarrow \varphi$. Then $\text{root}(\mathcal{D}) = \varphi \Rightarrow \varphi$. Our goals of (1) and (2) hold clearly.

For inductive step, first we show (1). Fix any $m, n \in \mathbb{N}$. Suppose for contradiction that $\text{root}(\mathcal{D}) = p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n$. Suppose $m = n = 0$. Then $\text{root}(\mathcal{D}) = \Rightarrow$, which is impossible since $G^-(S5)$ does not have (Cut) . Hence we suppose $m > 0$ or $n > 0$. We divide our argument depending on the last applied rule to obtain $\text{root}(\mathcal{D})$. We should only discuss cases of weakening, contraction and $(\Box_{S5})$.

- Let the last applied rule be $(w \Rightarrow)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ p^{m-1} \Rightarrow (\Box_a \neg \Box_a \neg p)^n \end{array}}{p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n} (w \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq p^{m-1} \Rightarrow (\Box_a \neg \Box_a \neg p)^n$. We have a contradiction.

- Let the last applied rule be $(\Rightarrow w)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^{n-1} \end{array}}{p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n} (\Rightarrow w).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^{n-1}$. We have a contradiction.

- Let the last applied rule be $(c \Rightarrow)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ p^{m+1} \Rightarrow (\Box_a \neg \Box_a \neg p)^n \end{array}}{p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n} (c \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq p^{m+1} \Rightarrow (\Box_a \neg \Box_a \neg p)^n$. We have a contradiction.

- Let the last applied rule be $(\Rightarrow c)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^{n+1} \end{array}}{p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^n} (\Rightarrow c).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq p^m \Rightarrow (\Box_a \neg \Box_a \neg p)^{n+1}$. We have a contradiction.

- Let the last applied rule be $(\Box_{S5})$. It follows that $m = 0$. Then we have the derivation as follows:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ \Rightarrow (\Box_a \neg \Box_a \neg p)^{n-1}, \neg \Box_a \neg p \end{array}}{\Rightarrow (\Box_a \neg \Box_a \neg p)^n} (\Box_{S5}).$$

By induction hypothesis of (2), $\text{root}(\mathcal{D}') \neq \Rightarrow (\Box_a \neg \Box_a \neg p)^{n-1}, \neg \Box_a \neg p$. We have a contradiction. It is noted that this case requires induction hypothesis of (2) (instead of (1)). This is why we conduct a simultaneous induction.

Next we show (2) $\forall a, b, c, d, e \in \mathbb{N}(\mathbf{G}^-(\mathbf{S5}) \not\models (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$. Fix any $a, b, c, d, e \in \mathbb{N}$. Suppose for contradiction that $\text{root}(\mathcal{D}) = (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$. Suppose $a = b = c = d = e = 0$. Then $\text{root}(\mathcal{D}) = \Rightarrow$, which is impossible. Hence we suppose $a \neq 0$ or $b \neq 0$ or $c \neq 0$ or $d \neq 0$ or $e \neq 0$. We divide our argument depending on the last applied rule to obtain $\text{root}(\mathcal{D})$. We should only discuss cases of weakening, contraction, $(\neg \Rightarrow)$, $(\Rightarrow \neg)$, $(\Box_a \Rightarrow)$ and $(\Box_{SS})$.

- Let the last applied rule be $(w \Rightarrow)$. We divide our argument depending on the principal formula, namely $(\neg p)^a$ and $(\Box_a \neg p)^b$.

- Let the principal formula be $(\neg p)^a$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^{a-1}, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (w \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^{a-1}, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$. We have a contradiction.

- Let the principal formula be $(\Box_a \neg p)^b$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^a, (\Box_a \neg p)^{b-1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (w \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^{b-1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$. We have a contradiction.

- Let the last applied rule be $(\Rightarrow w)$. Again we divide our argument depending on the principal formula, namely $(\Box_a \neg \Box_a \neg p)^c$, $(\neg \Box_a \neg p)^d$ and p^e .

- Let the principal formula be $(\Box_a \neg \Box_a \neg p)^c$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^{c-1}, (\neg \Box_a \neg p)^d, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow w).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^{c-1}, (\neg \Box_a \neg p)^d, p^e$.
We have a contradiction.

- Let the principal formula be $(\neg \Box_a \neg p)^d$:

$$\frac{\begin{array}{c} \vdots \\ \mathcal{D}' \end{array} \quad \frac{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^{d-1}, p^e}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow w)}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow w).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^{d-1}, p^e$.
We have a contradiction.

- Let the principal formula be p^e :

$$\frac{\begin{array}{c} \vdots \\ \mathcal{D}' \end{array} \quad \frac{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^{e-1}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow w)}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow w).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^{e-1}$.
We have a contradiction.

- Let the last applied rule be $(c \Rightarrow)$. Again we divide our argument depending on the principal formula, namely $(\neg p)^a$ and $(\Box_a \neg p)^b$.

- Let the principal formula be $(\neg p)^a$:

$$\frac{\begin{array}{c} \vdots \\ \mathcal{D}' \end{array} \quad \frac{(\neg p)^{a+1}, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (c \Rightarrow)}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (c \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^{a+1}, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$.
We have a contradiction.

- Let the principal formula be $(\Box_a \neg p)^b$:

$$\frac{\begin{array}{c} \vdots \\ \mathcal{D}' \end{array} \quad \frac{(\neg p)^a, (\Box_a \neg p)^{b+1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (c \Rightarrow)}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (c \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^{b+1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$.
We have a contradiction.

- Let the last applied rule be $(\Rightarrow c)$. Again we divide our argument depending on the principal formula, namely $(\Box_a \neg \Box_a \neg p)^c$, $(\neg \Box_a \neg p)^d$ and p^e .
 - Let the principal formula be $(\Box_a \neg \Box_a \neg p)^c$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^{c+1}, (\neg \Box_a \neg p)^d, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow c).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^{c+1}, (\neg \Box_a \neg p)^d, p^e$.
We have a contradiction.

- Let the principal formula be $(\neg \Box_a \neg p)^d$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^{d+1}, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow c).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^{d+1}, p^e$.
We have a contradiction.

- Let the principal formula be p^e :

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^{e+1} \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow c).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^{e+1}$.
We have a contradiction.

- Let the last applied rule be $(\neg \Rightarrow)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^{a-1}, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^{e+1} \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\neg \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^{a-1}, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^{e+1}$.
We have a contradiction.

- Let the last applied rule be $(\Rightarrow \neg)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^a, (\Box_a \neg p)^{b+1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^{d-1}, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Rightarrow \neg).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^a, (\Box_a \neg p)^{b+1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^{d-1}, p^e$. We have a contradiction.

- Let the last applied rule be $(\Box_a \Rightarrow)$:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\neg p)^{a+1}, (\Box_a \neg p)^{b-1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e \end{array}}{(\neg p)^a, (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e} (\Box_a \Rightarrow).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\neg p)^{a+1}, (\Box_a \neg p)^{b-1} \Rightarrow (\Box_a \neg \Box_a \neg p)^c, (\neg \Box_a \neg p)^d, p^e$. We have a contradiction.

- Let the last applied rule be $(\Box_{S5})$. It follows that $a = d = e = 0$. Then our derivation looks as follows:

$$\frac{\begin{array}{c} \vdots \mathcal{D}' \\ (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^{c-1}, \neg \Box_a \neg p \end{array}}{(\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^c} (\Box_{S5}).$$

By induction hypothesis, $\text{root}(\mathcal{D}') \neq (\Box_a \neg p)^b \Rightarrow (\Box_a \neg \Box_a \neg p)^{c-1}, \neg \Box_a \neg p$. We have a contradiction.

For all the cases, we get a desired contradiction and we conclude that Proposition 35 holds. $\square$

A.2 Proof of Lemma 40

Recall that a $\Box_a$ -formula is a formula of the form $\Box_a \varphi$ for some formula φ . Let φ be a formula and Ξ and Ω sets of formulas. Then, Ξ_φ (or Ξ_Ω) denotes the resulting set by deleting from Ξ all the occurrences of φ (or formulas in Ω , respectively). In what follows, when $\Gamma \Rightarrow \Delta$ is derivable in $G^a(S5)$, we denote $G^a(S5) \vdash \Gamma \Rightarrow \Delta$. We can establish the following.

Lemma 64 (Takano [44]). Suppose $\Xi \subseteq \text{Sub}(\Gamma, \Delta)$. If the sequent $\Phi, \Gamma \Rightarrow \Delta, \Psi$ is derivable in $G^a(\mathbf{S5})$ for every partition $\langle \Phi; \Psi \rangle$ of Ξ , then $\Gamma \Rightarrow \Delta$ has a derivation in $G^a(\mathbf{S5})$.

Proof. We prove this by induction on the number n of formulas in Ξ .

$\forall \Xi \forall \Gamma \forall \Delta (\text{if } |\Xi| = n \text{ and } \Xi \subseteq \text{Sub}(\Gamma, \Delta) \text{ then}$

$\forall \langle \Phi; \Psi \rangle : \text{partition of } \Xi (G^a(\mathbf{S5}) \vdash \Phi, \Gamma \Rightarrow \Delta, \Psi \text{ implies } G^a(\mathbf{S5}) \vdash \Gamma \Rightarrow \Delta)).$

1. Let $n = 0$. For $\langle ; \rangle$ is the only partition, Lemma 64 holds clearly.
2. Let $n > 0$. Fix any Ξ' such that $|\Xi'| = n + 1$ and $\Xi' \equiv \Xi \cup \{\varphi\}$, and fix any Γ and Δ such that $\Xi' \subseteq \text{Sub}(\Gamma, \Delta)$. Suppose $G^a(\mathbf{S5}) \vdash \Phi', \Gamma \Rightarrow \Delta, \Psi'$ for all partitions $\langle \Phi'; \Psi' \rangle$ of Ξ' . We show that $G^a(\mathbf{S5}) \vdash \Gamma \Rightarrow \Delta$. By induction hypothesis, it suffices to show that $G^a(\mathbf{S5}) \vdash \Phi, \Gamma \Rightarrow \Delta, \Psi$ for all partitions $\langle \Phi; \Psi \rangle$ of Ξ . Fix any partition $\langle \Phi; \Psi \rangle$ of Ξ . Consider the following partitions of Ξ' : $\langle \Phi, \varphi; \Psi \rangle$ and $\langle \Phi; \Psi, \varphi \rangle$. By supposition, $\Phi, \Gamma \Rightarrow \Delta, \Psi, \varphi$ and $\varphi, \Phi, \Gamma \Rightarrow \Delta, \Psi$ are derivable in $G^a(\mathbf{S5})$. We have the following derivaiton:

$$\frac{\frac{\Phi, \Gamma \Rightarrow \Delta, \Psi, \varphi \quad \varphi, \Phi, \Gamma \Rightarrow \Delta, \Psi}{\Phi, \Phi, \Gamma, \Gamma \Rightarrow \Delta, \Delta, \Psi, \Psi} (Cut)_\varphi}{\Phi, \Gamma \Rightarrow \Delta, \Psi} (c),$$

where $(Cut)_\varphi$ is acceptable (analytic) since $\varphi \in \Xi' \subseteq \text{Sub}(\Gamma, \Delta)$. We have shown that $G^a(\mathbf{S5}) \vdash \Phi, \Gamma \Rightarrow \Delta, \Psi$, as required. $\square$

In what follows, we provide a proof of Lemma 57, whose statement is the following:

Suppose that $(Ecut)_\psi$ is admissible in $G^a(\mathbf{S5})$ for all subformulas ψ of φ . If the sequent $\Box_a \Gamma \Rightarrow \Box_a \Delta$ is derivable in $G^a(\mathbf{S5})$ but with an application of $(\Box_{S5})$ for its lowest inference, then the sequent $\Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi$ is derivable in $G^a(\mathbf{S5})$.

Proof. Let $\mathcal{D}$ be such derivation of $\Box_a \Gamma \Rightarrow \Box_a \Delta$ in $G^a(\mathbf{S5})$ with an application K of $(\Box_{S5})$ for its lowest inference. We divide our argument into the following two cases: $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$ and $\Box_a \varphi \notin \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$. First, let $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$. Then, we have:

$$\frac{\frac{\vdots \mathcal{D} \quad \varphi \Rightarrow \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta \quad \Box_a \varphi \Rightarrow \varphi} (\Box_a \Rightarrow)}{\Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi} (Cut)_{\Box_a \varphi}^a$$

where $(Cut)_{\Box_a\varphi}^a$ is acceptable by assumption.

Second, let

$$(a) \Box_a\varphi \notin \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi)).$$

For any cut in $\mathcal{D}$

$$\frac{\Theta \Rightarrow \Omega, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Theta, \Pi \Rightarrow \Omega, \Sigma} (Cut)_\varphi$$

with $\varphi \in (\Theta, \Pi, \Omega, \Sigma)$, it can be replaced by applications of weakening rules. By (a) $\Box_a\varphi \notin \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi))$, we can establish the following claim.

Claim 1. (b) None of the sequents of $\mathcal{D}$ contain $\Box_a\varphi$ in its antecedent.

(Proof of Claim) We show that $\Box_a\varphi \notin \Pi$ for all sequents $\Pi \Rightarrow \Sigma$ in $\mathcal{D}$. Let the height of $\mathcal{D}$ be n . We prove the following statement by induction on $k \leq n$:

$$\forall \Pi \Rightarrow \Sigma \text{ in } \mathcal{D} \text{ (if the height of } \Pi \Rightarrow \Sigma \text{ in } \mathcal{D} \text{ is } n - k, \text{ then } \Box_a\varphi \notin \Pi).$$

For the base step, let $k = 0$. Our derivation $\mathcal{D}$ is of the following form:

$$\frac{\Box_a\Gamma \Rightarrow \Box_a\Delta', \psi}{\Box_a\Gamma \Rightarrow \Box_a\Delta} (\Box_{S5})$$

By (a) $\Box_a\varphi \notin \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi))$, $\Box_a\varphi \notin \Box_a\Gamma$ holds.

For inductive step, let $k > 0$ and fix any sequent $\Pi \Rightarrow \Sigma$ in $\mathcal{D}$ whose height is $n - k$. Then, we divide our argument depending on the last applied rule to obtain $\Pi \Rightarrow \Sigma$. First, we discuss the inference rules except (Cut) . For all right rules except (Cut) , our goal is immediate. So, we can focus on the remaining left rules of $G^a(S5)$.

- Let the last applied rule be $(w \Rightarrow)$:

$$\frac{\Pi' \Rightarrow \Sigma}{\psi, \Pi' \Rightarrow \Sigma} (w \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\varphi \notin \{\psi\} \cup \Pi'$, which implies $\Box_a\varphi \notin \Pi'$.

- Let the last applied rule be $(c \Rightarrow)$:

$$\frac{\psi, \psi, \Pi' \Rightarrow \Sigma}{\psi, \Pi' \Rightarrow \Sigma} (c \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\varphi \notin \{\psi\} \cup \Pi'$, which implies $\Box_a\varphi \notin \{\psi, \psi\} \cup \Pi'$.

- Let the last applied rule be $(\wedge \Rightarrow)$:

$$\frac{\psi, \Pi' \Rightarrow \Sigma}{\psi \wedge \gamma, \Pi' \Rightarrow \Sigma} (\wedge \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{\psi \wedge \gamma\} \cup \Pi'$. Here we divide the argument into the following two cases: $\Box_a \varphi \equiv \psi$ and $\Box_a \varphi \not\equiv \psi$.

- Suppose $\Box_a \varphi \equiv \psi$. Then $\psi \wedge \gamma \equiv \Box_a \varphi \wedge \gamma$. By the subformula property, $\Box_a \varphi \wedge \gamma \in \text{Sub}(\Box_a \Gamma, \Box_a \Delta)$. Since $\Box_a \varphi \wedge \gamma \not\equiv \Box_a \varphi$, we have $\Box_a \varphi \wedge \gamma \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$. Then $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$, which is contradicted to (a).
 - Suppose $\Box_a \varphi \not\equiv \psi$. It follows that $\Box_a \varphi \notin \{\psi\} \cup \Pi'$, which suffices our goal.
- Let the last applied rule be $(\vee \Rightarrow)$:

$$\frac{\psi, \Pi' \Rightarrow \Sigma \quad \gamma, \Pi' \Rightarrow \Sigma}{\psi \vee \gamma, \Pi' \Rightarrow \Sigma} (\vee \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{\psi \vee \gamma\} \cup \Pi'$. Again we divide the argument into the following three cases: $\Box_a \varphi \equiv \psi$, $\Box_a \varphi \equiv \gamma$ or $(\Box_a \varphi \not\equiv \psi \text{ and } \Box_a \varphi \not\equiv \gamma)$.

- Suppose $\Box_a \varphi \equiv \psi$. Then $\psi \vee \gamma \equiv \Box_a \varphi \vee \gamma$. By the subformula property, $\Box_a \varphi \vee \gamma \in \text{Sub}(\Box_a \Gamma, \Box_a \Delta)$. Since $\Box_a \varphi \vee \gamma \not\equiv \Box_a \varphi$, we have $\Box_a \varphi \vee \gamma \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$. Then $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$, which is contradicted to (a).
- Suppose $\Box_a \varphi \equiv \gamma$. Then $\psi \vee \gamma \equiv \psi \vee \Box_a \varphi$. By the subformula property, $\psi \vee \Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a \Delta)$. Since $\psi \vee \Box_a \varphi \not\equiv \Box_a \varphi$, we have $\psi \vee \Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$. Then $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a(\Delta_\varphi))$, which is contradicted to (a).
- Suppose $\Box_a \varphi \not\equiv \psi$ and $\Box_a \varphi \not\equiv \gamma$. It follows that $\Box_a \varphi \notin \{\psi\} \cup \Pi'$ and $\Box_a \varphi \notin \{\gamma\} \cup \Pi'$, which suffices our goal.

- Let the last applied rule be $(\rightarrow \Rightarrow)$:

$$\frac{\Pi_1 \Rightarrow \Sigma_1, \psi \quad \gamma, \Pi_2 \Rightarrow \Sigma_2}{\psi \rightarrow \gamma, \Pi_1, \Pi_2 \Rightarrow \Sigma_1, \Sigma_2} (\rightarrow \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{\psi \rightarrow \gamma\} \cup \Pi'$. Again we divide the argument into the following two cases: $\Box_a \varphi \equiv \gamma$ and $\Box_a \varphi \not\equiv \gamma$:

- Suppose $\Box_a \varphi \equiv \gamma$. Then $\psi \rightarrow \gamma \equiv \psi \rightarrow \Box_a \varphi$. By the subformula property, $\psi \rightarrow \Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a \Delta)$. Since $\psi \rightarrow \Box_a \varphi \not\equiv \Box_a \varphi$, we have $\psi \rightarrow \Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a (\Delta_\varphi))$. Then $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a (\Delta_\varphi))$, which is contradicted to (a).
 - Suppose $\Box_a \varphi \not\equiv \gamma$. It follows that $\Box_a \varphi \notin \Pi'$ and $\Box_a \varphi \notin \{\gamma\} \cup \Pi'$, which suffices our goal.
- Let the last applied rule be $(\neg \Rightarrow)$:

$$\frac{\Pi' \Rightarrow \Sigma, \psi}{\neg \psi, \Pi' \Rightarrow \Sigma} (\neg \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{\neg \psi\} \cup \Pi'$, which implies $\Box_a \varphi \notin \Pi'$.

- Let the last applied rule be $(\Box_a \Rightarrow)$:

$$\frac{\psi, \Pi' \Rightarrow \Sigma}{\Box_a \psi, \Pi' \Rightarrow \Sigma} (\Box_a \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a \varphi \notin \{\Box_a \psi\} \cup \Pi'$. Again we divide the argument into the following two cases: $\Box_a \varphi \equiv \psi$ and $\Box_a \varphi \not\equiv \psi$.

- Suppose $\Box_a \varphi \equiv \psi$. Then $\Box_a \psi \equiv \Box_a \Box_a \varphi$. By the subformula property, $\Box_a \Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a \Delta)$. Since $\Box_a \Box_a \varphi \not\equiv \Box_a \varphi$, we have $\Box_a \Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a (\Delta_\varphi))$. Then $\Box_a \varphi \in \text{Sub}(\Box_a \Gamma, \Box_a (\Delta_\varphi))$, which is contradicted to (a).
- Suppose $\Box_a \varphi \not\equiv \psi$. It follows that $\Box_a \varphi \notin \{\psi\} \cup \Pi'$, which suffices our goal.

Second, let the last applied rule to obtain $\Pi \Rightarrow \Sigma$ be $(Cut)^a$.

$$\frac{\Pi_1 \Rightarrow \Sigma_1, \psi \quad \psi, \Pi_2 \Rightarrow \Sigma_2}{\Pi_1, \Pi_2 \Rightarrow \Sigma_1, \Sigma_2} (Cut)^a_\psi$$

By induction hypothesis, we have $\Box_a \varphi \notin \Pi_1 \cup \Pi_2$. We divide our argument into the following two cases: $\Box_a \varphi \equiv \psi$ and $\Box_a \varphi \not\equiv \psi$. Suppose $\Box_a \varphi \not\equiv \psi$. Then $\Box_a \varphi \notin \Pi_1$

and $\Box_a\varphi \notin \psi, \Pi_2$ hold clearly. So, let us suppose $\Box_a\varphi \equiv \psi$. We show that this case is impossible. Then, the derivation under consideration is of the following form:

$$\frac{\Pi_1 \Rightarrow \Sigma_1, \Box_a\varphi \quad \Box_a\varphi, \Pi_2 \Rightarrow \Sigma_2}{\Pi_1, \Pi_2 \Rightarrow \Sigma_1, \Sigma_2} (Cut)_{\Box_a\varphi}^a.$$

Since $(Cut)_{\Box_a\varphi}^a$ is an a -cut, $\Box_a\varphi \in \text{Sub}(\Pi_1, \Pi_2, \Sigma_1, \Sigma_2)$. By the subformula property, $\Pi_1 \cup \Pi_2 \cup \Sigma_1 \cup \Sigma_2 \subseteq \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi), \varphi)$ hence $\text{Sub}(\Pi_1, \Pi_2, \Sigma_1, \Sigma_2) \subseteq \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi), \varphi)$. Thus, $\Box_a\varphi \in \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi), \varphi)$. Since (a) $\Box_a\varphi \notin \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi))$, we obtain $\Box_a\varphi \in \text{Sub}(\varphi)$, which is impossible.

(QED of Claim)

Define Ξ to be a sequence of all formulas except $\Box_a\varphi$ which occur in the lower sequents of some applications of $(\Box_{ss})$ in $\mathcal{D}$. It follows that Ξ only consist $\Box_a$ -formulas. By the subformula property, we have $\Xi \subseteq \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi), \varphi)$. Our goal is to show that $\Box_a\Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi$ is derivable in $G^a(\mathbf{S5})$. By Lemma 64, it suffices to show that:

(2.1) For every partition $\langle \Box_a\Phi; \Box_a\Psi \rangle$ of Ξ , $\Box_a\Phi, \Box_a\Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a\Psi$ is derivable in $G^a(\mathbf{S5})$.

Fix any partition $\langle \Box_a\Phi; \Box_a\Psi \rangle$ of Ξ . Then, we divide our argument into the following two cases:

(i) $(\Box_a\Phi \cap \Box_a(\Delta_\varphi) \neq \emptyset \text{ or } \Box_a\Gamma \cap \Box_a\Psi \neq \emptyset)$ or (ii) $(\Box_a\Phi \cap \Box_a(\Delta_\varphi) = \emptyset \text{ and } \Box_a\Gamma \cap \Box_a\Psi = \emptyset)$.

First, suppose that (i) $(\Box_a\Phi \cap \Box_a(\Delta_\varphi) \neq \emptyset \text{ or } \Box_a\Gamma \cap \Box_a\Psi \neq \emptyset)$. Then, either $\Box_a\Phi \Rightarrow \Box_a(\Delta_\varphi)$ or $\Box_a\Gamma \Rightarrow \Box_a\Psi$ is derivable in $G^a(\mathbf{S5})$ by weakening rules. Hence $\Box_a\Phi, \Box_a\Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a\Psi$ is derivable in $G^a(\mathbf{S5})$ by weakening rules.

Second, suppose that (ii) $(\Box_a\Phi \cap \Box_a(\Delta_\varphi) = \emptyset \text{ and } \Box_a\Gamma \cap \Box_a\Psi = \emptyset)$. By definition of Ξ , we have $\Box_a\Gamma \subseteq \Xi$ and $\Box_a(\Delta_\varphi) \subseteq \Xi$. Since $\langle \Box_a\Phi; \Box_a\Psi \rangle$ is a partition of Ξ , it follows from (ii) that

(c) $\Box_a\Gamma \subseteq \Box_a\Phi$ and $\Box_a(\Delta_\varphi) \subseteq \Box_a\Psi$.

Consider the uppermost application K' of $(\Box_{ss})$ in $\mathcal{D}$:

$$\frac{\Box_a\Pi \Rightarrow \Box_a\Sigma, \psi}{\Box_a\Pi \Rightarrow \Box_a\Sigma, \Box_a\psi} (\Box_{ss})K'$$

such that

$$(d) \Box_a \Pi \subseteq \Box_a \Phi \text{ and } \Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}.$$

By the subformula property, it is noted that $\Box_a \Pi \subseteq \Box_a \Gamma$ and $\Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Delta$. Such an application exists certainly, because, in the lowermost application ($\Box_{SS}$) in the derivation, by (c) $\Box_a \Gamma \subseteq \Box_a \Phi$ and $\Box_a(\Delta_\varphi) \subseteq \Box_a \Psi$ hence $\Box_a \Delta \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$. Our goal is to show:

$$(2.1') \Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi \text{ is derivable in } G^a(S5).$$

It suffices to show that:

(2.2) For every sequent $\Theta \Rightarrow \Omega$ which lies above the lower sequent of K' , the sequent $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$ is derivable in $G^a(S5)$.

Suppose (2.2) holds. By applying (2.2) to the upper sequent $\Box_a \Pi \Rightarrow \Box_a \Sigma, \psi$ of K' , it follows that $\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \psi, \Box_a \Psi$ is derivable in $G^a(S5)$. To show (2.1), we divide the argument into the following two cases: $\psi \equiv \varphi$ and $\psi \not\equiv \varphi$. First, let $\psi \equiv \varphi$. Then, (2.1') holds by weakening and contraction rules:

$$\frac{\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \varphi, \Box_a \Psi}{\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi}$$

since (d) $\Box_a \Pi \subseteq \Box_a \Phi$ and $\Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$ hence $\Box_a(\Sigma_\varphi) \subseteq \Box_a \Psi$. Second, let $\psi \not\equiv \varphi$. Then, (2.1') holds by weakening and contraction rules:

$$\frac{\frac{\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \psi, \Box_a \Psi}{\Box_a \Phi, \Box_a \Pi \Rightarrow \Box_a(\Sigma_\varphi), \Box_a \psi, \Box_a \Psi} (\Box_{SS})}{\Box_a \Phi, \Box_a \Gamma \Rightarrow \Box_a(\Delta_\varphi), \varphi, \Box_a \Psi}$$

since (d) $\Box_a \Pi \subseteq \Box_a \Phi$ and $\Box_a \Sigma \cup \{\Box_a \psi\} \subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$ hence $\Box_a(\Sigma_\varphi) \cup \{\Box_a \psi\} \subseteq \Box_a \Psi$.

In what follows, we establish (2.2). Fix any sequent $\Theta \Rightarrow \Omega$ which lies above the lower sequent of K' .

Let n be the height of derivation above the lower sequent of K' . To show (2.2), we prove the following statement by induction on $k \leq n$:

$\forall \Theta \Rightarrow \Omega$ above the lower sequent of K' (if the height of $\Theta \Rightarrow \Omega$ is k ,

then $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$ is derivable in $G^a(S5)$).

For the base step, let $k = 0$. Then $\Theta \Rightarrow \Omega \equiv \psi \Rightarrow \psi$ and $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi \equiv \Box_a \Phi, \psi \Rightarrow \psi, \Box_a \Psi$. By Claim 1, $\psi \not\equiv \Box_a \varphi$, then $\Box_a \Phi, \psi \Rightarrow \psi, \Box_a \Psi$ can be derived by structure rules.

For inductive step, let $k > 0$ and fix any sequent $\Theta \Rightarrow \Omega$ in the derivation above the lower sequent of K' whose height is k . Then, we divide our argument depending on the last applied rule to obtain $\Theta \Rightarrow \Omega$.

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of an a-cut. Then we have the last derivation as follows:

$$\frac{\Theta' \Rightarrow \Omega', \psi \quad \psi, \Theta'' \Rightarrow \Omega''}{\Theta', \Theta'' \Rightarrow \Omega', \Omega''} (Cut)_\psi,$$

where $\psi \in \text{Sub}(\Theta', \Theta'', \Omega', \Omega'')$. By induction hypothesis, $\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi, \psi$ and $\psi, \Box_a \Phi, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi$ are derivable in $G^a(\mathbf{S5})$. We divide the argument into the following two cases: $\psi \in \text{Sub}(\Box_a \varphi)$ and $\psi \notin \text{Sub}(\Box_a \varphi)$

- Suppose $\psi \in \text{Sub}(\Box_a \varphi)$. Since $\psi \neq \Box_a \varphi$ by Claim 1, we have $\psi \in \text{Sub}(\varphi)$. We have the following derivation:

$$\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi, \psi \quad \psi, \Box_a \Phi, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi} (Cut)_\psi.$$

This (Cut) is removable by our assumption of Lemma 40.

- Suppose $\psi \notin \text{Sub}(\Box_a \varphi)$. It follows that $\psi \in \text{Sub}(\Theta', \Theta'', \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi})$. Then, our goal is obtained by:

$$\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi, \psi \quad \psi, \Box_a \Phi, \Theta'' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} (Cut)_\psi$$

$$\frac{\Box_a \Phi, \Box_a \Phi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi}{\Box_a \Phi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi} (c).$$

- Let $\Box_a \Theta \Rightarrow \Box_a \Omega$ be the lower sequent of $(\Box_{SS})$. Our goal is to show $\Box_a \Phi, \Box_a \Theta \Rightarrow \Box_a(\Omega_\varphi), \Box_a \Psi$ is derivable in $G^a(\mathbf{S5})$. By Claim 1 and the definition of Ξ , we have the following claim.

Claim 2. (e) $\Box_a \Theta \cap \Box_a \Psi \neq \emptyset$ or $\Box_a(\Omega_\varphi) \cap \Box_a \Phi \neq \emptyset$.

(Proof of Claim) Since $\Box_a \Theta \Rightarrow \Box_a \Omega$ does not satisfy (d), i.e., $\Box_a \Theta \not\subseteq \Box_a \Psi$ or $\Box_a \Omega \cup \{\Box_a \psi\} \not\subseteq \Box_a \Psi \cup \{\Box_a \varphi\}$. Then, we have $\exists \alpha (\alpha \in \Box_a \Theta \text{ and } \alpha \notin \Box_a \Phi)$ or $\exists \beta (\beta \in \Box_a \Omega \text{ and } \beta \notin \Box_a \Psi \cup \{\Box_a \varphi\})$.

First, fix some α such that ($\alpha \in \Box_a\Theta$ and $\alpha \notin \Box_a\Phi$). By (b), $\alpha \not\equiv \Box_a\varphi$. By definition of Ξ , $\alpha \in \Box_a\Theta \subseteq \Xi$. Since $\alpha \notin \Box_a\Phi$, we have $\alpha \in \Box_a\Psi$, which implies $\Box_a\Theta \cap \Box_a\Psi \neq \emptyset$.

Second, fix some β such that ($\beta \in \Box_a\Omega$ and $\beta \notin \Box_a\Psi \cup \{\Box_a\varphi\}$). Here we divide the argument into the following two cases: $\beta \equiv \Box_a\varphi$ and $\beta \not\equiv \Box_a\varphi$.

Let $\beta \equiv \Box_a\varphi$. Then $\beta \in \{\Box_a\varphi\} \subseteq \Box_a\Psi \cup \{\Box_a\varphi\}$, we have a contradiction with $\beta \notin \Box_a\Psi \cup \{\Box_a\varphi\}$.

Let $\beta \not\equiv \Box_a\varphi$. Then $\beta \in \Box_a(\Omega_\varphi)$ and $\beta \notin \Box_a\Psi$. By definition of Ξ , $\beta \in \Box_a(\Omega_\varphi) \subseteq \Xi$. Then by $\beta \notin \Box_a\Psi, \beta \in \Box_a\Phi$. Then $\Box_a(\Omega_\varphi) \cup \Box_a\Phi \neq \emptyset$.
(QED of Claim)

By Claim 2, $\Box_a\Phi, \Box_a\Theta \Rightarrow \Box_a(\Omega_\varphi), \Box_a\Psi$ is derivable by structure rules.

Before we discuss the cases where $\Theta \Rightarrow \Omega$ is the lower sequent of the other remaining rules, let us establish the following claim on ψ occurring in the application K' of ($\Box_5$):

Claim 3. $\psi \not\equiv \Box_a\varphi$.

(Proof of Claim) Suppose for contradiction that $\psi \equiv \Box_a\varphi$. It follows from subformula property of $\mathcal{D}$ that $\Box_a\Box_a\varphi \equiv \Box_a\psi \in \text{Sub}(\Box_a\Gamma, \Box_a\Delta)$ and $\psi \in \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi), \varphi)$. Since $\psi \equiv \Box_a\varphi \notin \text{Sub}(\varphi)$, we get $\Box_a\varphi \in \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi))$, which contradicts (a) $\Box_a\varphi \notin \text{Sub}(\Box_a\Gamma, \Box_a(\Delta_\varphi))$.
(QED of Claim)

- Let $\Theta \Rightarrow \Omega$ be an axiom $\rho \Rightarrow \rho$. Since by 1 $\rho \not\equiv \Box_a\varphi$, our goal $\rho, \Box_a\Phi \Rightarrow \Box_a\Psi, \rho$ can be obtained by weakening rules clearly.
- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\wedge \Rightarrow)$. We have the derivation as follows:

$$\frac{\rho, \Theta' \Rightarrow \Omega}{\rho \wedge \gamma, \Theta' \Rightarrow \Omega} (\wedge \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\Phi, \rho, \Theta' \Rightarrow \Omega_{\Box_a\varphi}, \Box_a\Psi$. Then our goal is derivable by:

$$\frac{\Box_a\Phi, \rho, \Theta' \Rightarrow \Omega_{\Box_a\varphi}, \Box_a\Psi}{\Box_a\Phi, \rho \wedge \gamma, \Theta' \Rightarrow \Omega_{\Box_a\varphi}, \Box_a\Psi} (\wedge \Rightarrow).$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\Rightarrow \wedge)$. We have the derivation as follows:

$$\frac{\Theta \Rightarrow \Omega, \rho \quad \Theta \Rightarrow \Omega, \gamma}{\Theta \Rightarrow \Omega, \rho \wedge \gamma} (\Rightarrow \wedge).$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \rho, \Box_a \Psi$ and $\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \gamma, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \rho, \Box_a \Psi \quad \Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \gamma, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \rho \wedge \gamma, \Box_a \Psi} (\Rightarrow \wedge)$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\vee \Rightarrow)$. We have the derivation as follows:

$$\frac{\rho, \Theta \Rightarrow \Omega \quad \gamma, \Theta \Rightarrow \Omega}{\rho \vee \gamma, \Theta \Rightarrow \Omega} (\vee \Rightarrow)$$

By induction hypothesis, we obtain $\Box_a \Phi, \rho, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$ and $\Box_a \Phi, \gamma, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \rho, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi \quad \Box_a \Phi, \gamma, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \rho \vee \gamma, \Theta \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi} (\vee \Rightarrow)$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\Rightarrow \vee)$. We have the derivation as follows:

$$\frac{\Theta \Rightarrow \Omega', \rho}{\Theta \Rightarrow \Omega', \rho \vee \gamma} (\Rightarrow \vee)$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \rho, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \rho, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \rho \vee \gamma, \Box_a \Psi} (\Rightarrow \vee)$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\rightarrow \Rightarrow)$. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \Omega', \rho \quad \gamma, \Theta'' \Rightarrow \Omega''}{\rho \rightarrow \gamma, \Theta', \Theta'' \Rightarrow \Omega', \Omega''} (\rightarrow \Rightarrow)$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \rho, \Box_a \Psi$ and $\Box_a \Phi, \gamma, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \rho, \Box_a \Psi \quad \Box_a \Phi, \gamma, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, \Theta', \Theta'', \rho \rightarrow \gamma \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} (\rightarrow \Rightarrow)}{\Box_a \Phi, \rho \rightarrow \gamma, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi} (c)$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\Rightarrow \rightarrow)$. We have the derivation as follows:

$$\frac{\rho, \Theta \Rightarrow \Omega', \gamma}{\Theta \Rightarrow \Omega', \rho \rightarrow \gamma} (\Rightarrow \rightarrow)$$

By induction hypothesis, we obtain $\Box_a \Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \gamma, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \gamma, \Box_a \Psi}{\Box_a \Phi, \Theta \Rightarrow \Omega'_{\Box_a \varphi}, \rho \rightarrow \gamma, \Box_a \Psi} (\Rightarrow \rightarrow)$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\neg \Rightarrow)$. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \Omega, \rho}{\neg \rho, \Theta' \Rightarrow \Omega} (\neg \Rightarrow)$$

By induction hypothesis, we obtain $\Box_a \Phi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \rho, \Box_a \Psi$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \rho, \Box_a \Psi}{\Box_a \Phi, \neg \rho, \Theta' \Rightarrow \Omega_{\Box_a \varphi}, \Box_a \Psi} (\neg \Rightarrow).$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\Rightarrow \neg)$. We have the derivation as follows:

$$\frac{\rho, \Theta \Rightarrow \Omega'}{\Theta \Rightarrow \neg\rho, \Omega'} (\Rightarrow \neg).$$

By induction hypothesis, we obtain $\Box_a\Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a\varphi}, \Box_a\Psi$. Then our goal is derivable by:

$$\frac{\Box_a\Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a\varphi}, \Box_a\Psi}{\Box_a\Phi, \Theta \Rightarrow \neg\rho, \Omega'_{\Box_a\varphi}, \Box_a\Psi} (\Rightarrow \neg).$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\Box_S)$. We have the derivation as follows:

$$\frac{\Box_a\Theta \Rightarrow \Box_a\Omega, \rho}{\Box_a\Theta \Rightarrow \Box_a\Omega, \Box_a\rho} (\Box_S).$$

By 2, either $\Box_a\Theta \cap \Box_a\Psi \neq \emptyset$ or $\Box_a(\Omega_\varphi) \cap \Box_a\Phi \neq \emptyset$. Then $\Box_a\Phi, \Box_a\Theta \Rightarrow \Box_a(\Omega_\varphi), \Box_a\Psi$ can be obtained by weakening rules.

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of $(\Box_a \Rightarrow)$. We have the derivation as follows:

$$\frac{\rho, \Theta \Rightarrow \Omega}{\Box_a\rho, \Theta \Rightarrow \Omega} (\Box_a \Rightarrow).$$

By induction hypothesis, we obtain $\Box_a\Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a\varphi}, \Box_a\Psi$. Then our goal is derivable by:

$$\frac{\Box_a\Phi, \rho, \Theta \Rightarrow \Omega'_{\Box_a\varphi}, \Box_a\Psi}{\Box_a\Phi, \Box_a\rho, \Theta \Rightarrow \Omega'_{\Box_a\varphi}, \Box_a\Psi} (\Box_a \Rightarrow).$$

- Let $\Theta \Rightarrow \Omega$ be the lower sequent of an acceptable cut. We have the derivation as follows:

$$\frac{\Theta' \Rightarrow \Omega', \rho \quad \rho, \Theta'' \Rightarrow \Omega''}{\Theta', \Theta'' \Rightarrow \Omega', \Omega''} (Cut)_\rho,$$

where $\rho \in \text{Sub}(\Theta', \Theta'', \Omega', \Omega'')$. By induction hypothesis, we obtain $\Box_a\Phi, \Theta' \Rightarrow \Omega'_{\Box_a\varphi}, \Box_a\Psi, \rho$ and $\rho, \Box_a\Psi, \Theta'' \Rightarrow \Omega''_{\Box_a\varphi}, \Box_a\Psi$. We divide the argument into the following two cases: $\rho \in \text{Sub}(\Box_a\varphi)$ and $\rho \notin \text{Sub}(\Box_a\varphi)$.

- Suppose $\rho \in \text{Sub}(\Box_a \varphi)$. By Claim 3, $\rho \not\equiv \Box_a \varphi$. It follows that $\rho \in \text{Sub}(\varphi)$. Then, our goal is derivable by:

$$\frac{\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi, \rho \quad \rho, \Box_a \Phi, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Phi, \Box_a \Phi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi, \Box_a \Psi} (Ecut)}{\Box_a \Phi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi} (c),$$

where $(Ecut)$ above is admissible by the initial supposition of the lemma. Therefore, $\Box_a \Psi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi$ is derivable in $G^a(\mathbf{S5})$.

- Suppose $\rho \notin \text{Sub}(\Box_a \varphi)$. It follows that $\rho \in \text{Sub}(\Theta', \Theta'', \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi})$. Then our goal is derivable by:

$$\frac{\Box_a \Phi, \Theta' \Rightarrow \Omega'_{\Box_a \varphi}, \Box_a \Psi, \rho \quad \rho, \Box_a \Psi, \Theta'' \Rightarrow \Omega''_{\Box_a \varphi}, \Box_a \Psi}{\Box_a \Psi, \Theta', \Theta'' \Rightarrow \Omega'_{\Box_a \varphi}, \Omega''_{\Box_a \varphi}, \Box_a \Psi} (Cut)_\rho,$$

which is acceptable.

Therefore, we have finished establishing Lemma 40. $\square$